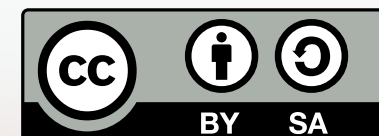


De l'Importance du Contexte

École Thématique #2 Métabolomique et Santé Humaine
Centre Paul-Langevin - CAES du CNRS, Aussois

Adriano Rutz,  2026-10-07



D BIOL



Strategic Focus Area

Personalized Health and Related Technologies

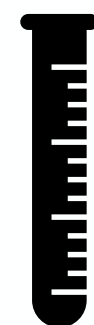
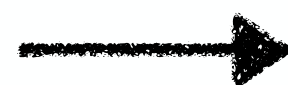


Strategic Focus Area

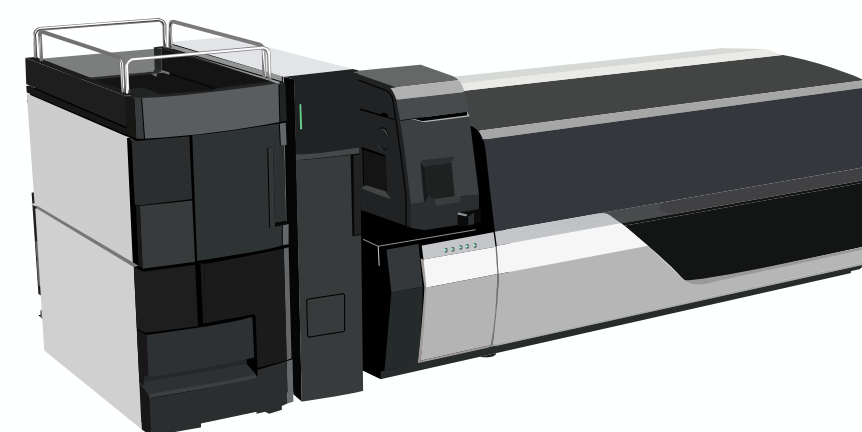
Personalized Health and Related Technologies



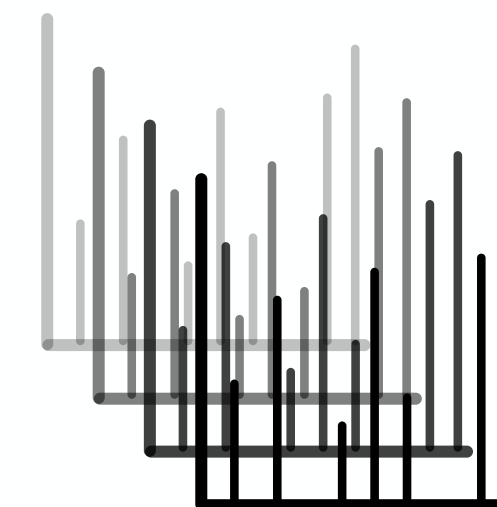
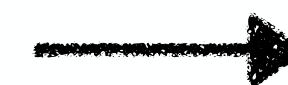
Humains



Échantillon



LC-MS



Spectres

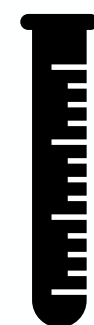
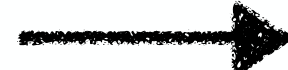


Strategic Focus Area

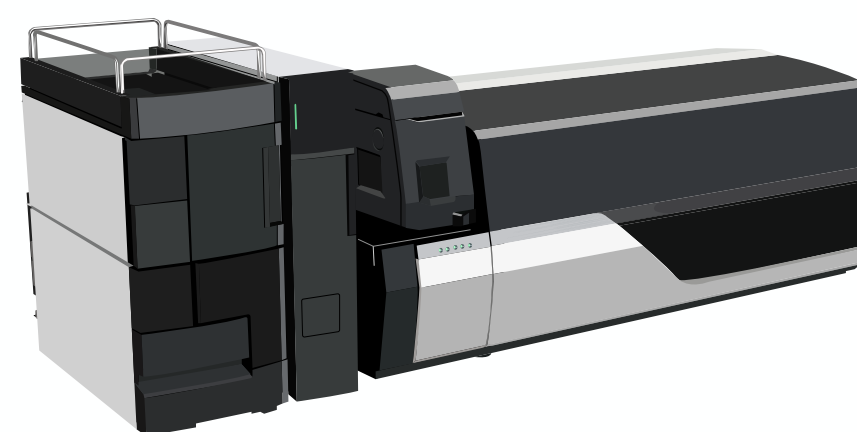
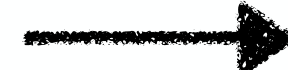
Personalized Health and Related Technologies



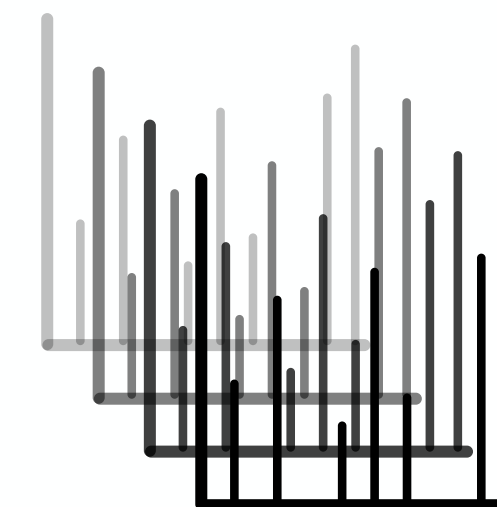
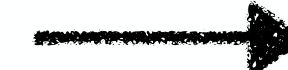
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Échantillon



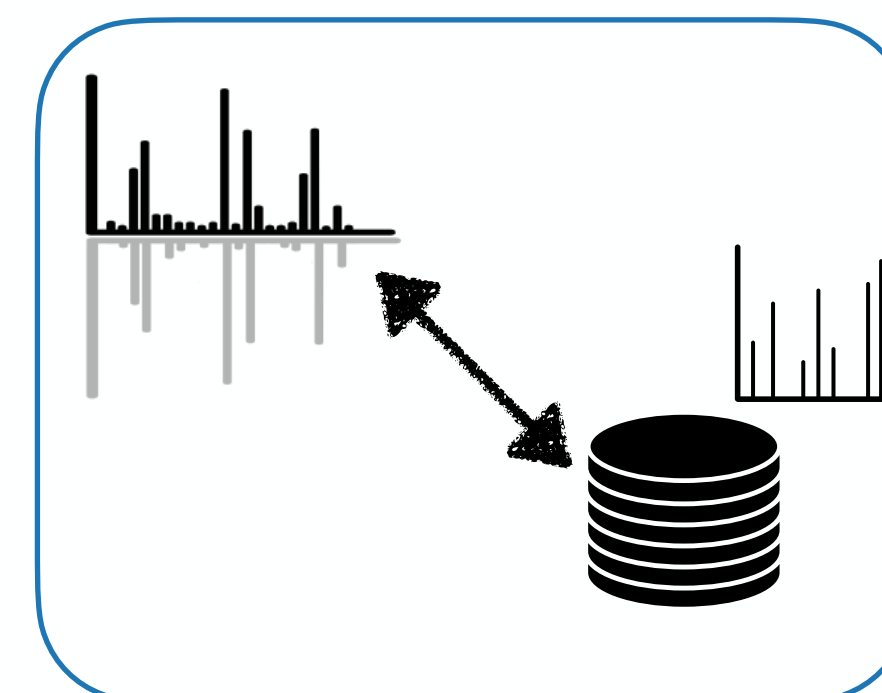
LC-MS



Spectres



Étape 1:
Libraries spectrales



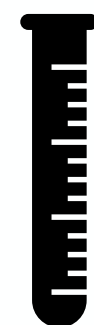
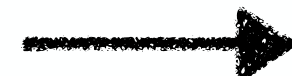


Strategic Focus Area

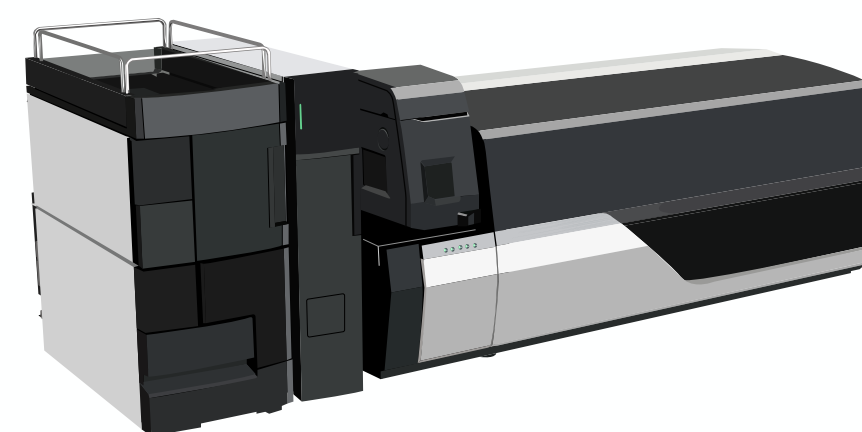
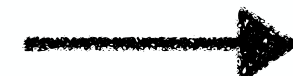
Personalized Health and Related Technologies



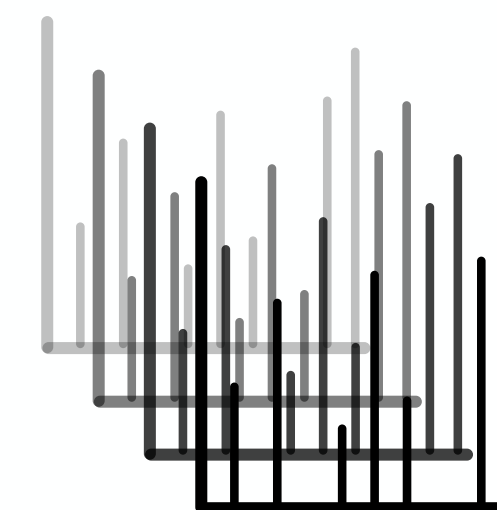
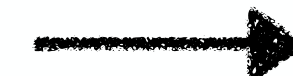
Humains



Échantillon



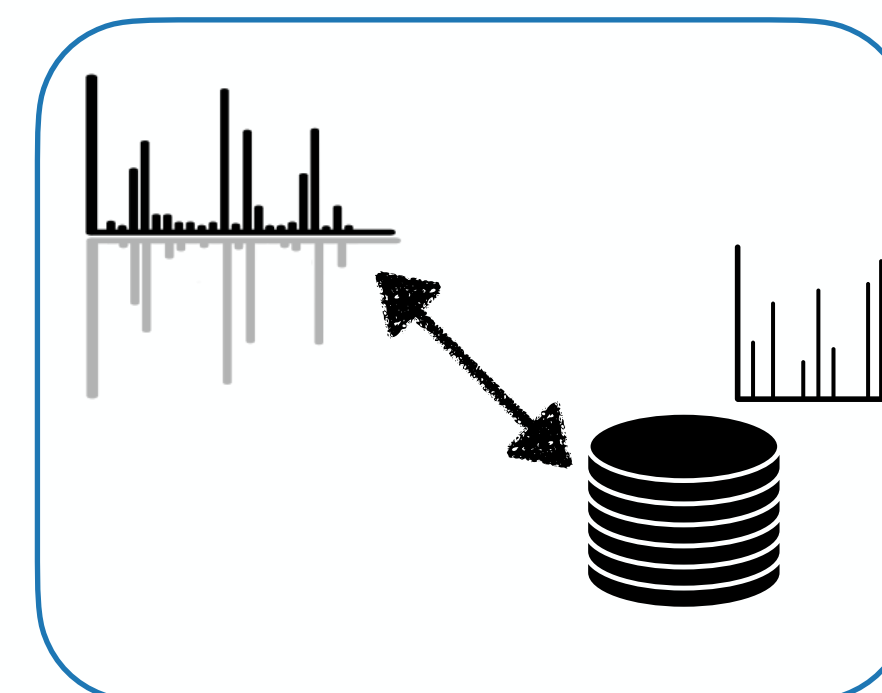
LC-MS



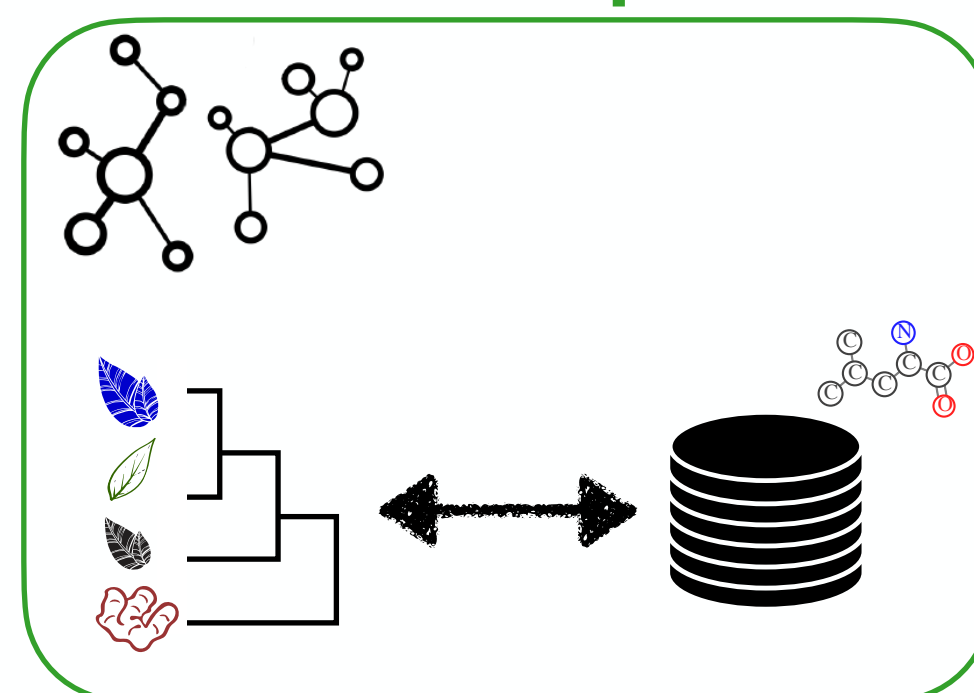
Spectres



Étape 1:
Libraries spectrales



Étape 2:
Espace chimique &
connaissances préalables



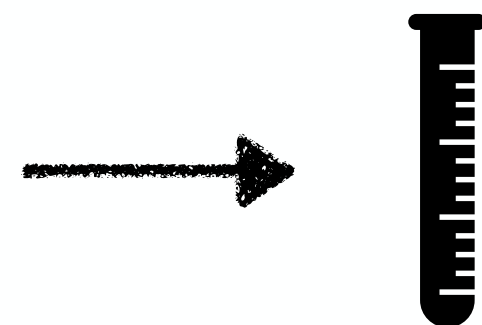


Strategic Focus Area

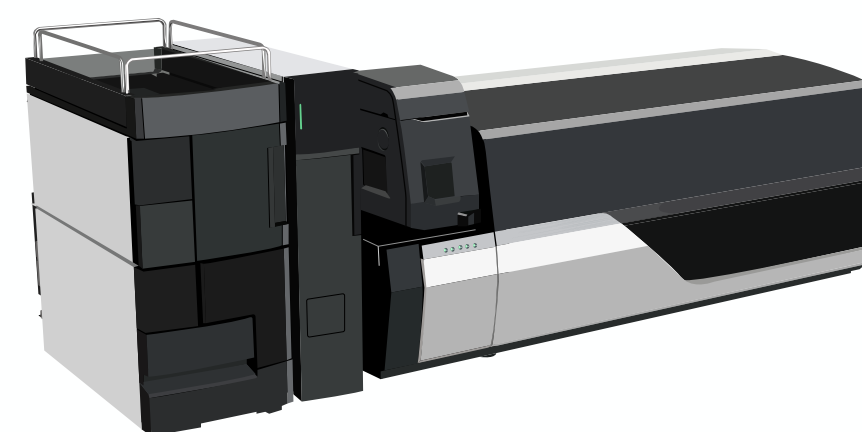
Personalized Health and Related Technologies



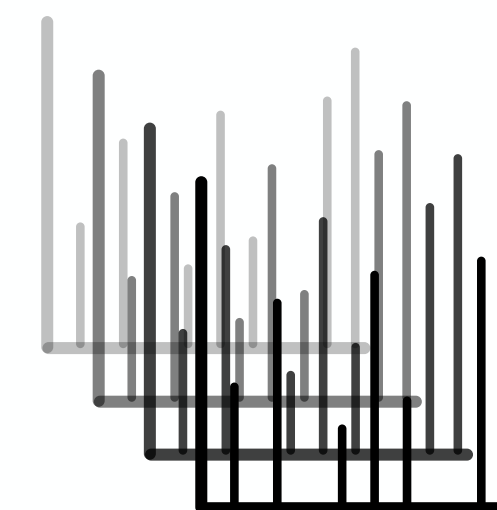
Humains



Échantillon

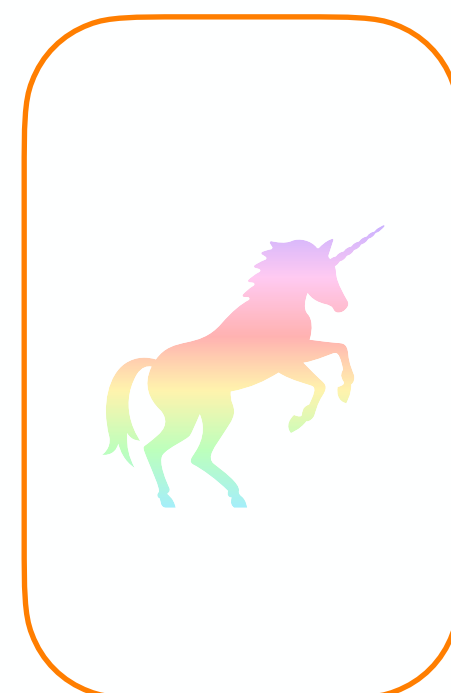


LC-MS

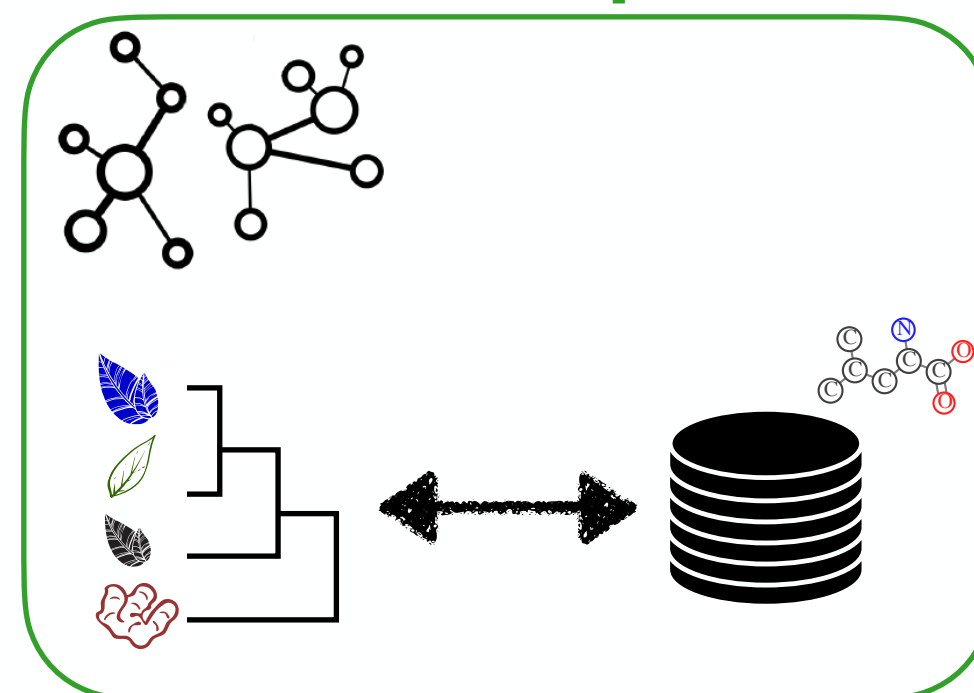


Spectres

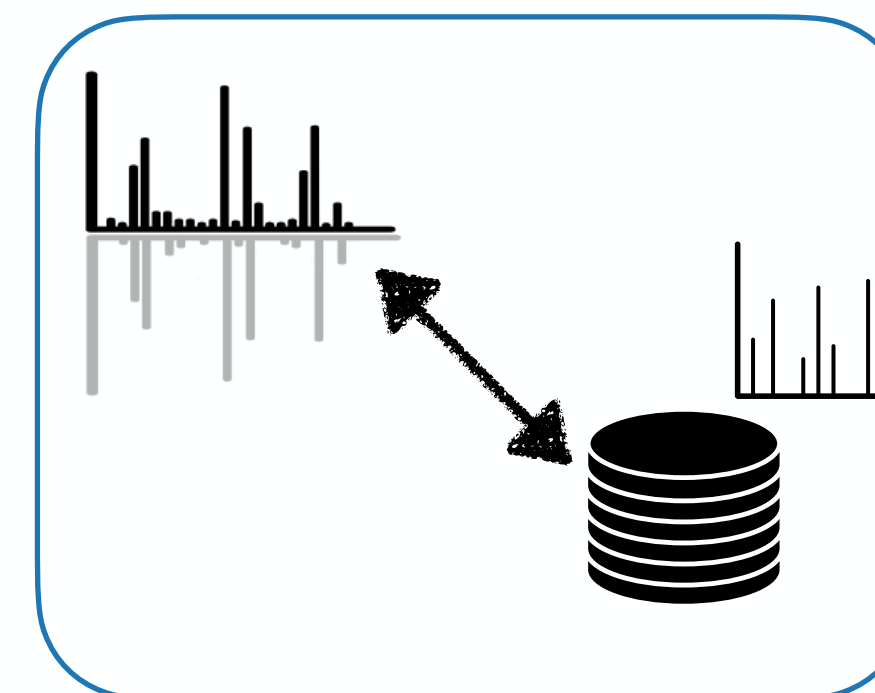
Étape 3:
Plongée dans
l'inconnu



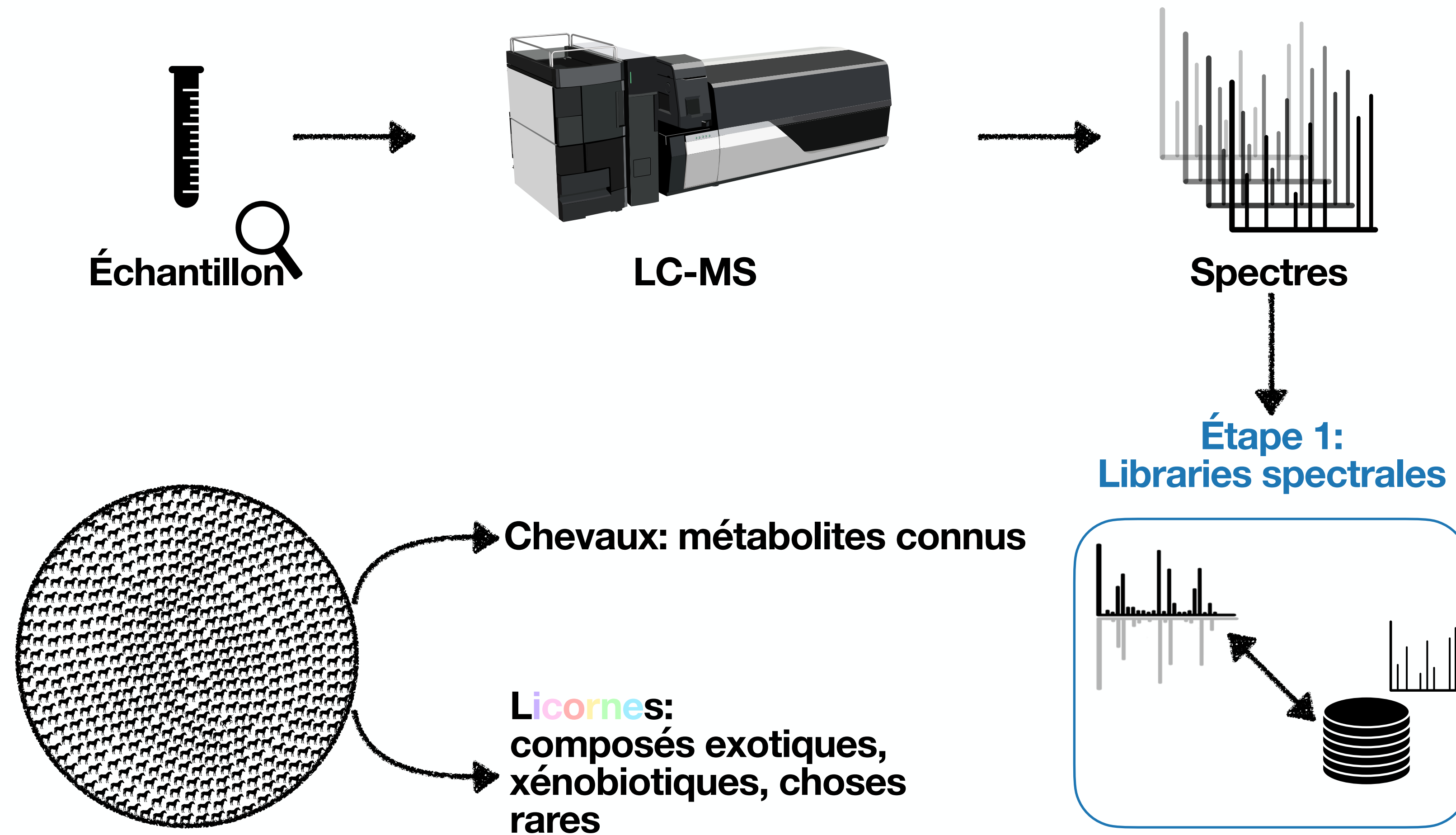
Étape 2:
Espace chimique &
connaissances préalables



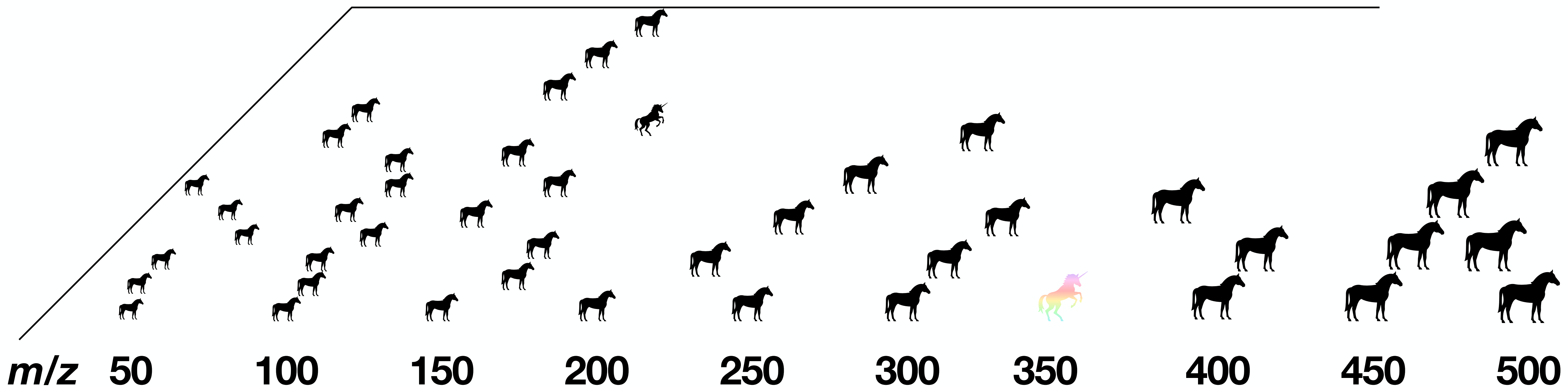
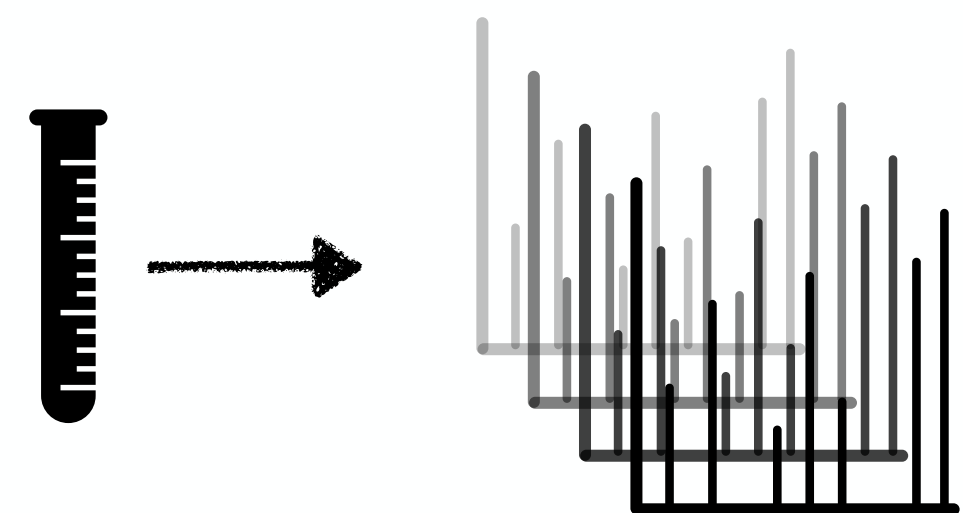
Étape 1:
Libraries spectrales



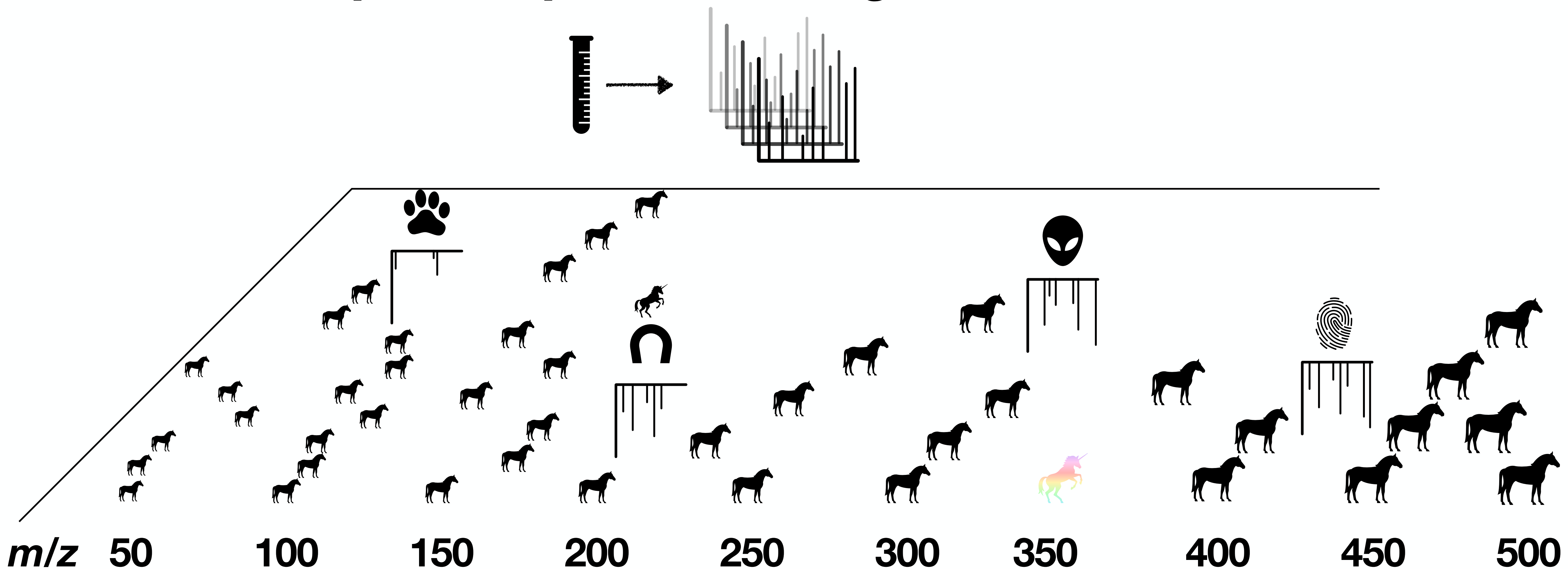
Chaque échantillon contient des chevaux et des licornes



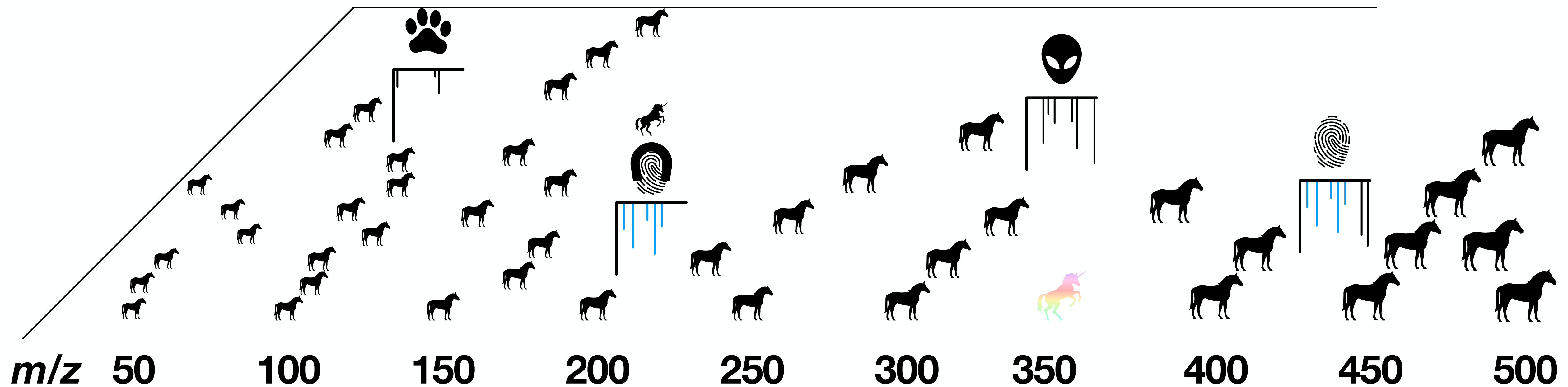
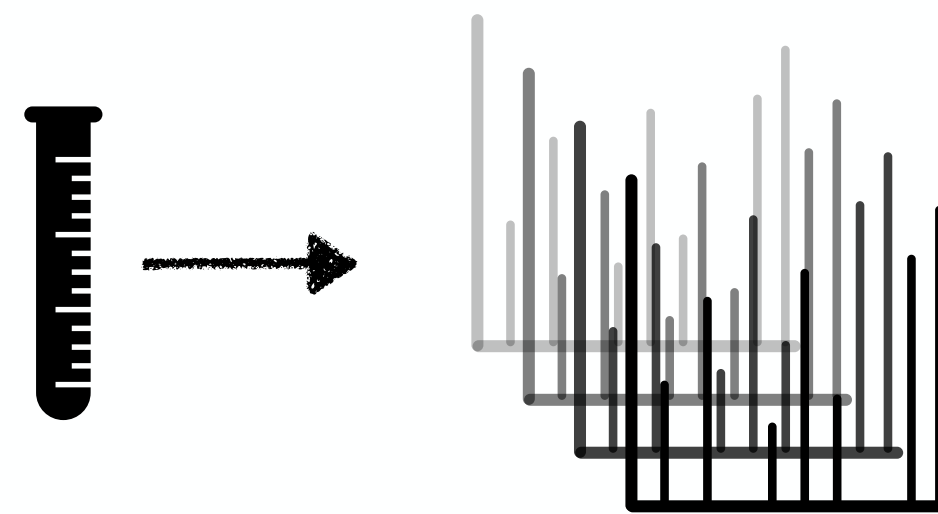
La LC-MSⁿ pour séparer le bon grain de l'ivraie



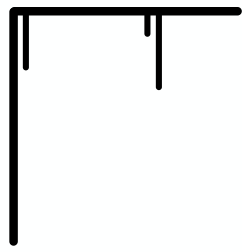
La LC-MSⁿ pour séparer le bon grain de l'ivraie



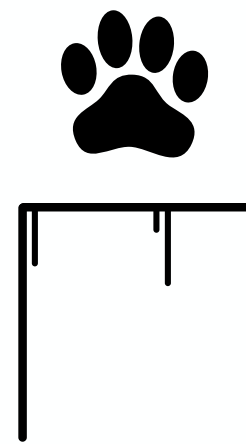
La LC-MSⁿ pour séparer le bon grain de l'ivraie



Étape 1: Libraries spectrales

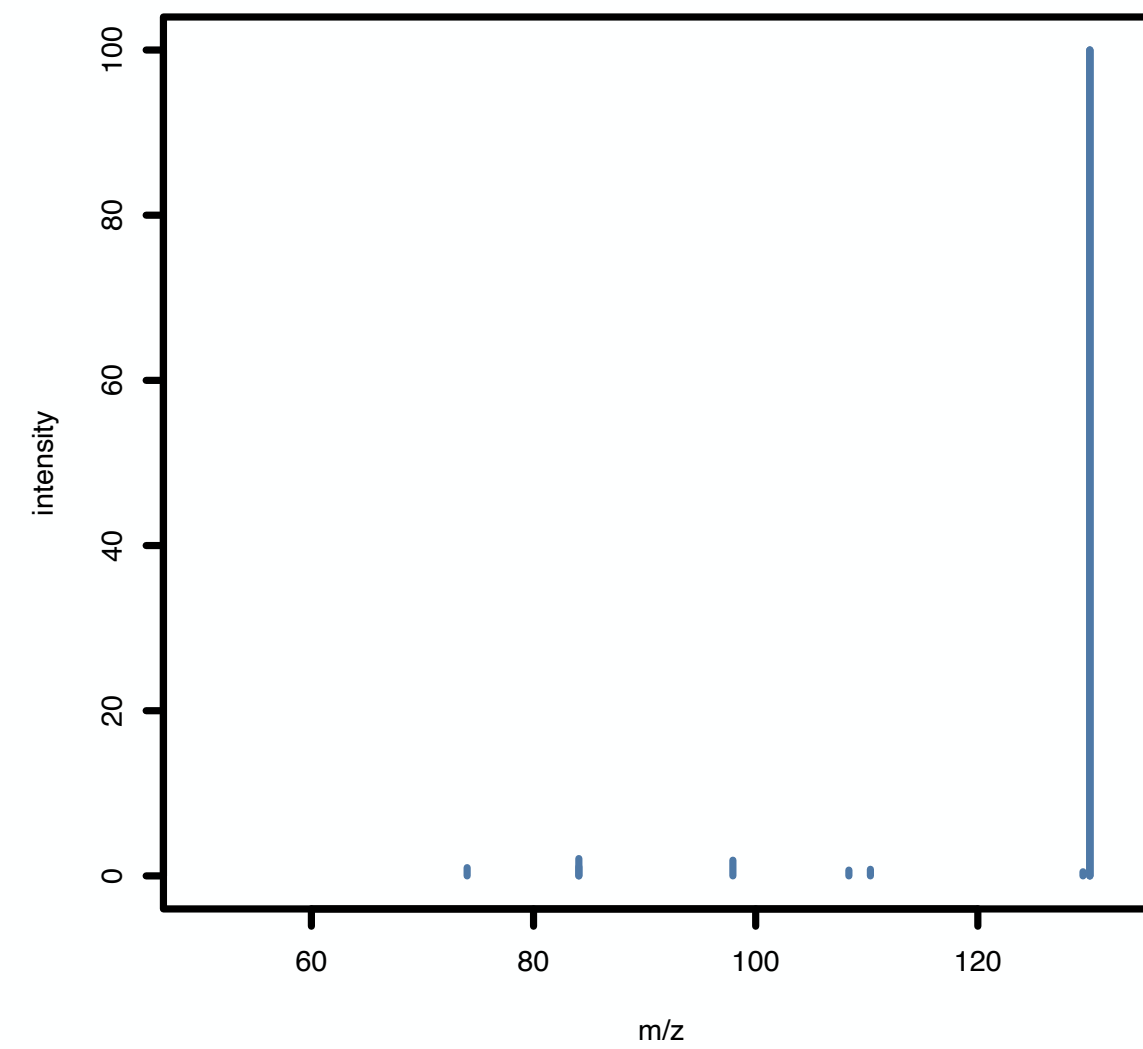


Étape 1: Libraries spectrales

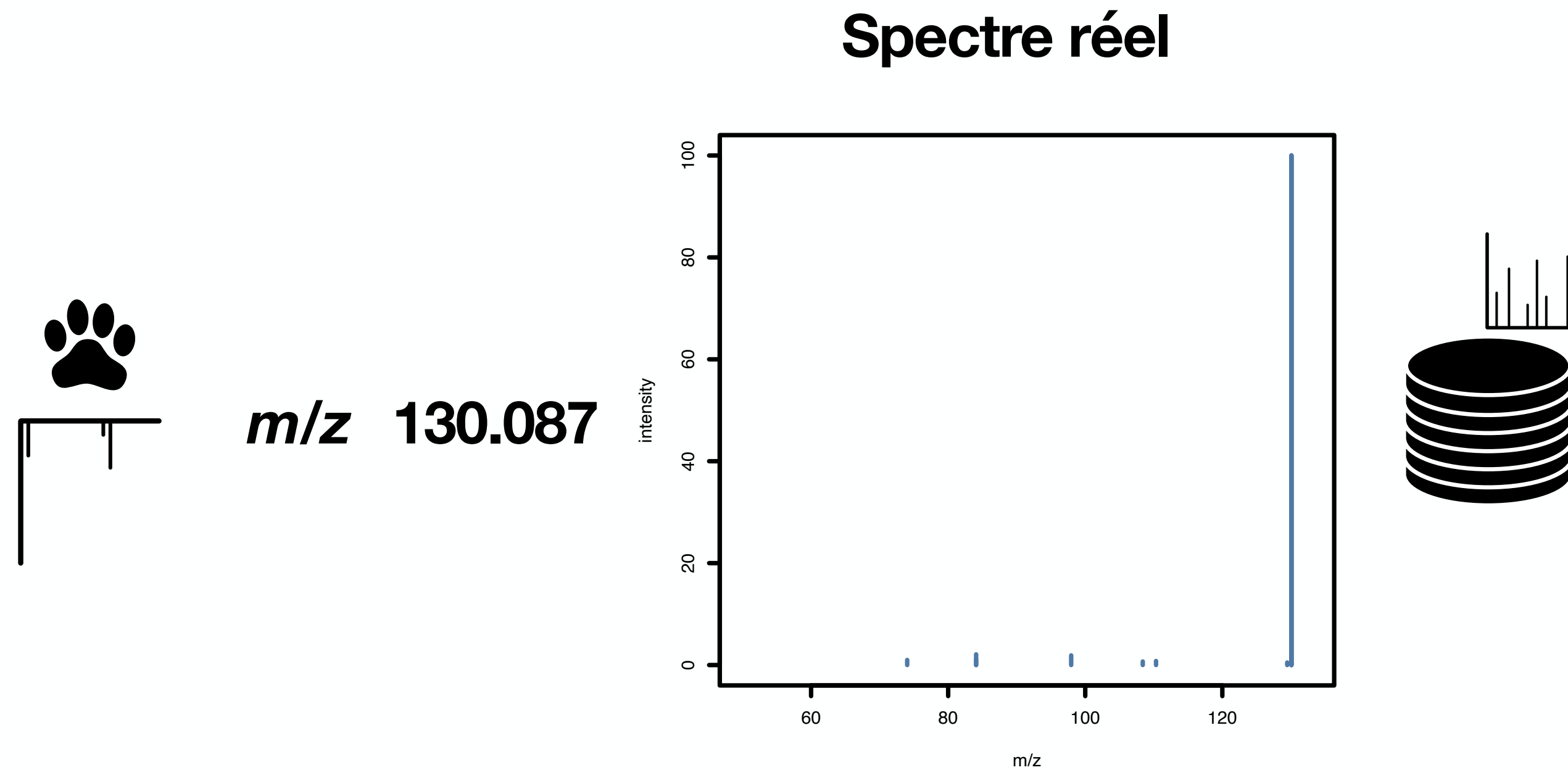


***m/z* 130.087**

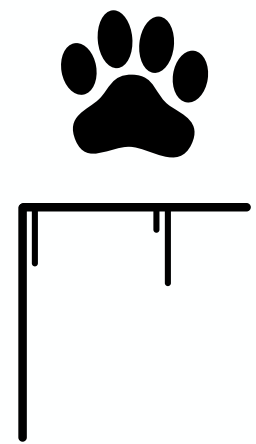
Spectre réel



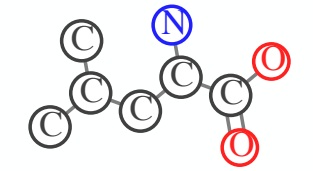
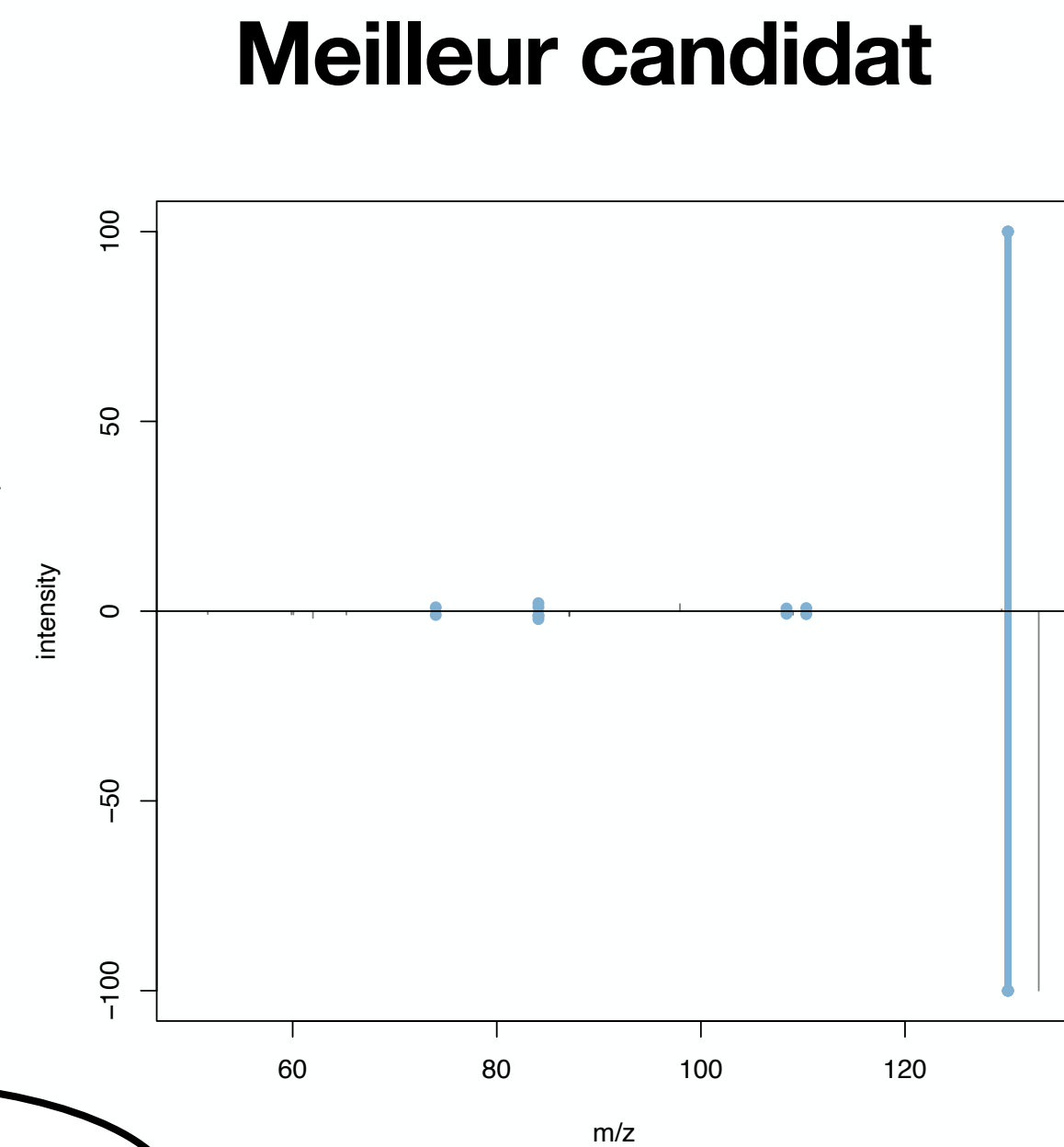
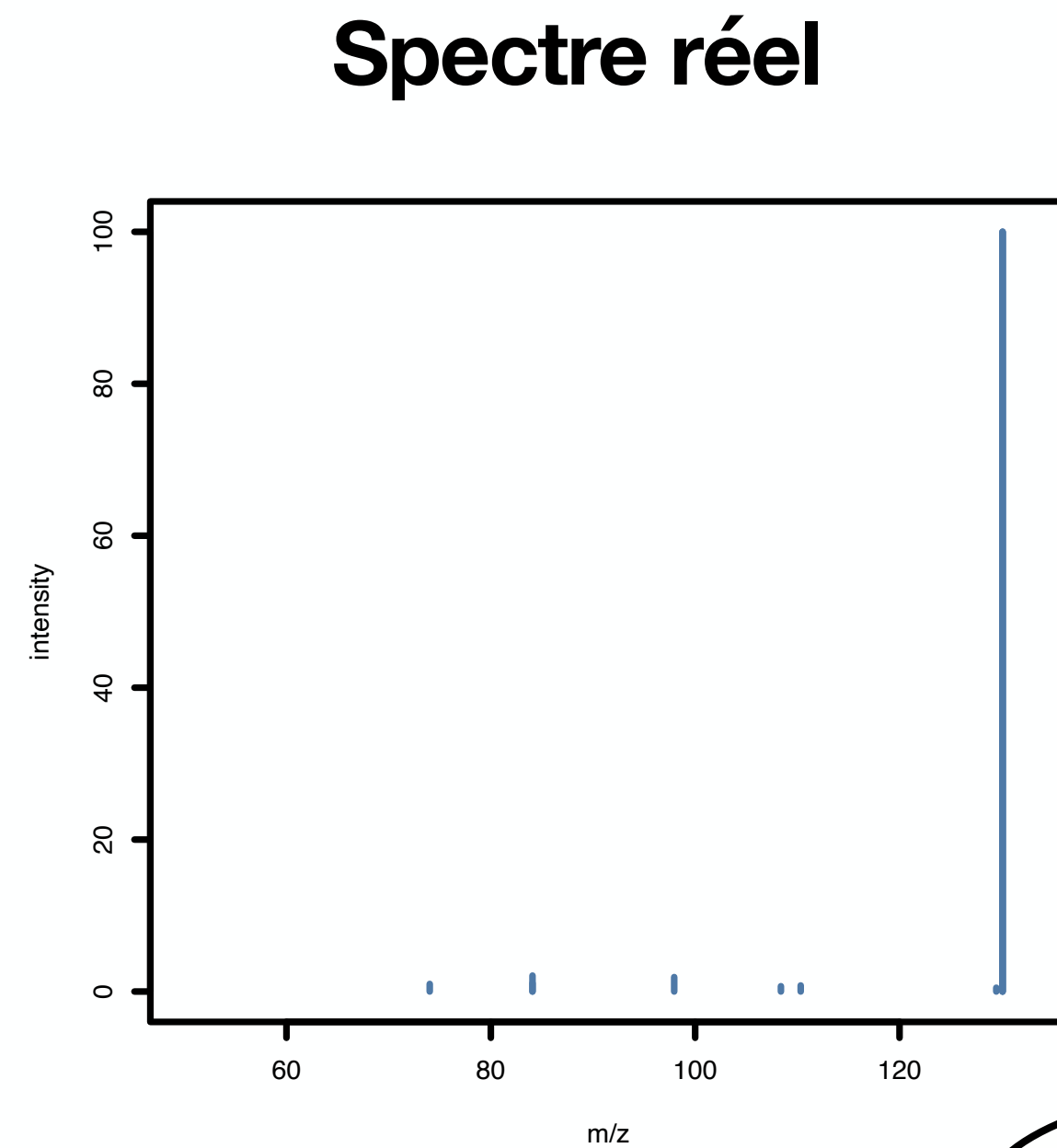
Étape 1: Libraries spectrales



Étape 1: Libraries spectrales

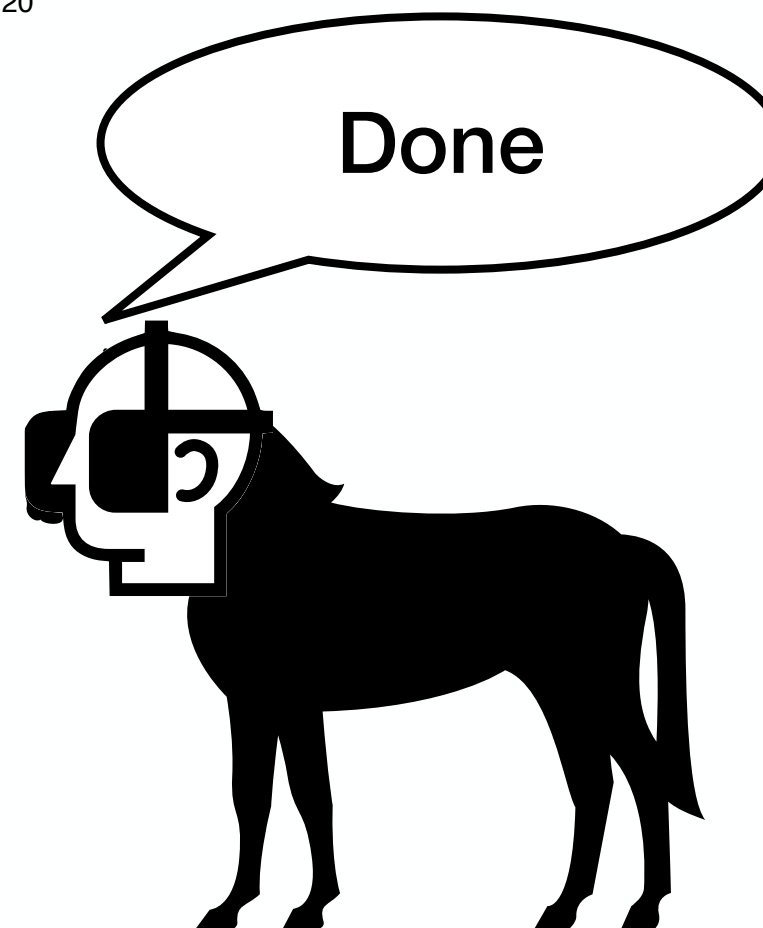


m/z 130.087

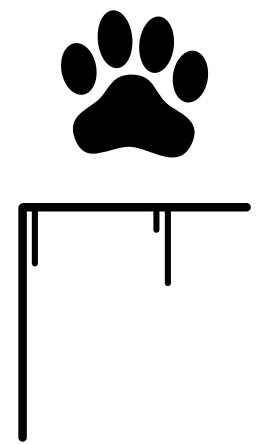


Leucine

0.64 de similarité

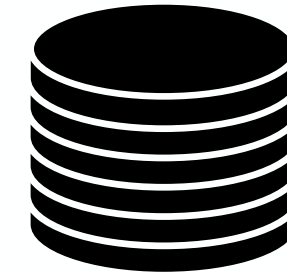
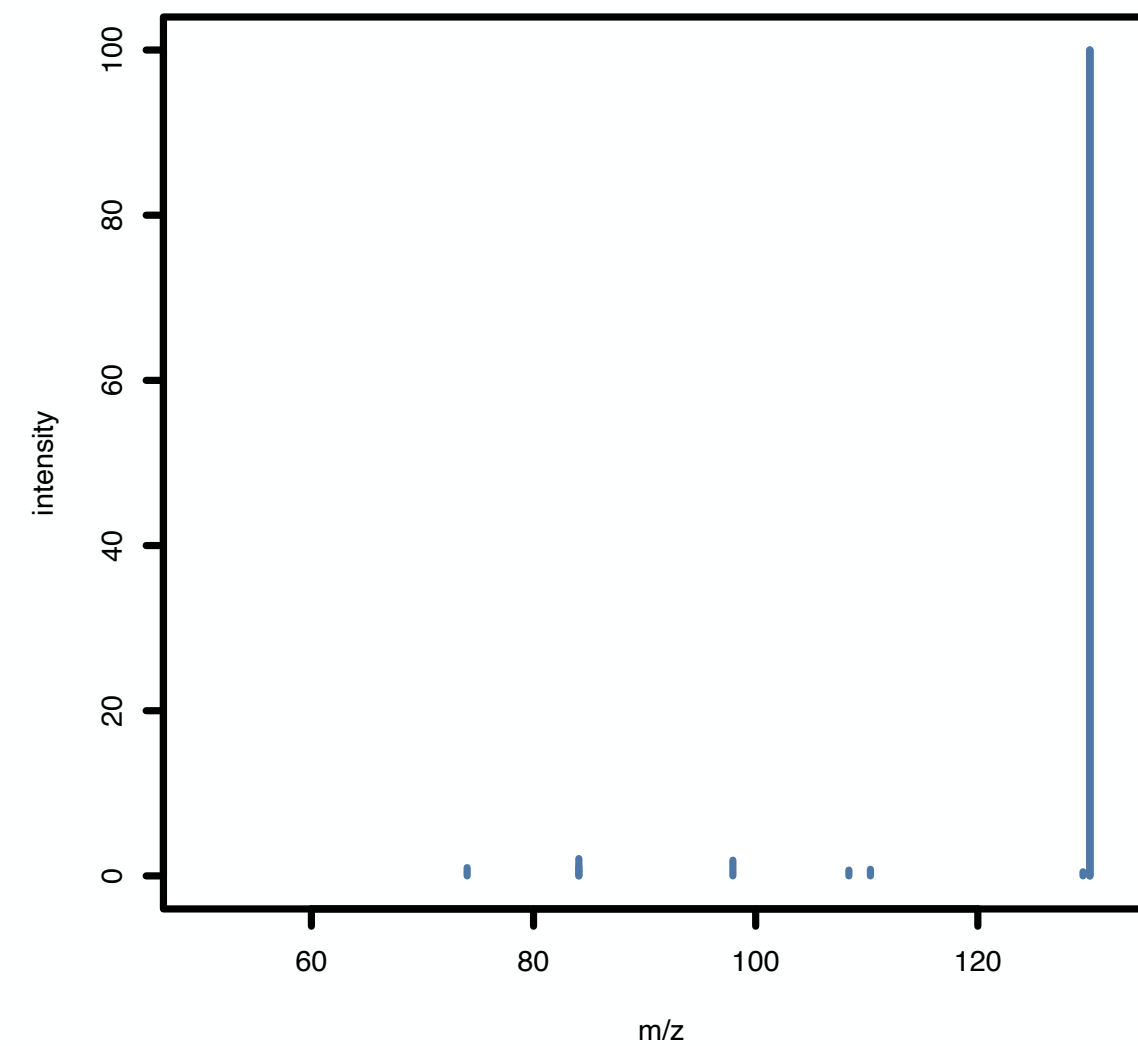


Étape 1: Gare aux bons candidats!



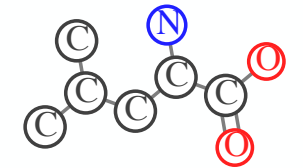
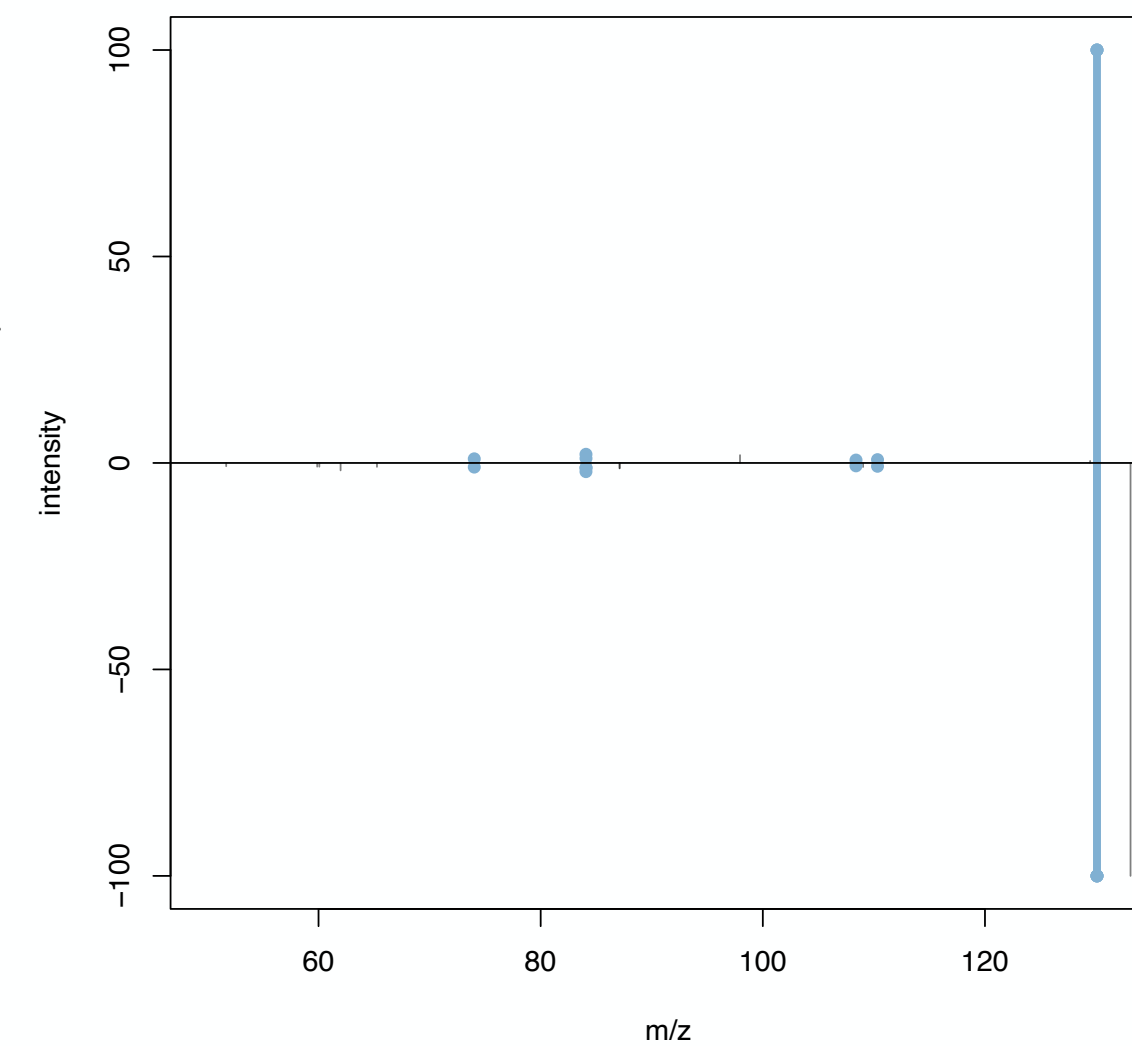
m/z 130.087

Spectre réel



Ma librairie couvre-t-elle le bon espace chimique?

Meilleur candidat

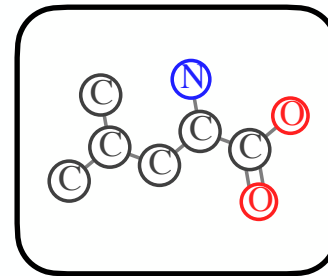


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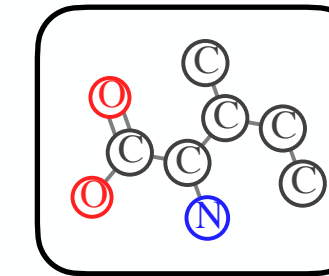
0.64 de similarité

Ces spectres sont-ils informatifs?

Étape 1: le cas de l'(iso)leucine

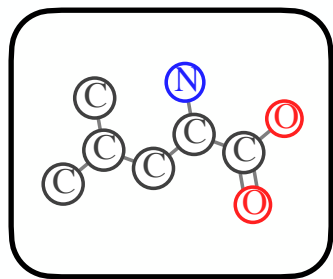


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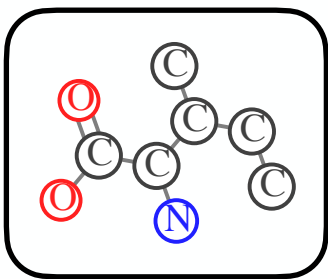


Isoleucine

Étape 1: le cas de l'(iso)leucine



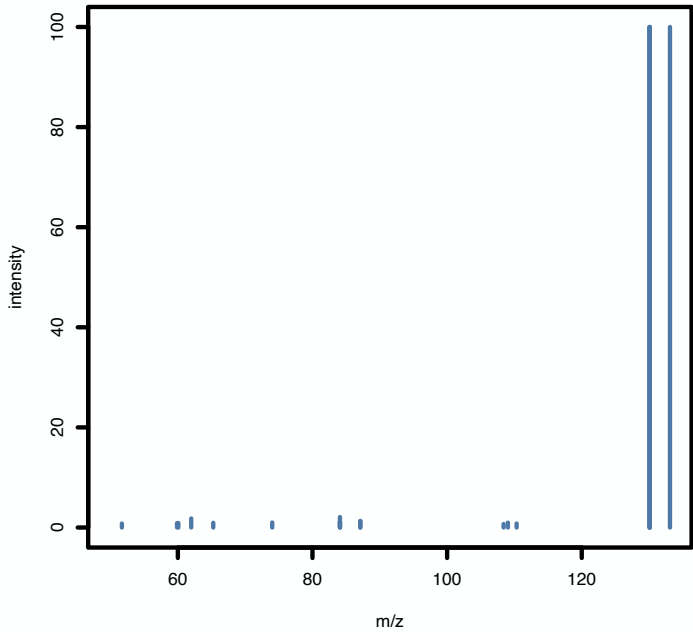
Leucine



Isoleucine

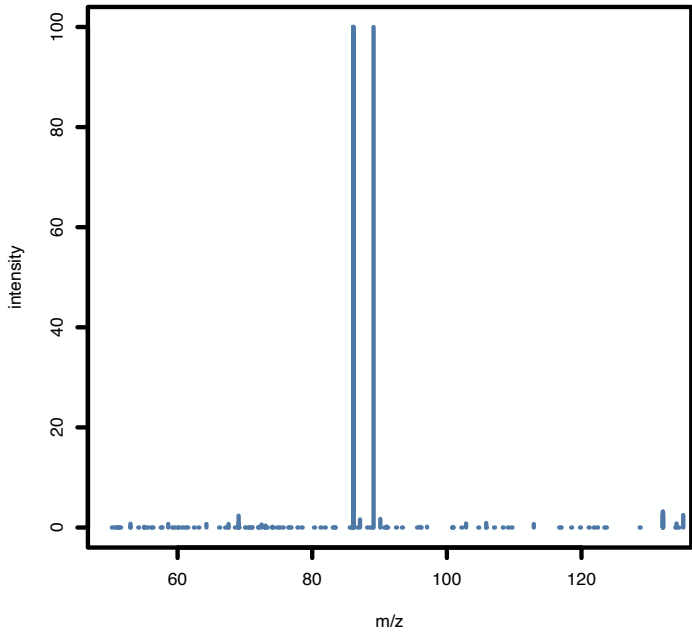
Négatif

ROHFNLRQFUQHCH 1 spectra



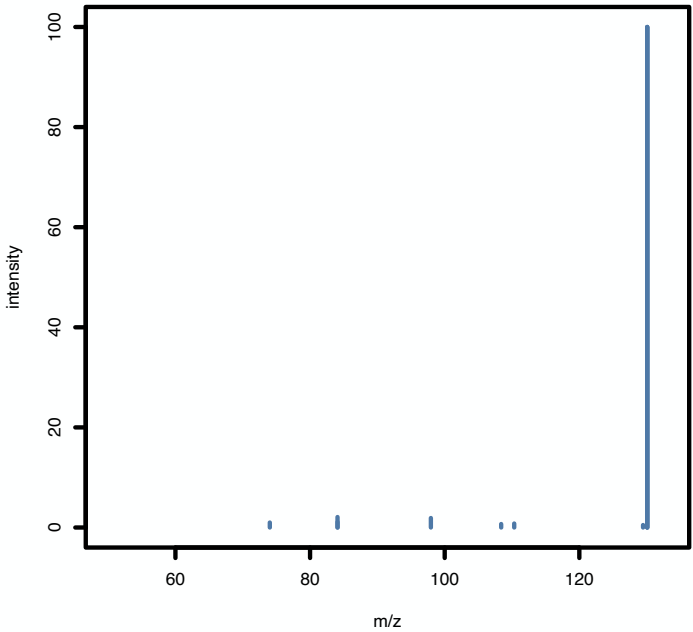
Positif

ROHFNLRQFUQHCH 1 spectra



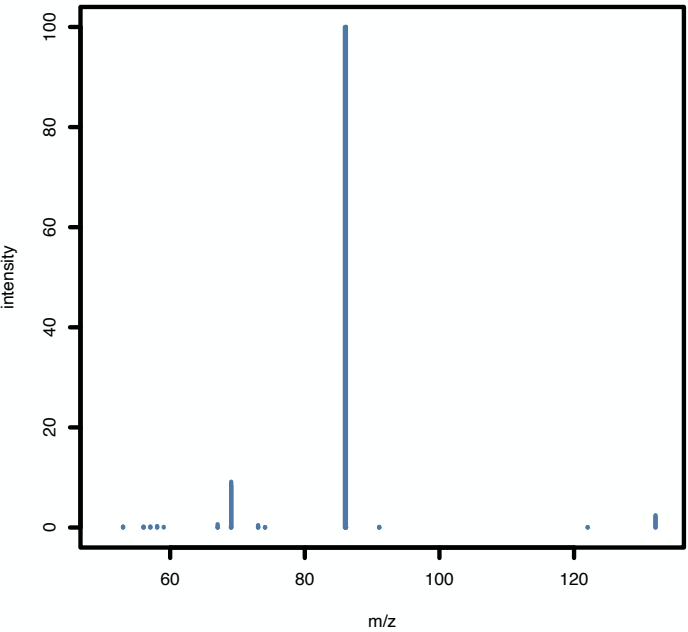
Négatif

AGPKZVBTJJNPAG 1 spectra

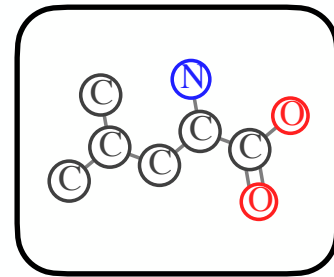


Positif

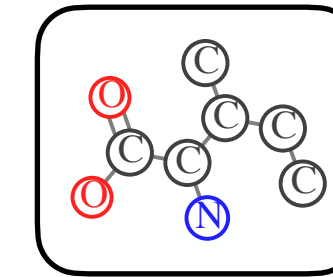
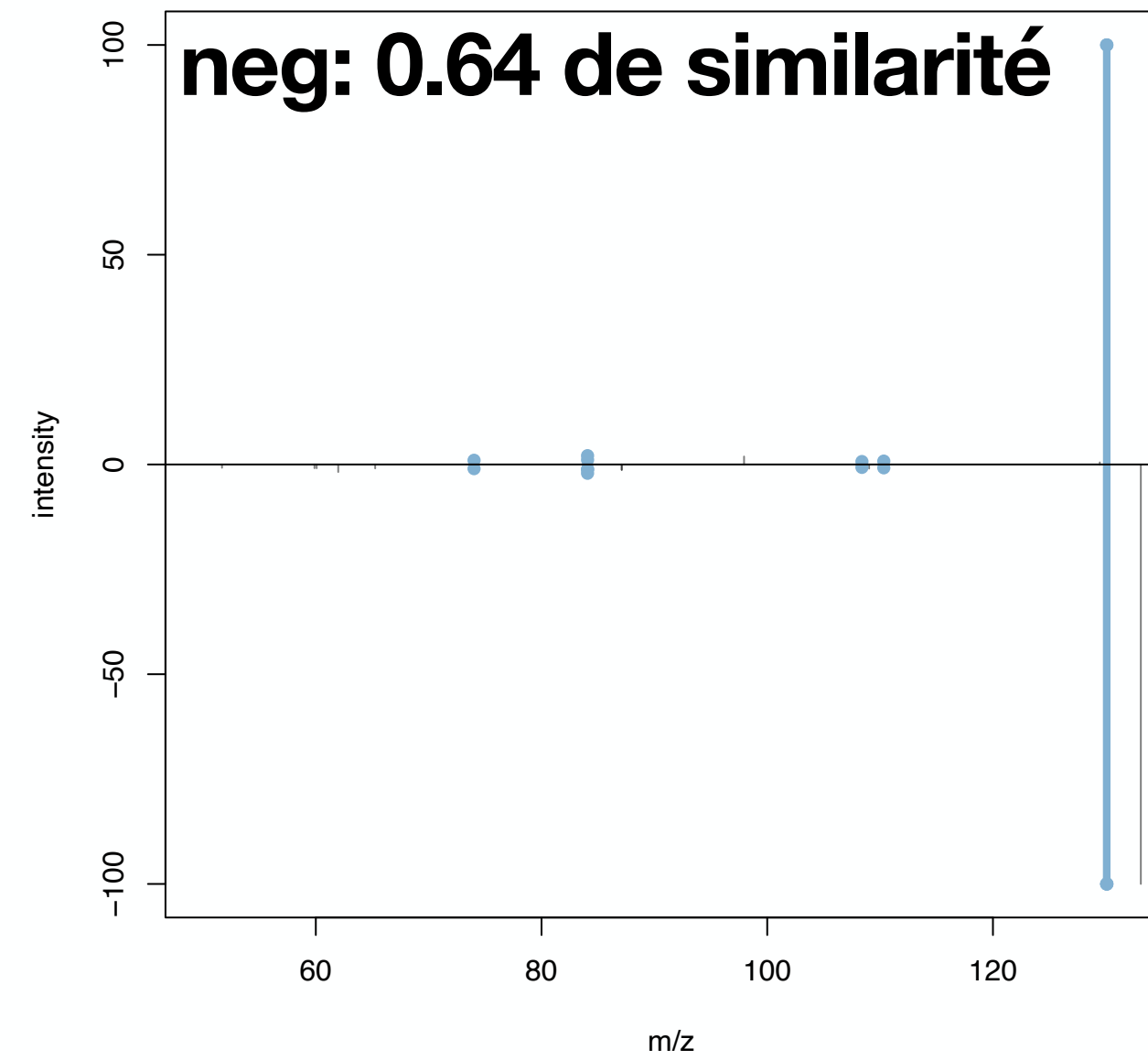
AGPKZVBTJJNPAG 1 spectra



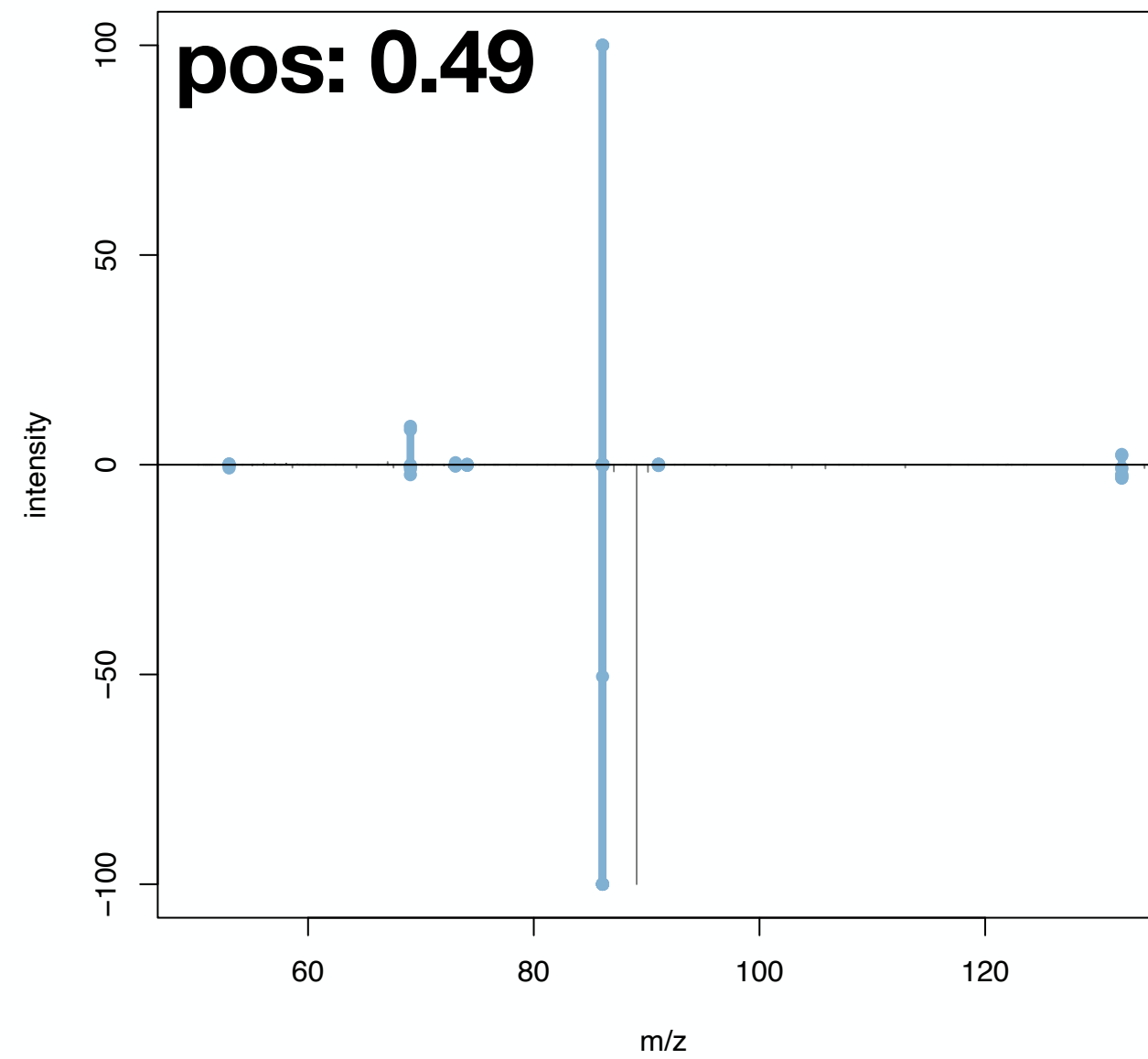
Étape 1: le cas de l'(iso)leucine



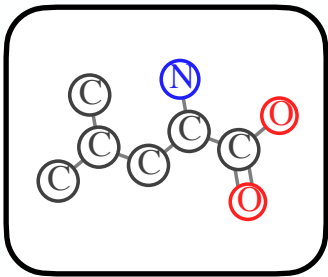
Leucine



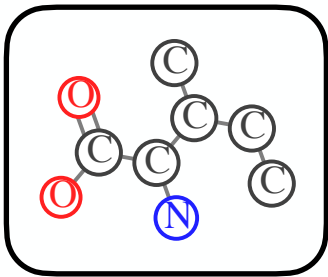
Isoleucine



Étape 1 - Information spectrale



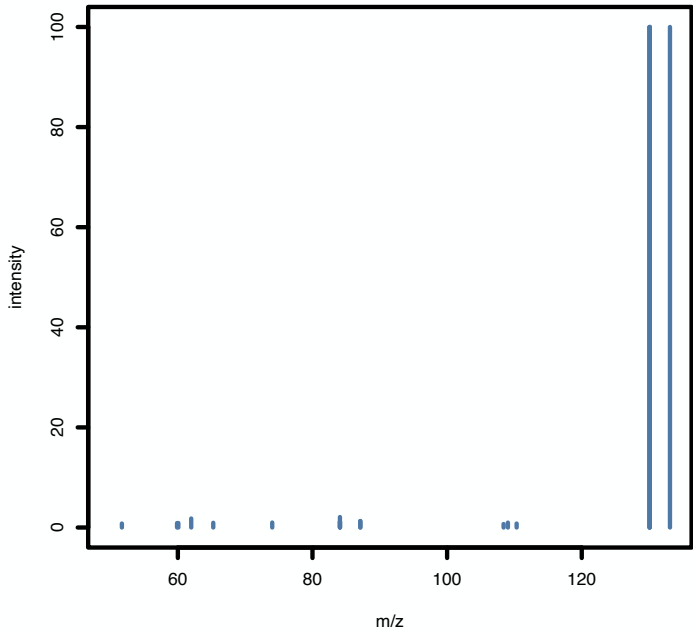
Leucine



Isoleucine

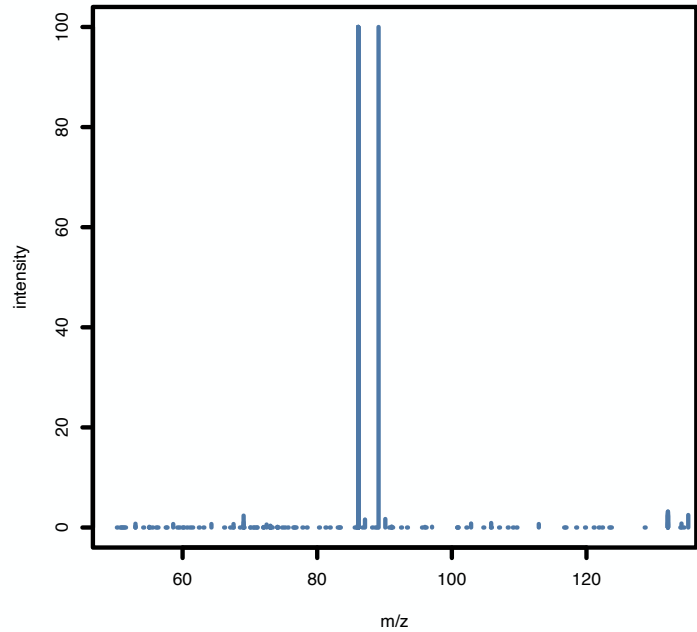
Négatif

ROHFNLRQFUQHCH 1 spectra



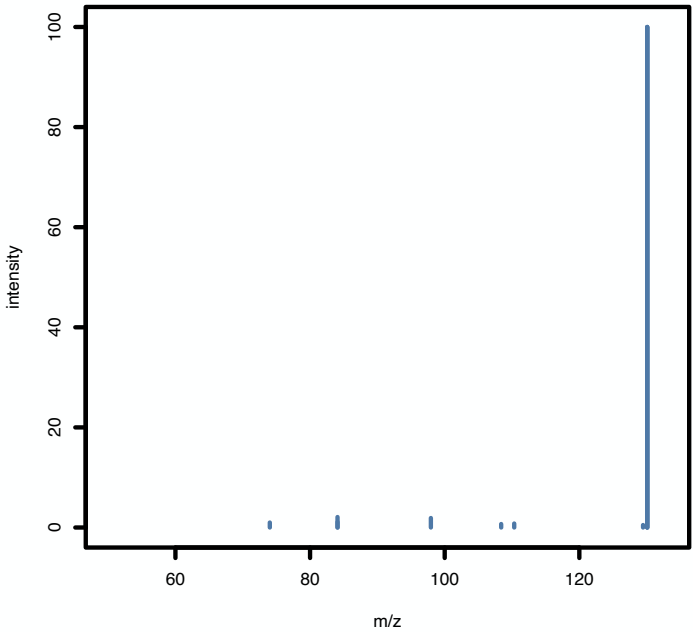
Positif

ROHFNLRQFUQHCH 1 spectra



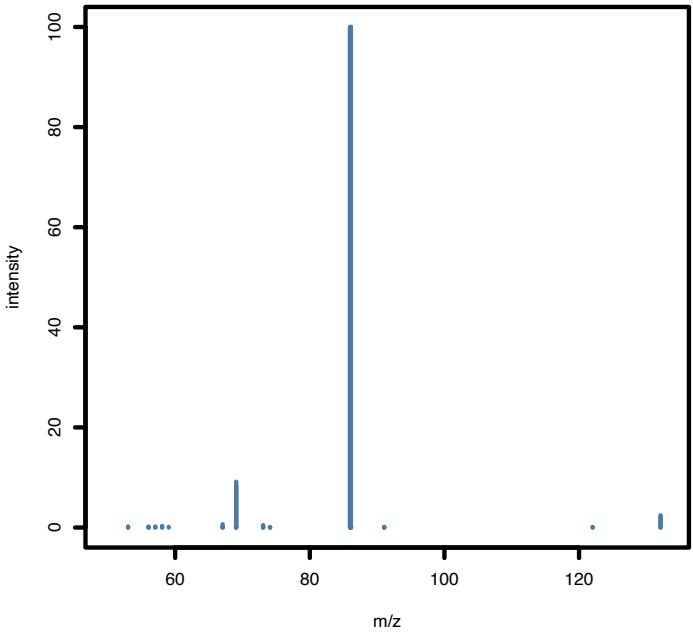
Négatif

AGPKZVBTJJNPAG 1 spectra



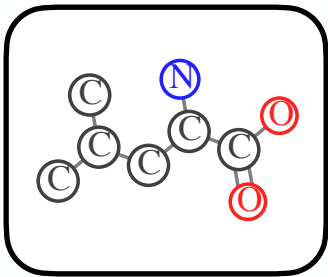
Positif

AGPKZVBTJJNPAG 1 spectra

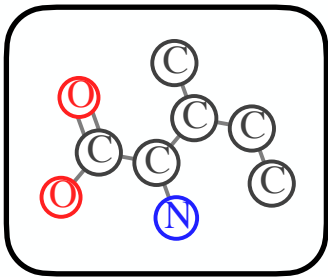


Peu
d'info

Étape 1 - Information spectrale

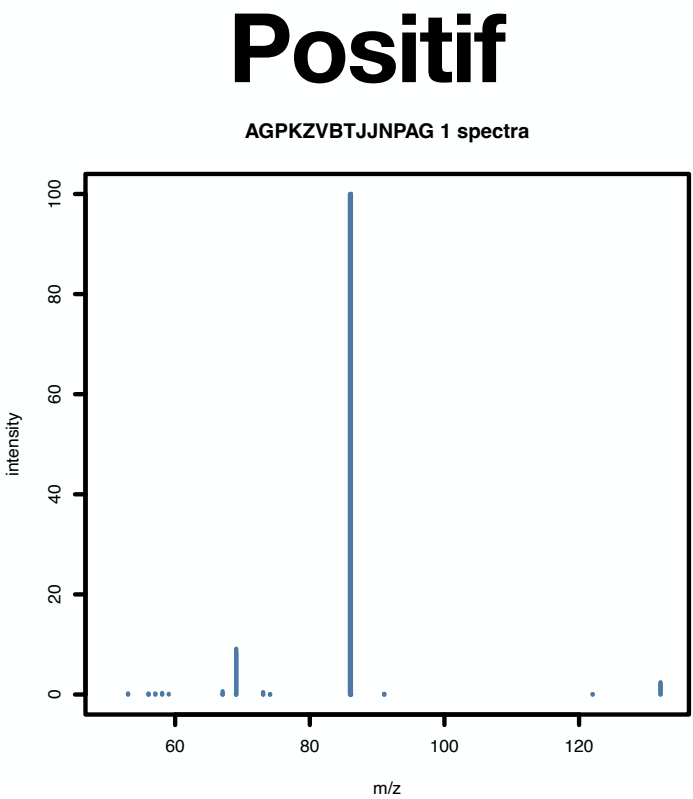
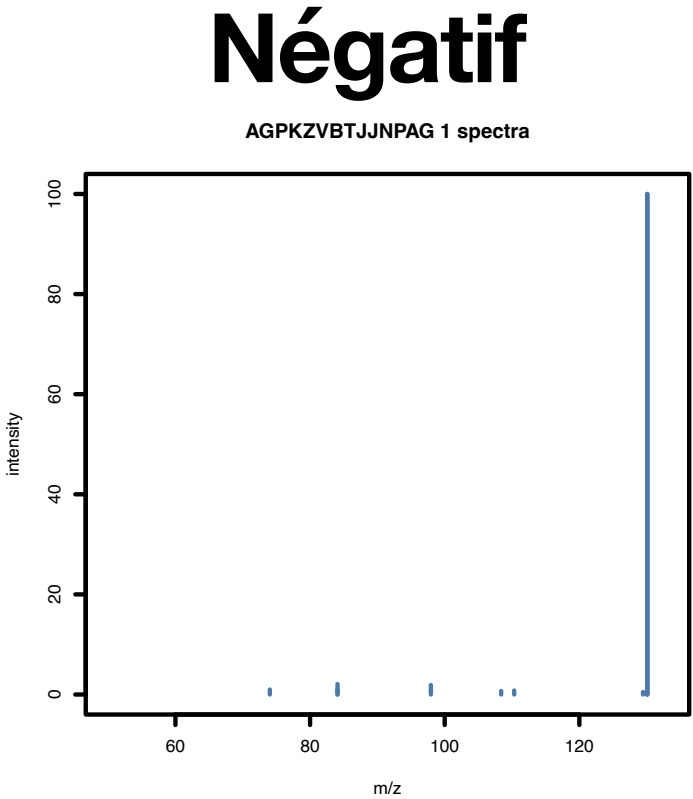
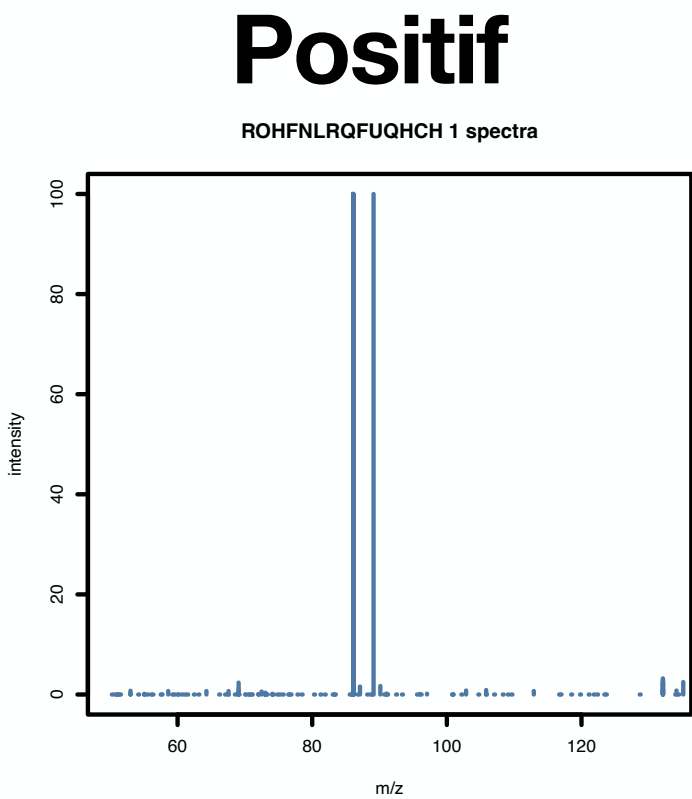
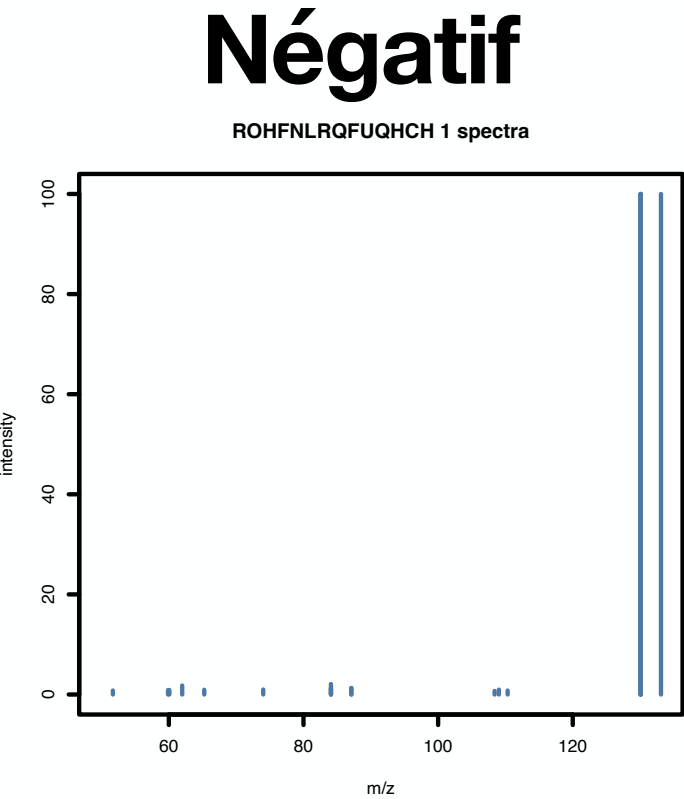


Leucine

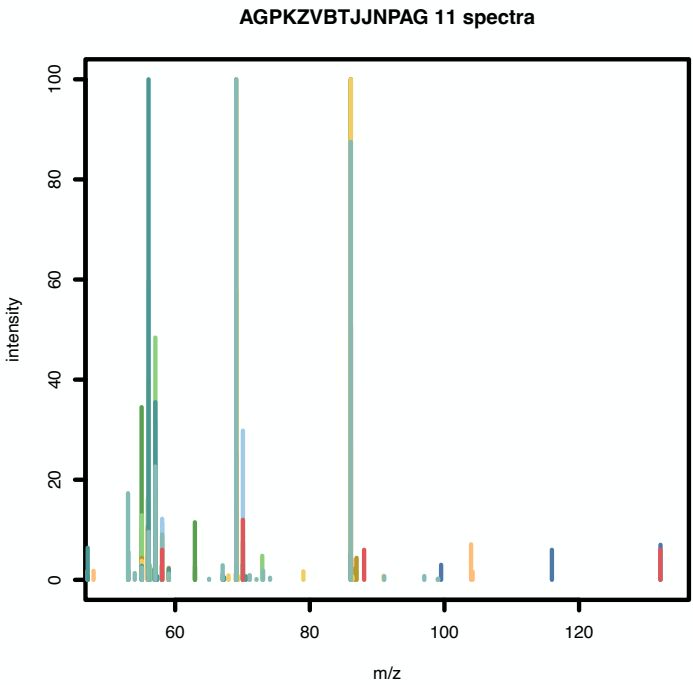
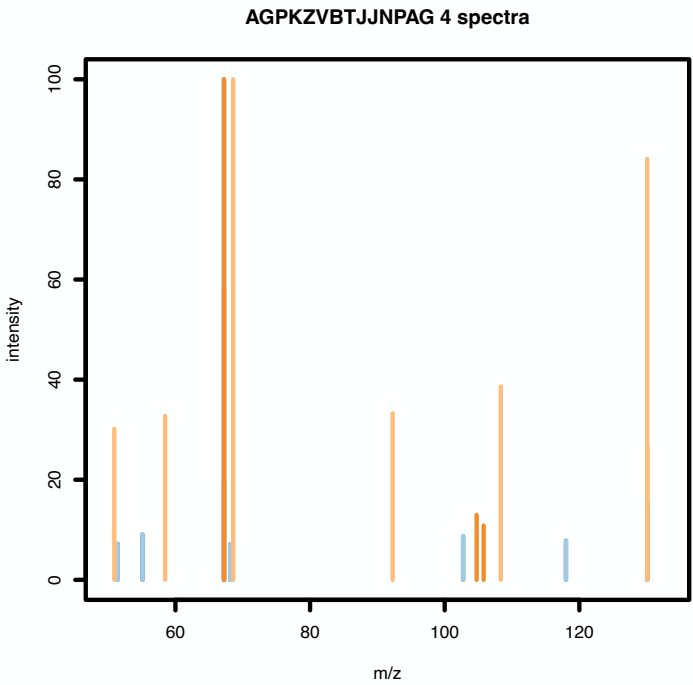
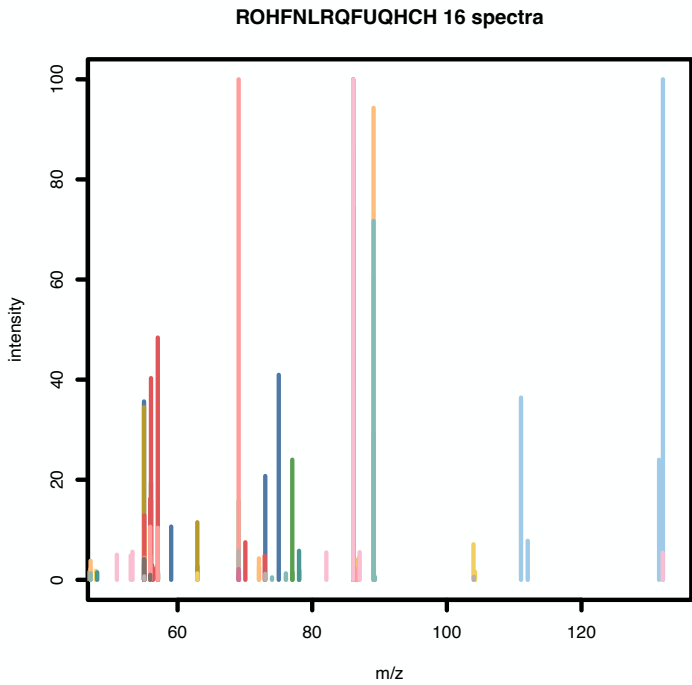
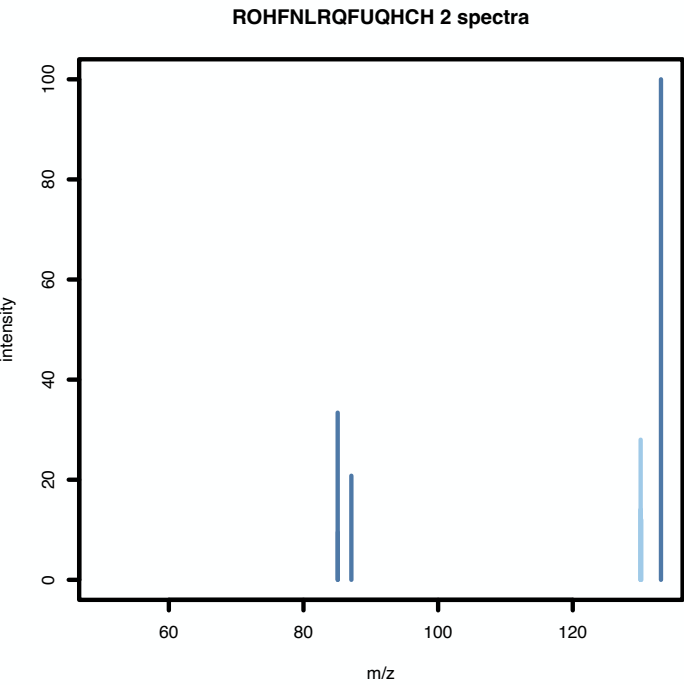


Isoleucine

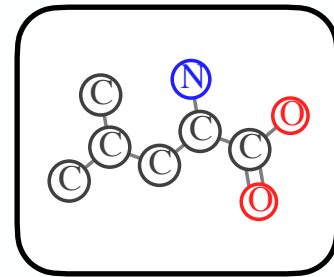
Peu
d'info



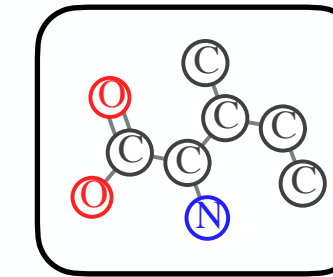
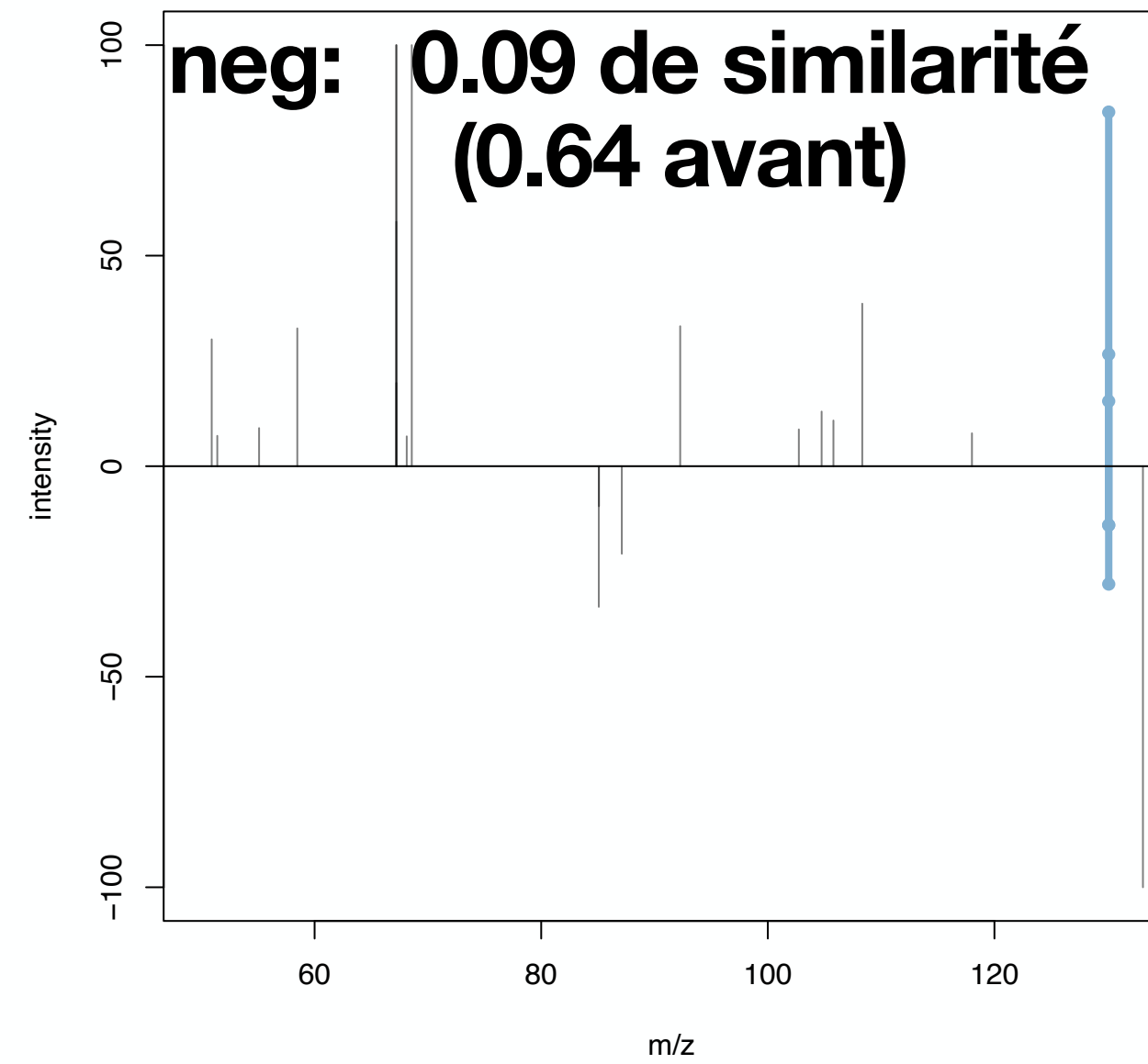
Bcp
d'info



Étape 1 - Information spectrale

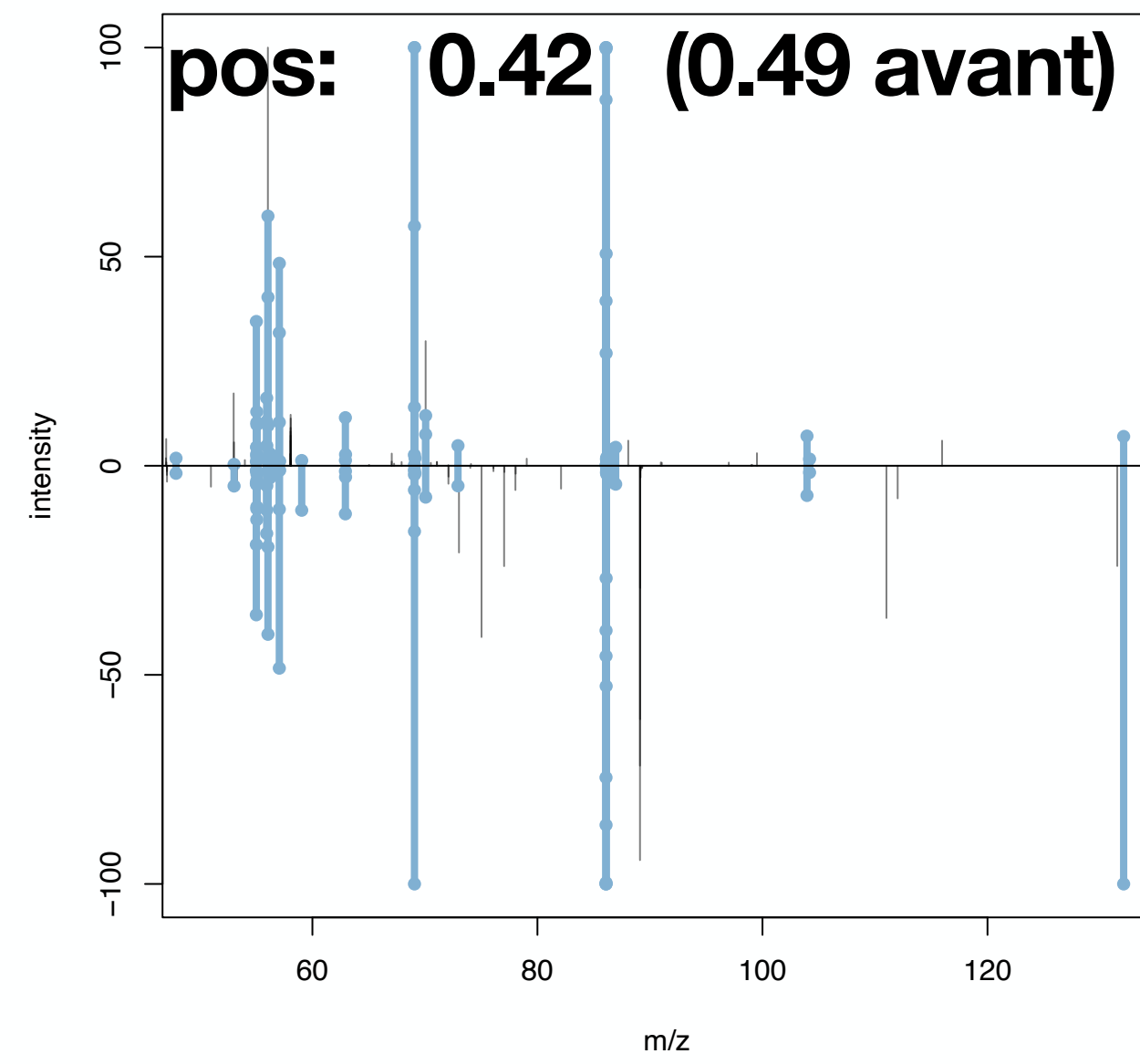


Leucine



Isoleucine

Clairement différent!

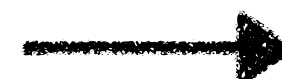
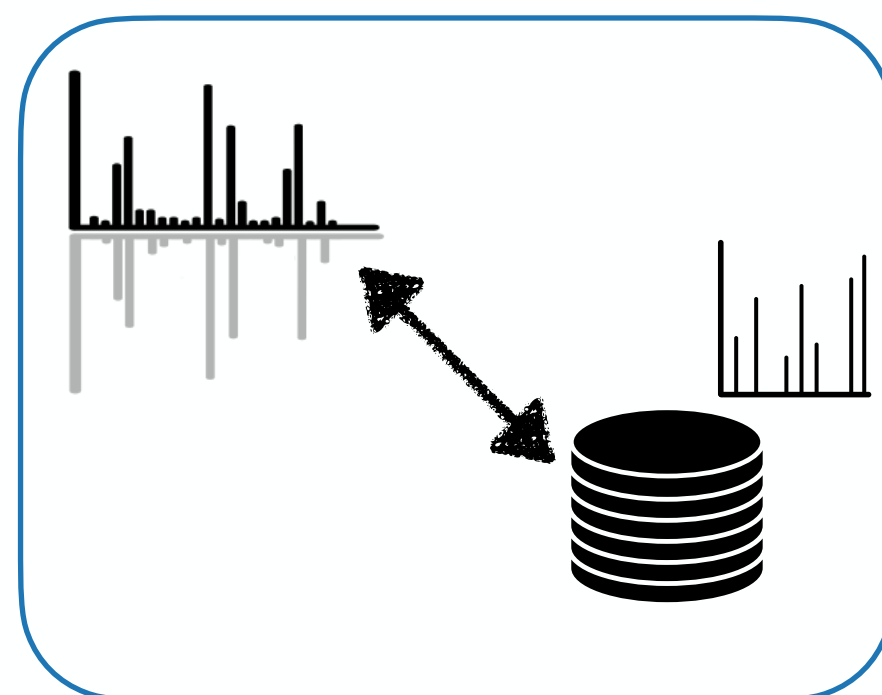


Toujours compliqué!

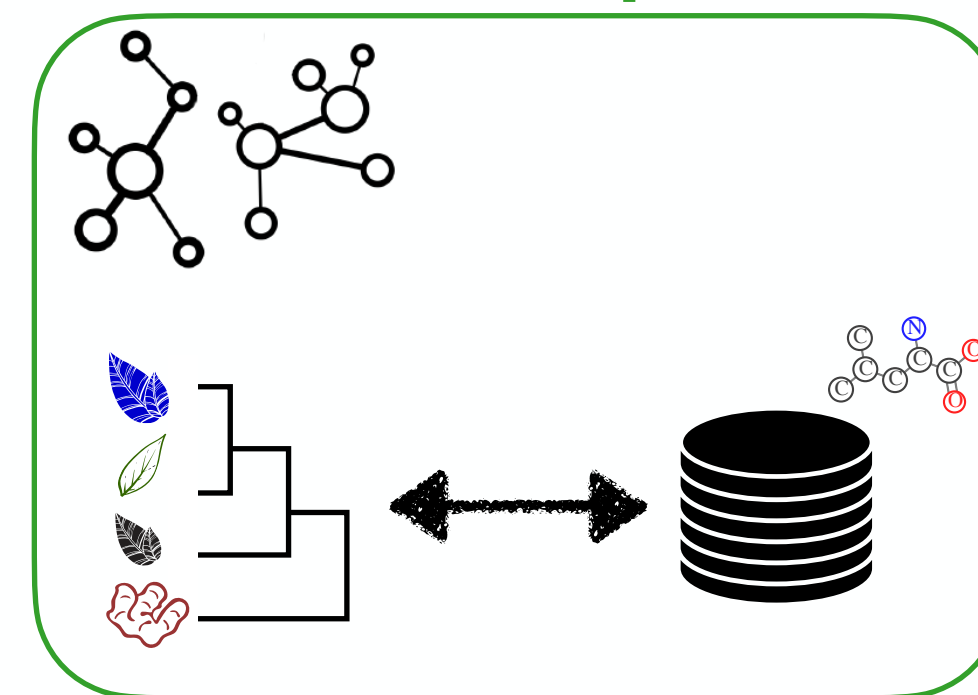
Étape 1 - Messages importants

- Gare aux meilleurs candidats spectraux!
- Attention à l'information (et la qualité) spectrale!
- L'espace chimique des libraries spectral est très restreint!

Étape 1:
Libraries spectrales



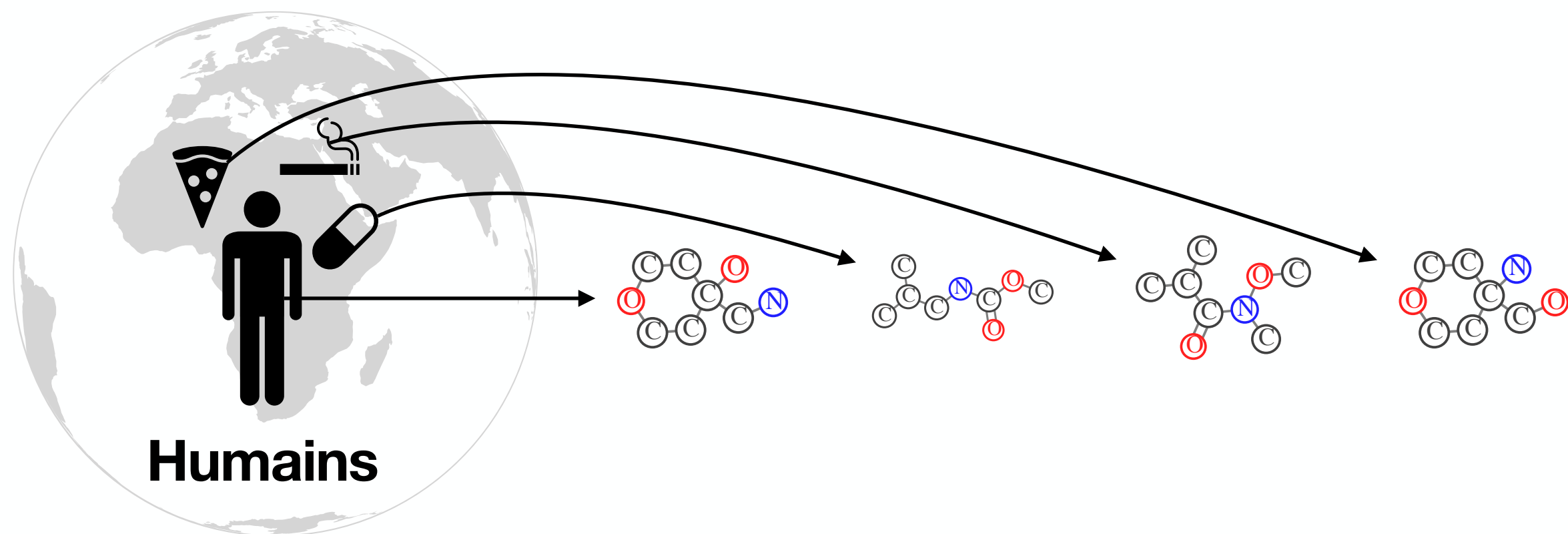
Étape 2:
Espace chimique &
connaissances préalables



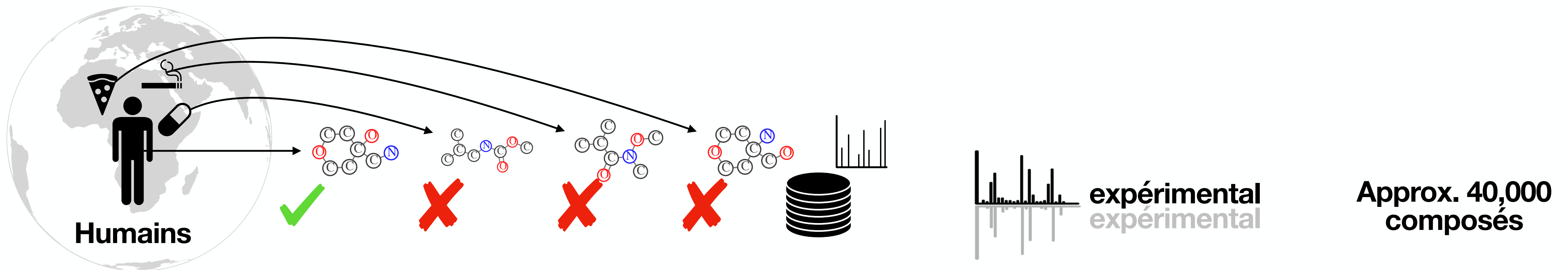
Étape 2 - Pourquoi diversifier son espace chimique?



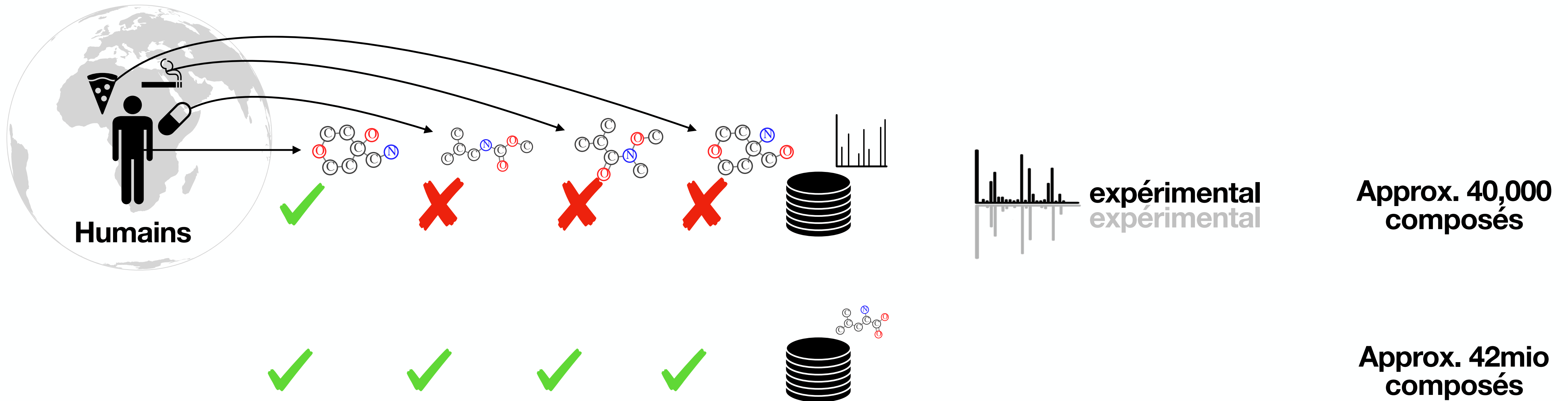
Étape 2 - Pourquoi diversifier son espace chimique?



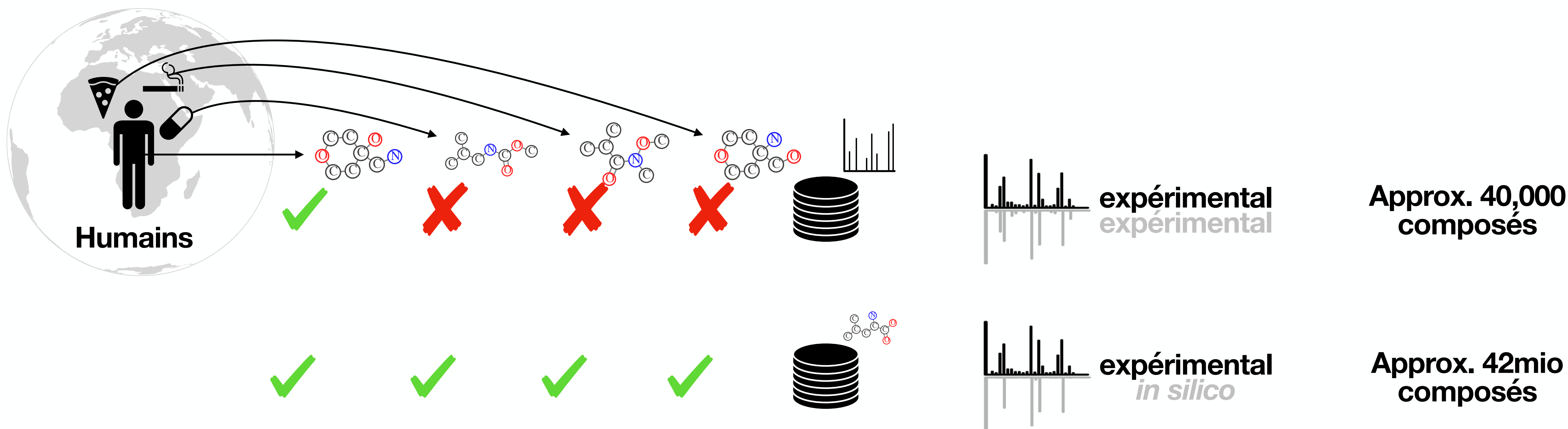
Étape 2 - Pourquoi diversifier son espace chimique?



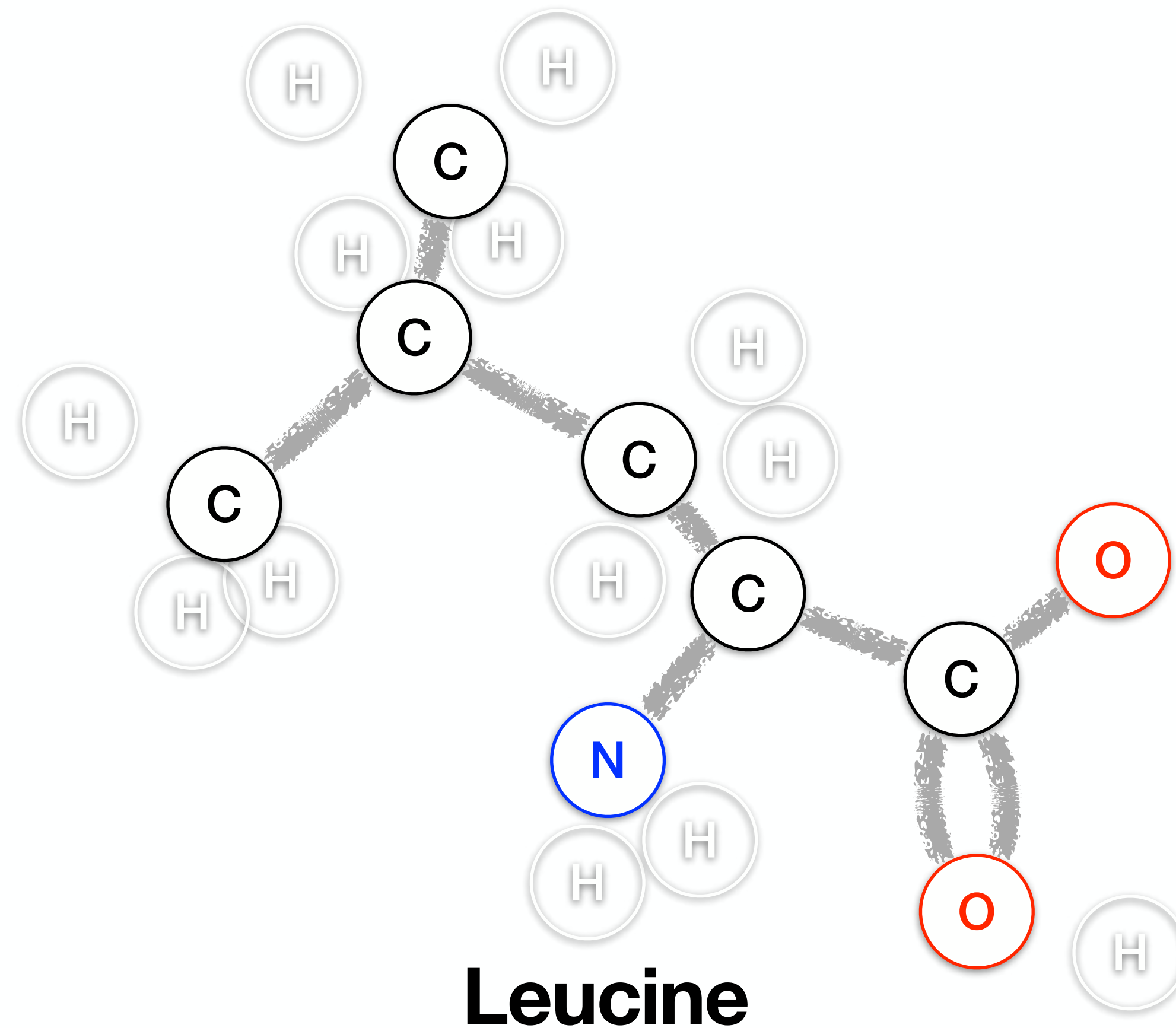
Étape 2 - Pourquoi diversifier son espace chimique?



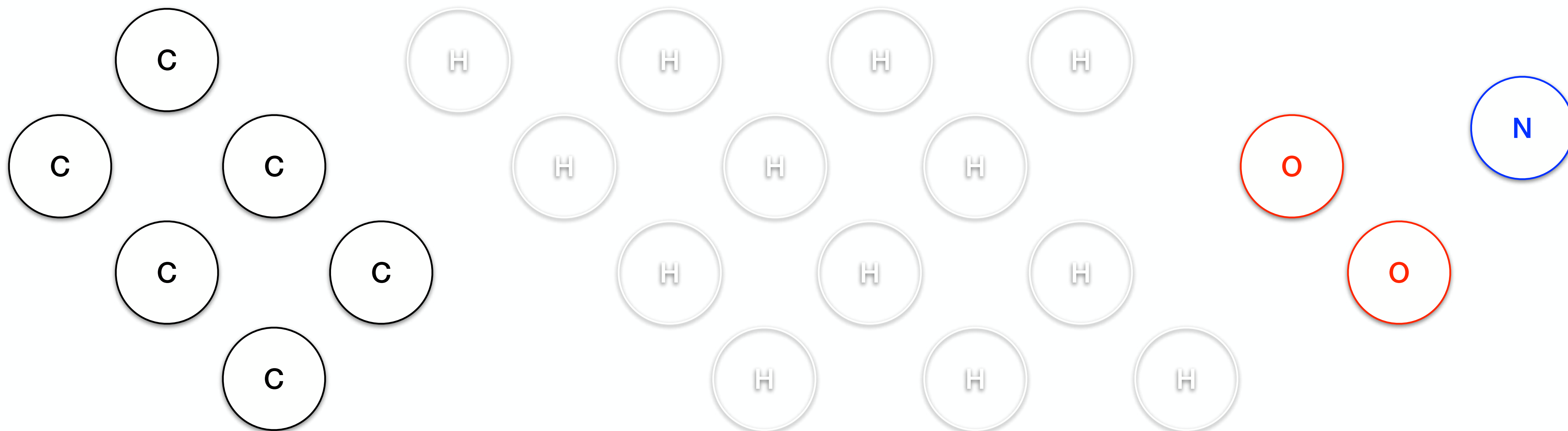
Étape 2 - Pourquoi diversifier son espace chimique?



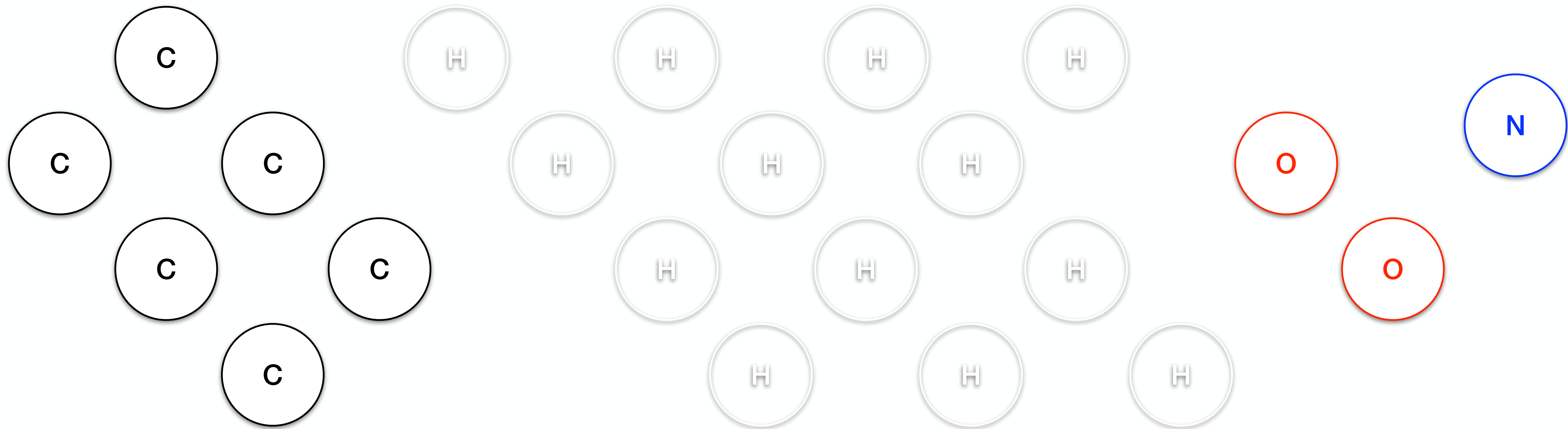
Étape 2 - Le risque avec l'espace chimique des petites molécules



Étape 2 - Le risque avec l'espace chimique des petites molécules

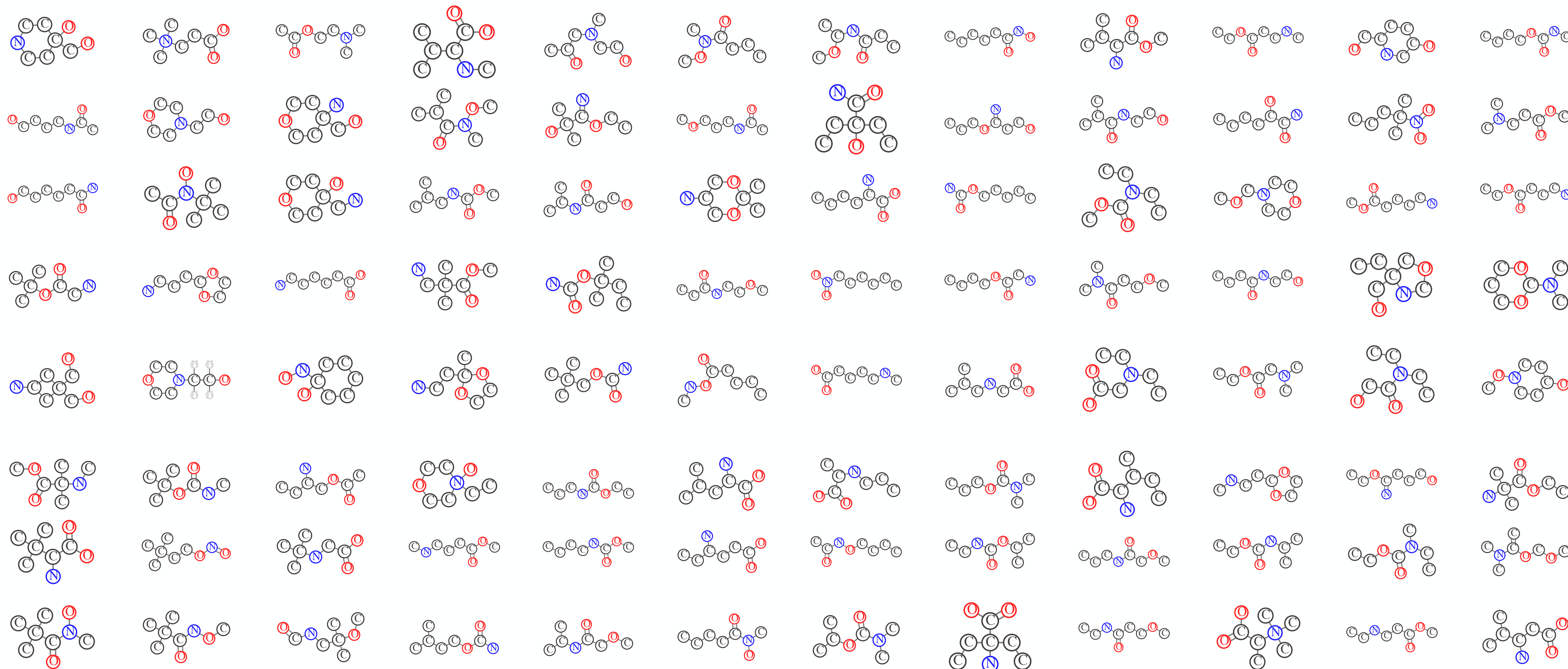


Étape 2 - Le risque avec l'espace chimique des petites molécules



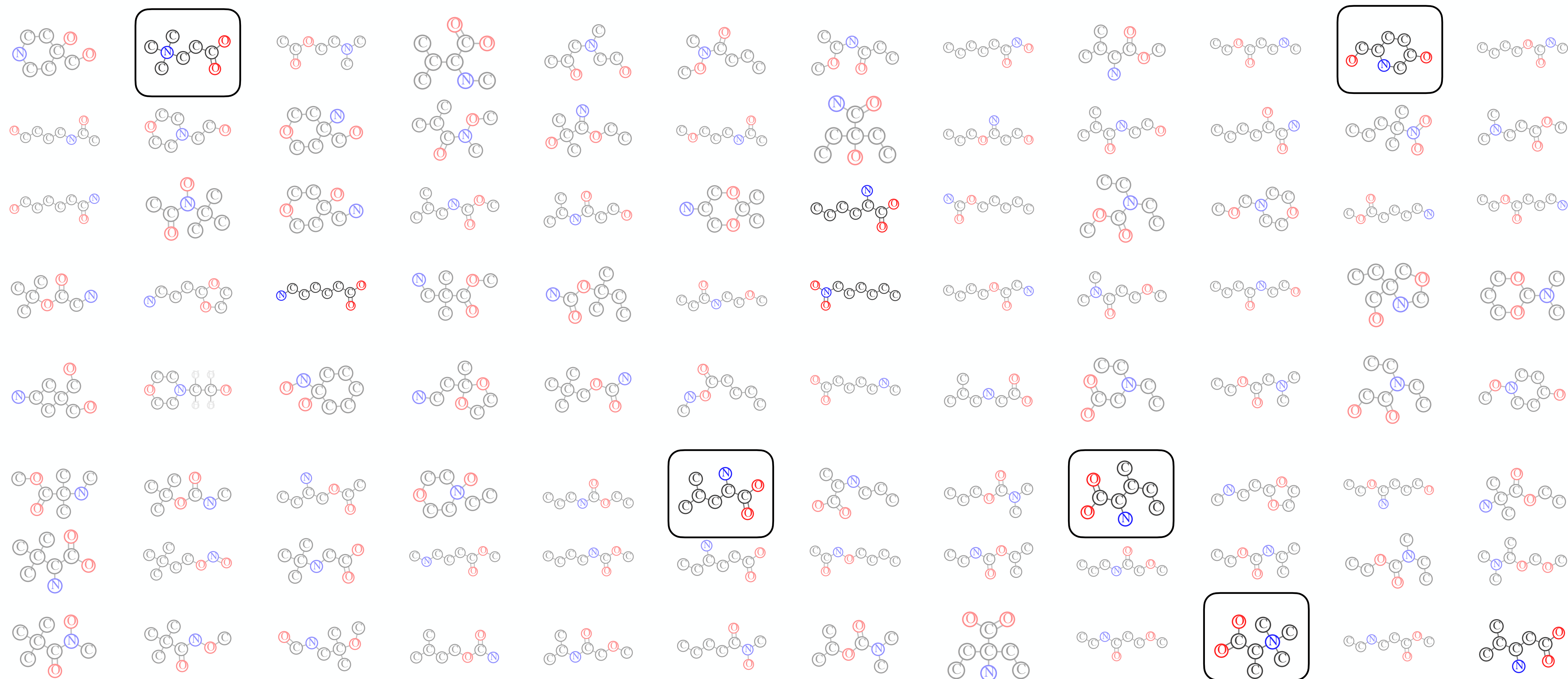
23,946 isomères possibles

Étape 2 - Un espace chimique guidé par les connaissances préalables



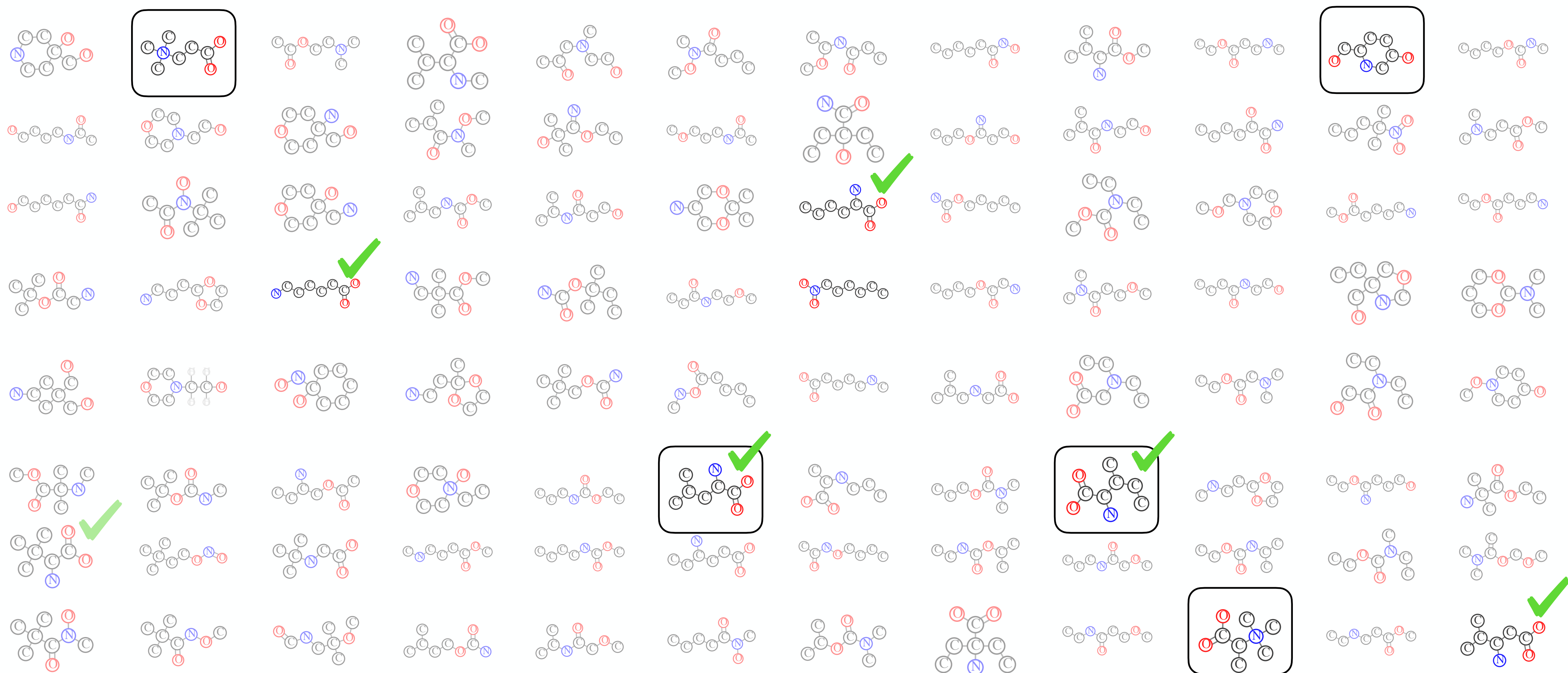
23,946 isomères possibles $\Rightarrow$ 109 isomères reportés

Étape 2 - Un espace chimique guidé par les connaissances préalables



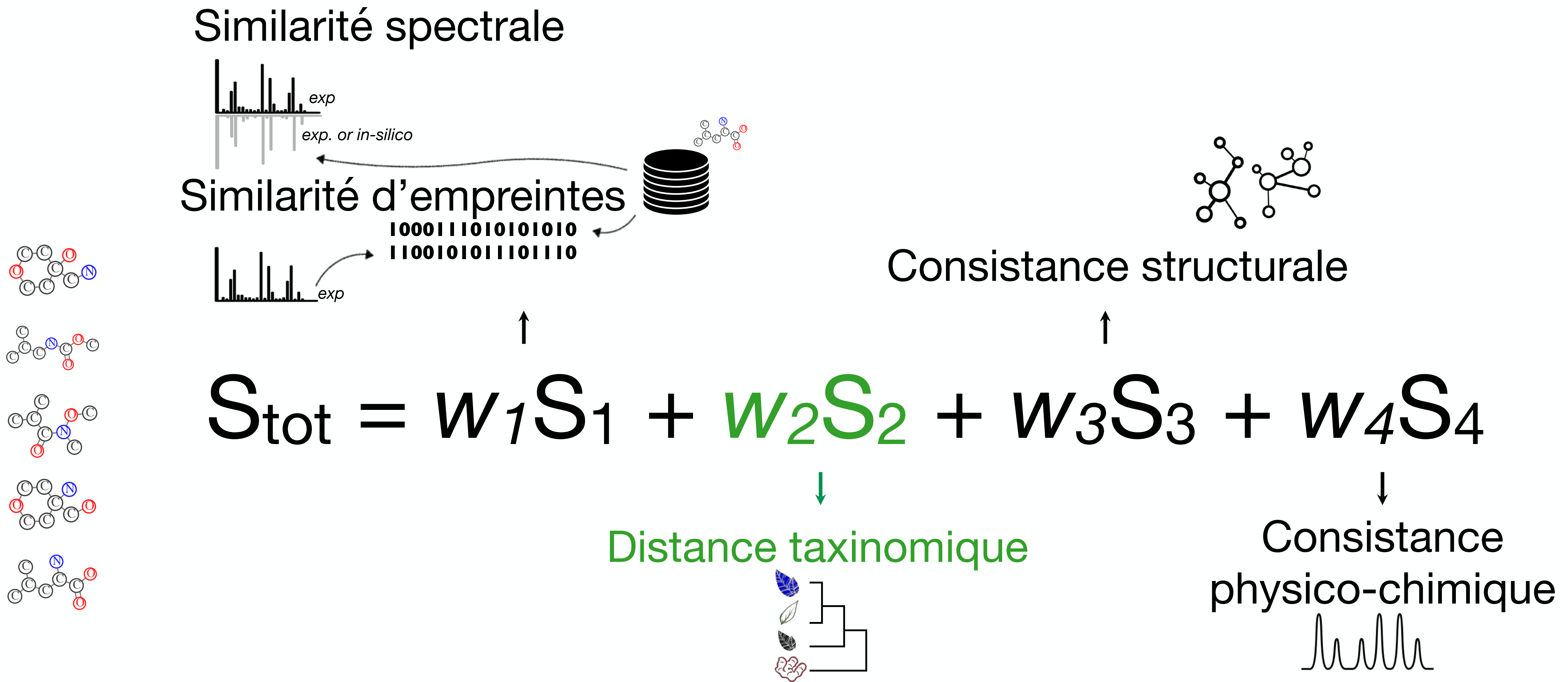
23,946 isomères possibles $\Rightarrow$ 109 isomères reportés $\Rightarrow$ 6 biologiques

Step 2 - Prior knowledge-informed chemical space



**23,946 isomères possibles $\Rightarrow$ 109 isomères reportés $\Rightarrow$ 6 biologiques
9 avec spectre associé ✓**

Étape 2 - Un système d'annotation éclairé pour repérer les chevaux



Étape 2 - À la recherche des chevaux avec les librairies publiques



9,818 spectres expérimentaux

Étape 2 - À la recherche des chevaux avec les bibliothèques publiques



9,818 spectres expérimentaux



Étape 2 - À la recherche des chevaux avec les librairies publiques



9,818 spectres expérimentaux



Correct candidates [%]

75
50
25
0

0

5

10

15

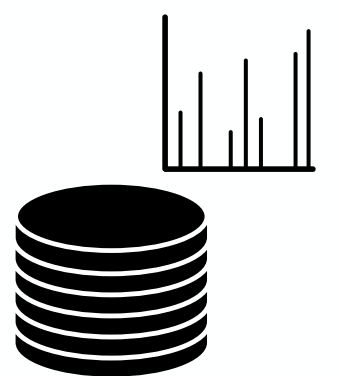
20

25

Rank

~10% candidats corrects au rang 1

Approche spectrale
classique (Étape 1)



Étape 2 - À la recherche des chevaux avec les librairies publiques

 **GNPS**
9,818 spectres expérimentaux



Humains



Correct candidates [%]

75

50

25

0

0

5

10

15

20

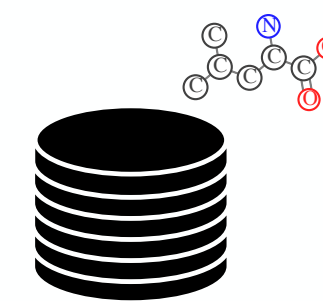
25

Rank

~45%

~10% candidats corrects au rang 1

Espace chimique et connaissances
préalables (Étape 2)



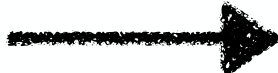
Approche spectrale
classique (Étape 1)



Étape 2 - À la recherche des chevaux avec les bibliothèques publiques



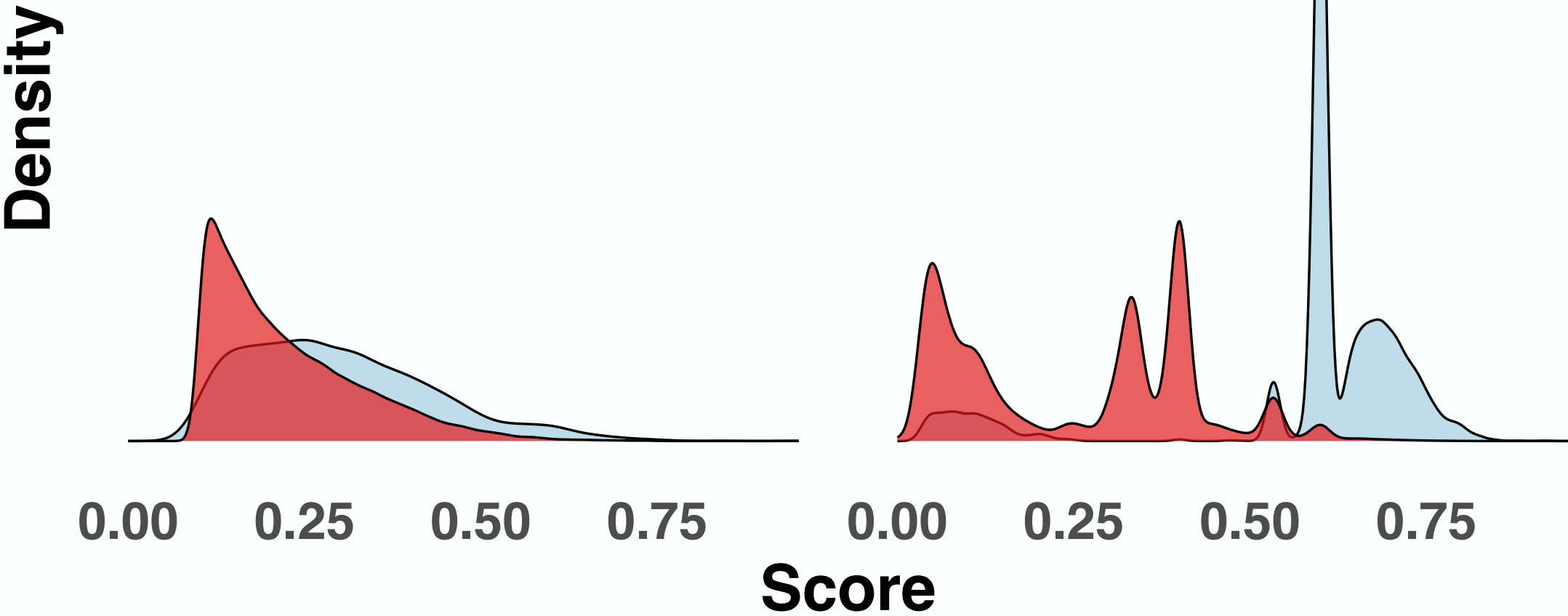
9,818 spectres expérimentaux



Humains

Étape 1

Étape 2



Correct candidates [%]

75
50
25
0

~45%

~10% candidats corrects au rang 1

Espace chimique et connaissances
préalables (Étape 2)

Approche spectrale
classique (Étape 1)

Rank

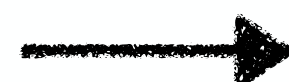
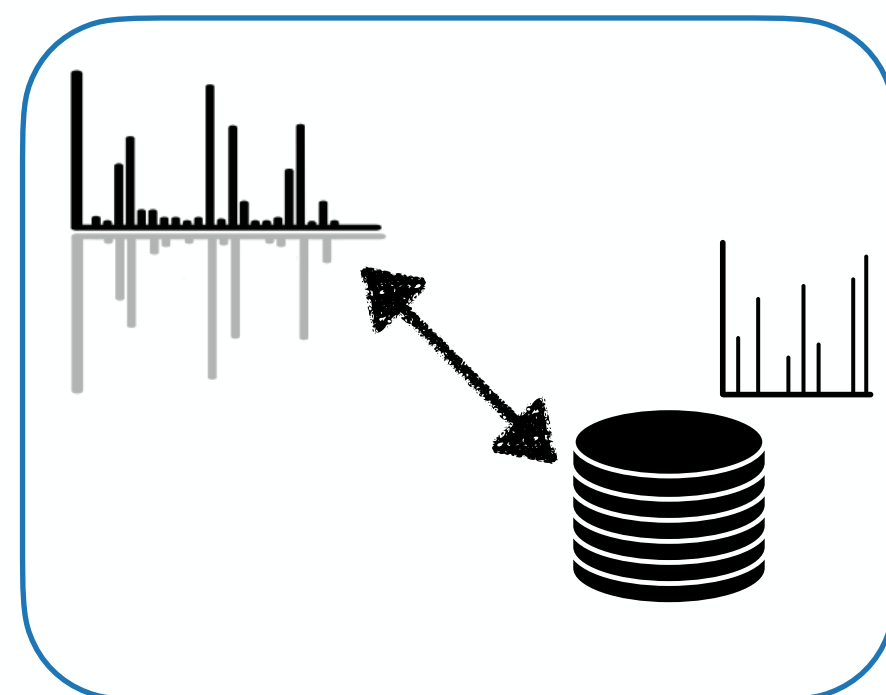
0 5 10 15 20 25



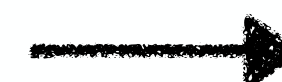
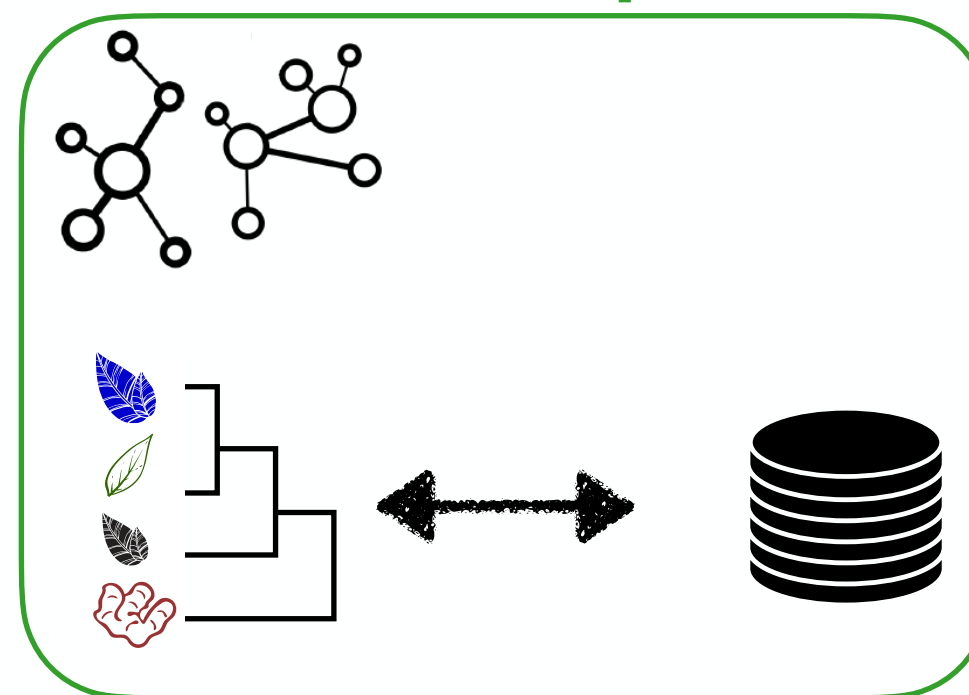
Étape 2 - Messages importants

- Les librairies chimiques peuvent aider à compenser la faible couverture de l'espace chimique des librairies
- Les connaissances préalables à propos des composés et de leurs occurrences peuvent guider l'annotation efficacement
- Nous restons très loin d'une annotation certaine et compréhensive
- Cela peut aider à écrémer les chevaux mais pas à trouver nos **licornes**!

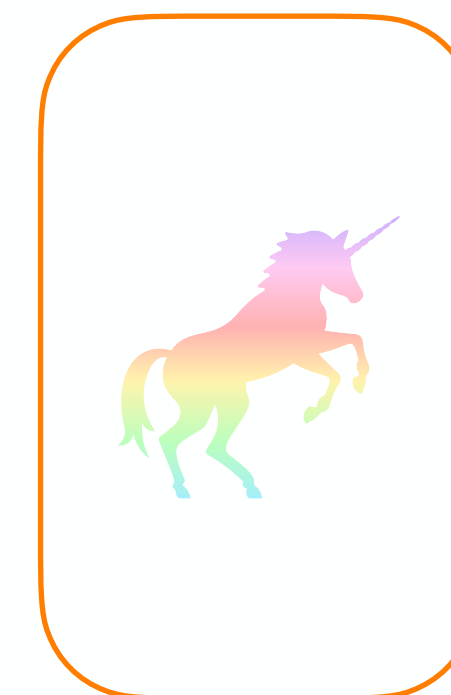
Étape 1:
Libraries spectrales



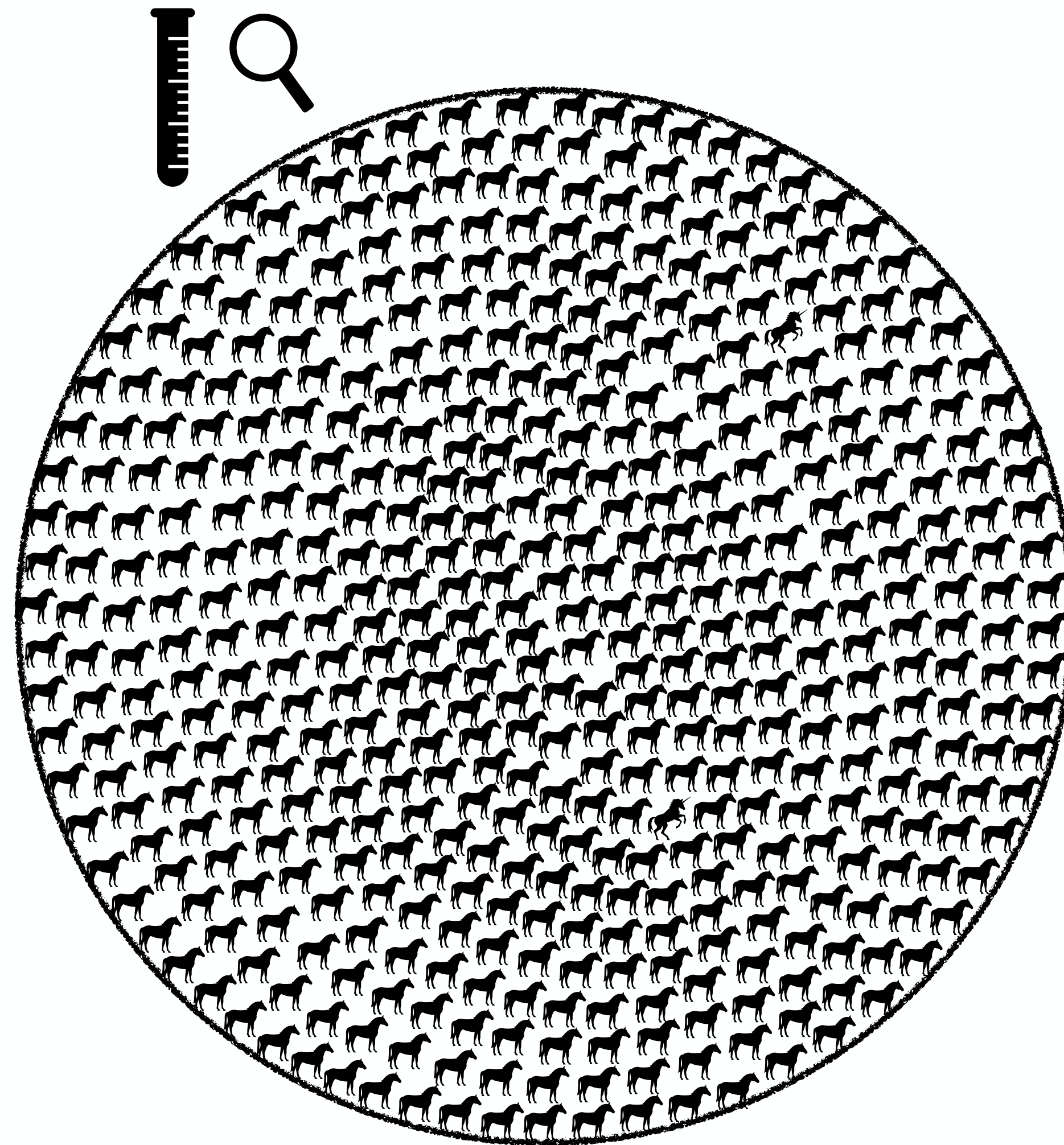
Étape 2:
Espace chimique &
connaissances préalables



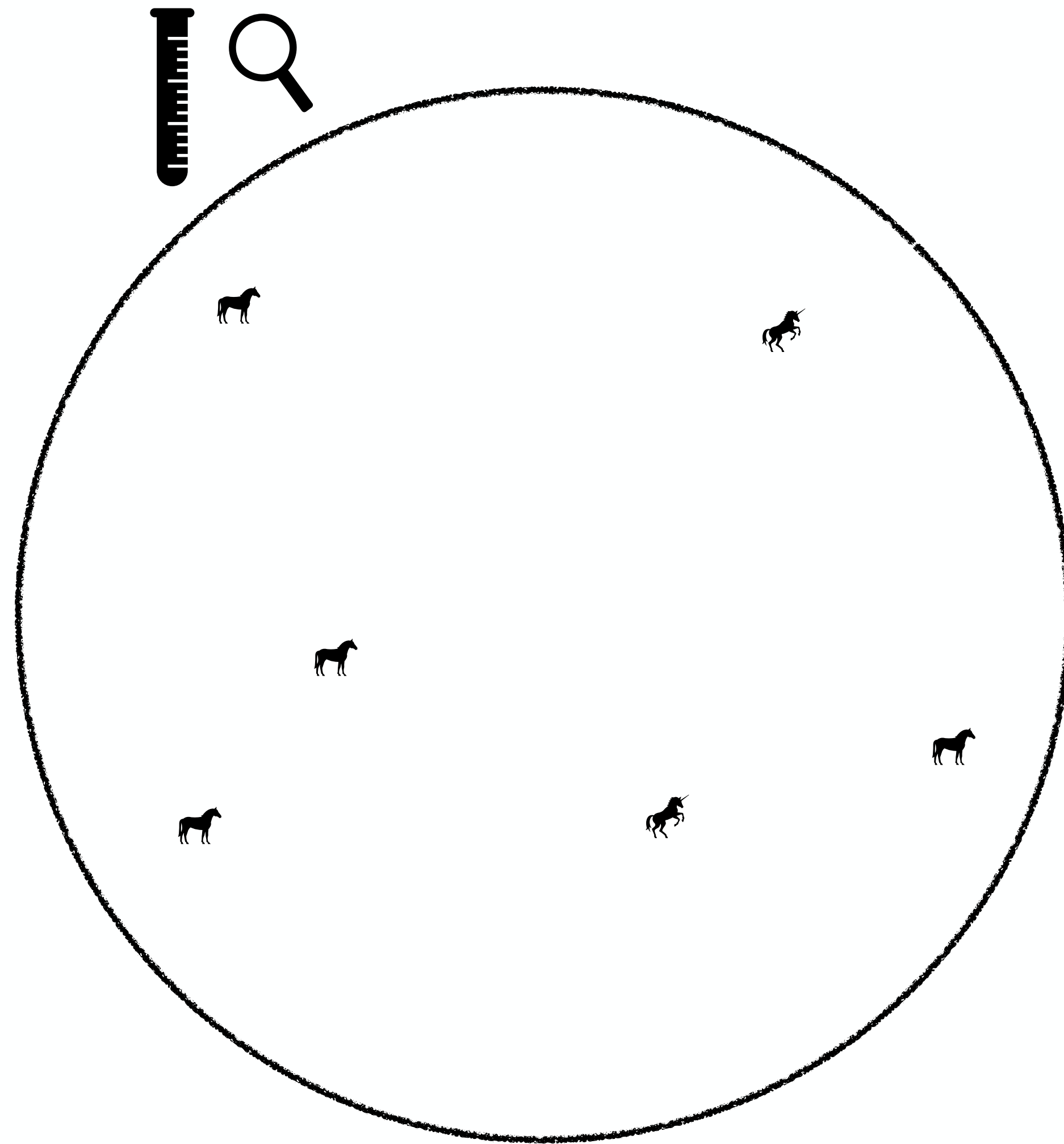
Étape 3:
Plongée dans
l'inconnu



Étapes précédentes: enlever le connu...

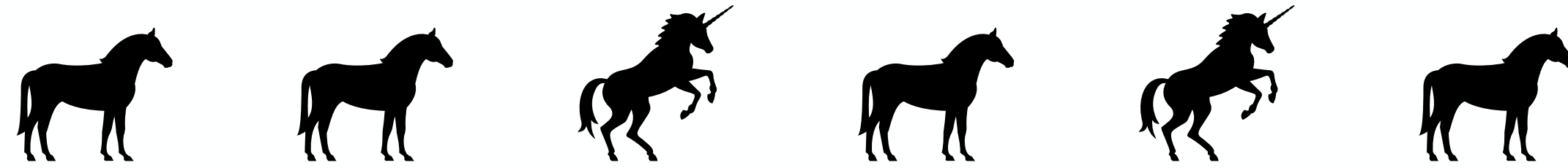


Étapes précédentes: enlever le connu...

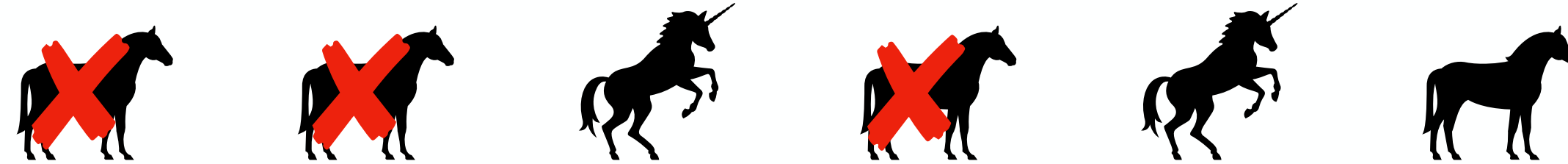


... pour mieux trouver les *potentielles* licornes!

Étape 3 - Comment faire avec les licornes?



Étape 3 - Comment chevaucher des licornes?



Recherche dans une base de données contenant des milliards d'entrées en quelques secondes



*A-t-on déjà mesuré
un spectre similaire
au mien?*

Oui

Non

Étape 3 - Comment chevaucher des licornes?



Recherche dans une base de données contenant des milliards d'entrées en quelques secondes



A-t-on déjà mesuré un spectre similaire au mien?

Oui → À creuser

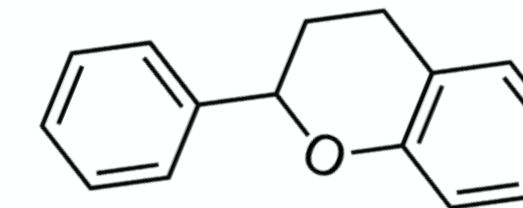
Non

Prédiction de classe chimique



À quelle classe chimique pourrait correspondre mon spectre?

Flavonoïde?



Étape 3 - Comment chevaucher des licornes?

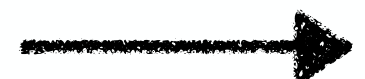


Recherche dans une base de données contenant des milliards d'entrées en quelques secondes



A-t-on déjà mesuré un spectre similaire au mien?

Oui

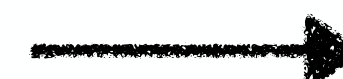


À creuser

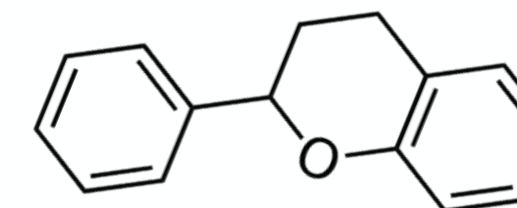
Prédiction de classe chimique



À quelle classe chimique pourrait correspondre mon spectre?

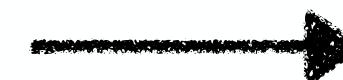


Flavonoïde?

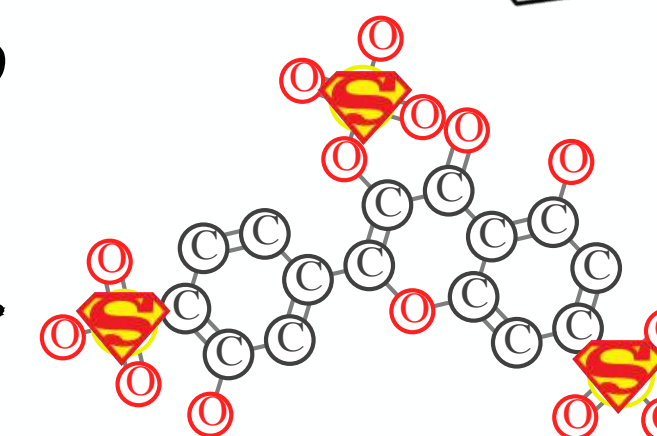


Non

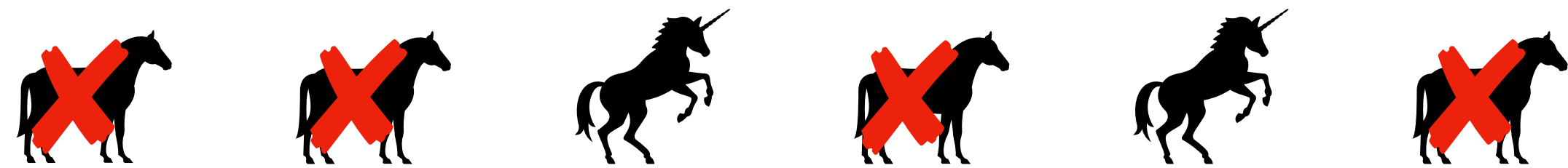
Génération de structure *de novo*



Dessine moi un mouton



Étape 3 - Comment chevaucher des licornes?



Recherche dans une base de données contenant des milliards d'entrées en quelques secondes



A-t-on déjà mesuré un spectre similaire au mien?

Oui

À creuser

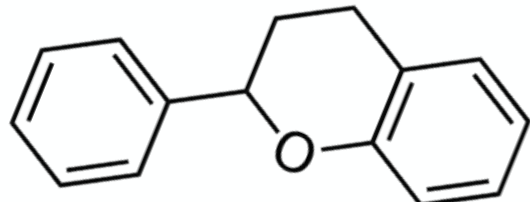
Revenir à des classes connues (trouvées à l'étape 2)

Prédiction de classe chimique

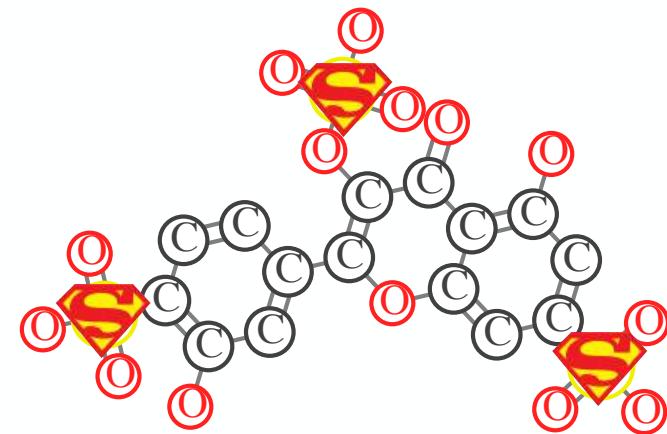
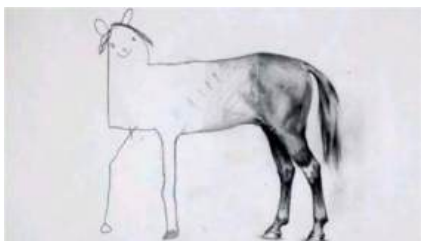


À quelle classe chimique pourrait correspondre mon spectre?

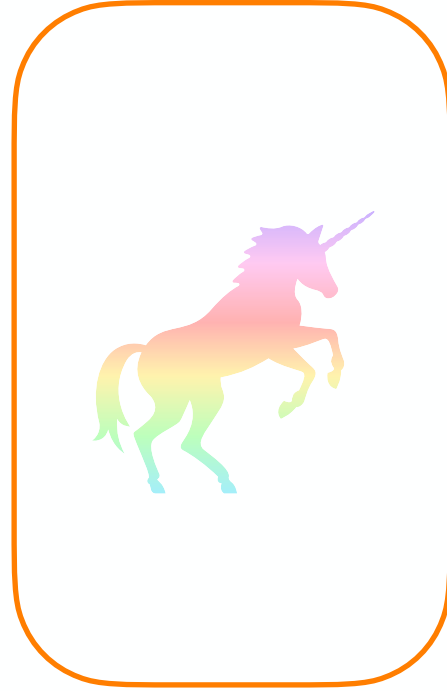
Flavonoïde?



Génération de structure de novo



Dessine moi un mouton



~~Je suis super nul lorsque'il s'agit d'écrire un rapport~~

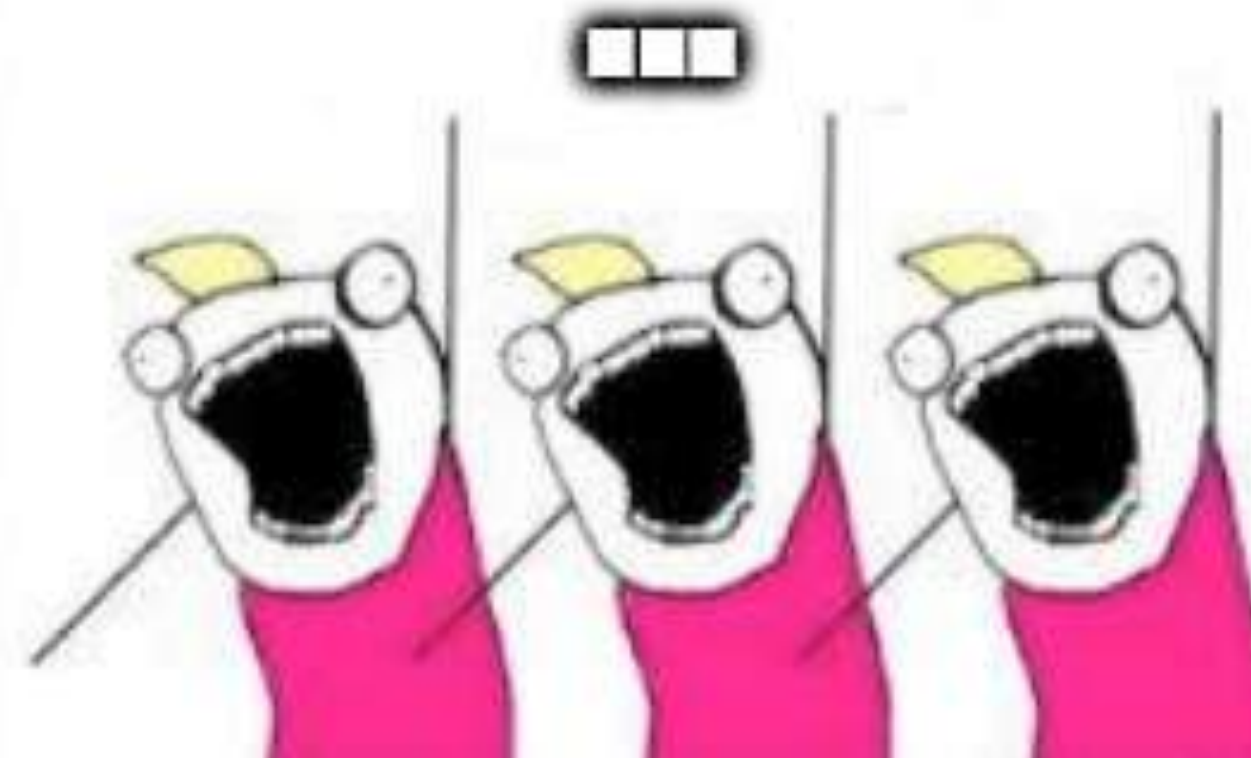
WHAT DO WE WANT?!



**CONFIDENT
METABOLITE ANNOTATION**



**WHO WANTS TO THEN
REPORT IT PROPERLY?**

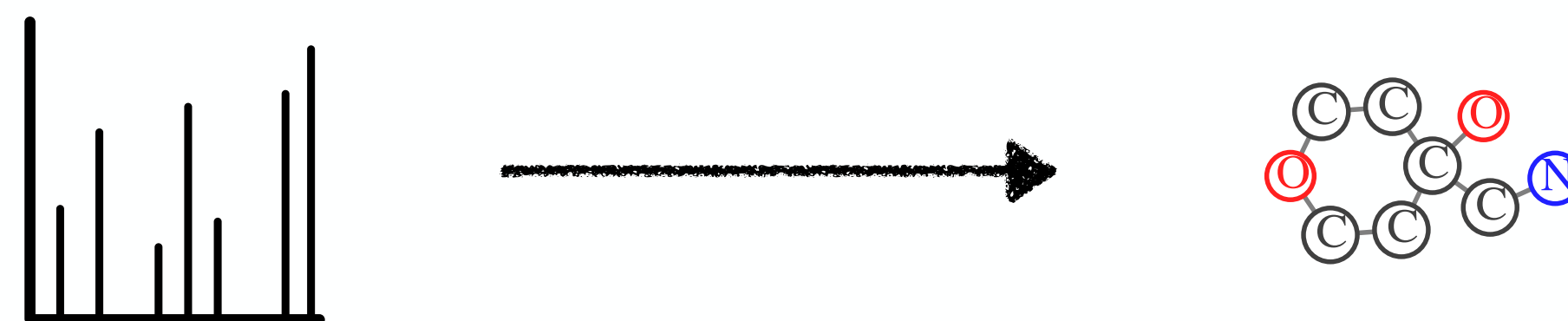


Situation actuelle



**Pluralité de résultats,
pas mauvais en soi**

Situation actuelle



**Pluralité de résultats,
pas mauvais en soi**

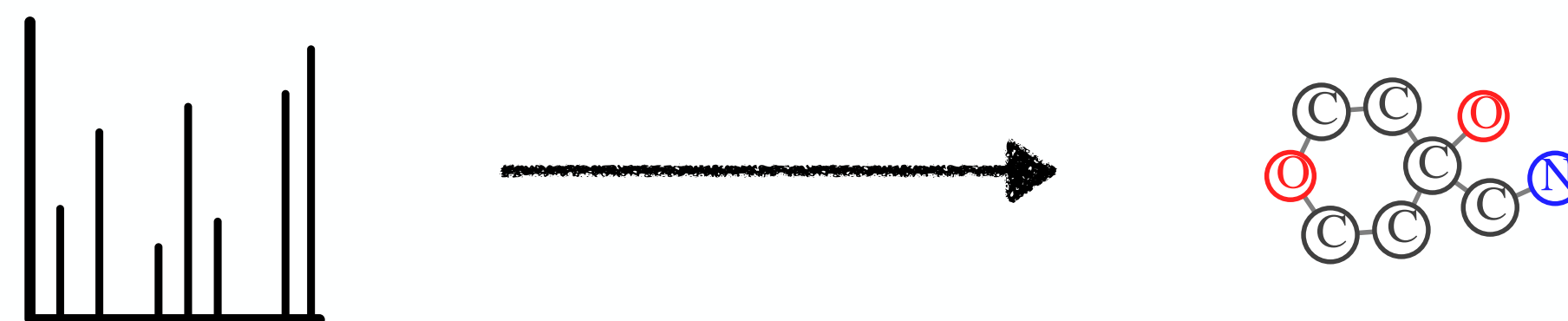
Sumner, L.W., Amberg, A., Barrett, D. et al. Proposed minimum reporting standards for chemical analysis. *Metabolomics* 3, 211–221 (2007). <https://doi.org/10.1007/s11306-007-0082-2>

> 4,300 citations

E L. Schymanski, J Jeon, R Gulde, K Fenner, M Ruff, H P. Singer, and J Hollender
Identifying Small Molecules via High Resolution Mass Spectrometry: Communicating Confidence
Environmental Science & Technology 2014 48 (4), 2097-2098 <https://doi.org/10.1021/es5002105>

> 2,900 citations

Situation actuelle



**Pluralité de résultats,
pas mauvais en soi**

Qui les utilise?

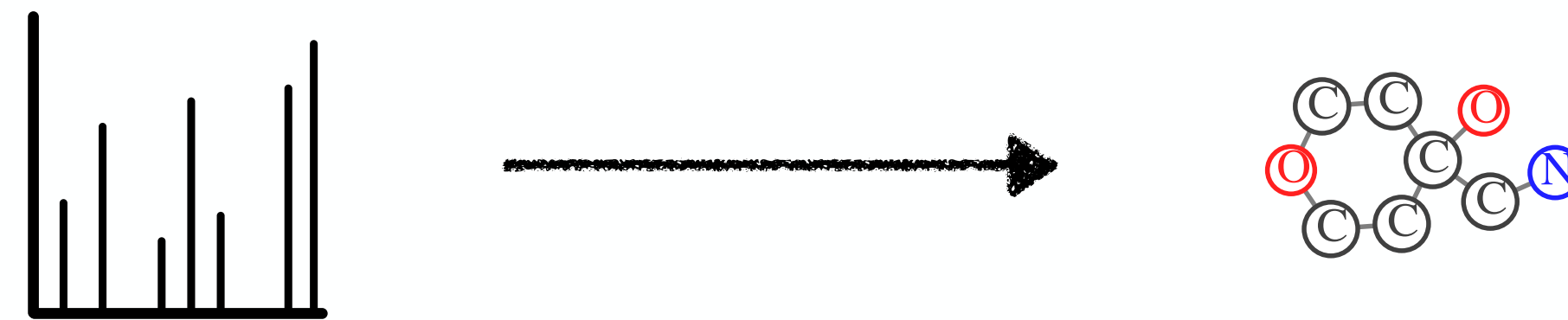
Sumner, L.W., Amberg, A., Barrett, D. et al. Proposed minimum reporting standards for chemical analysis. *Metabolomics* 3, 211–221 (2007). <https://doi.org/10.1007/s11306-007-0082-2>

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E L. Schymanski, J Jeon, R Gulc
Identifying Small Molecules via High Res
Environmental Science & Technology 2



emical analysis.
2

der
Confidence
es5002105

> 4,300 citations

> 2,900 citations

Situation actuelle



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pas mauvais en soi

Qui les utilise?

Sumner, L.W., Amberg, A., Barrett, D. et al. *PI Metabolomics* 3, 211–221 (2006)

E L. Schymanski, J Jeon, R Gulc
Identifying Small Molecules via High Res
Environmental Science & Technology 2



emical analysis.
2

der
Confidence
es5002105

> 4,300 citations

> 2,900 citations

Situation actuelle



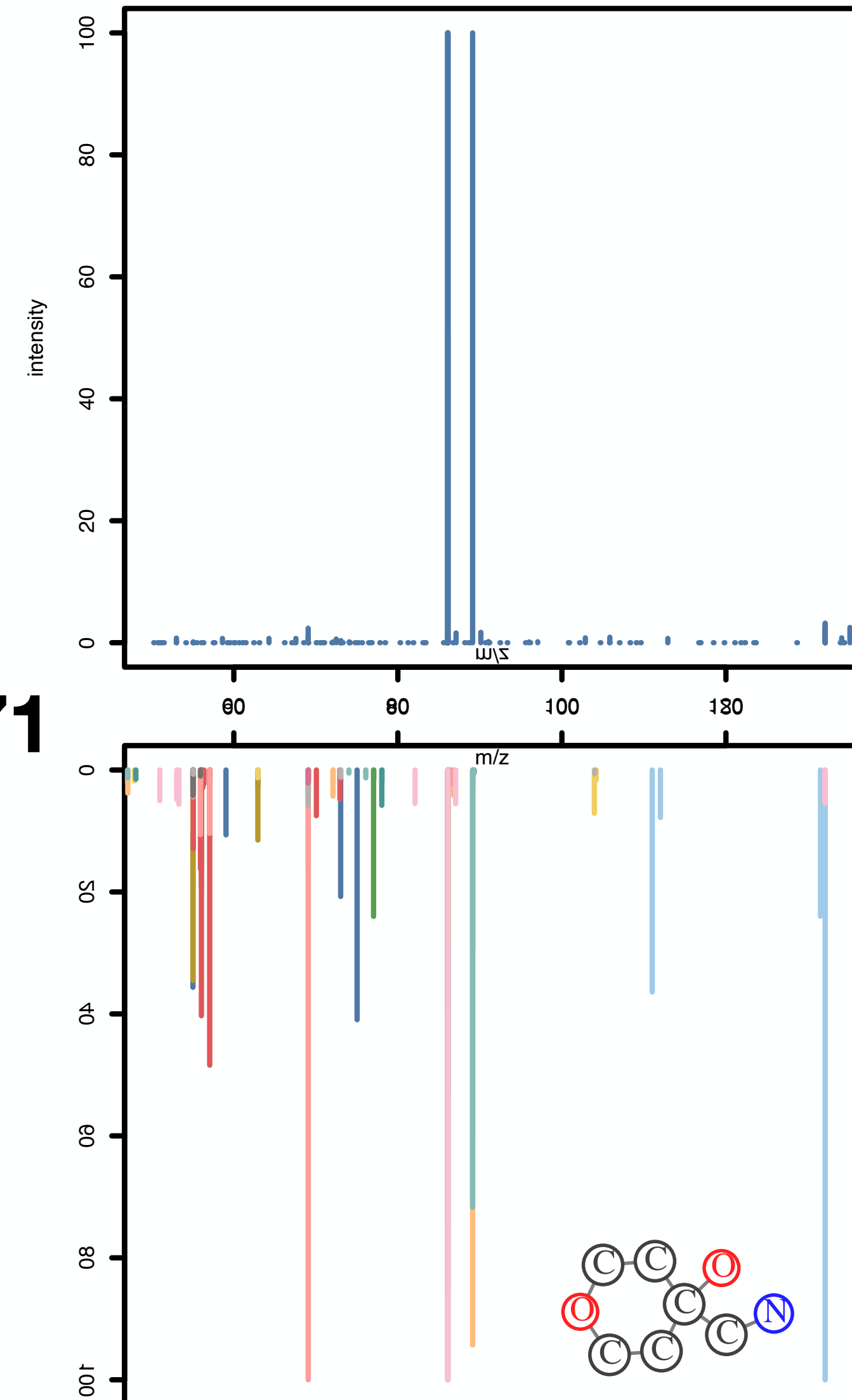
Nos méthodes
d'annotation ont évolué plus rapidement
que nos rapports

Situation actuelle



Nos méthodes
d'annotation ont évolué plus rapidement
que nos rapports

Cosine: 0.71

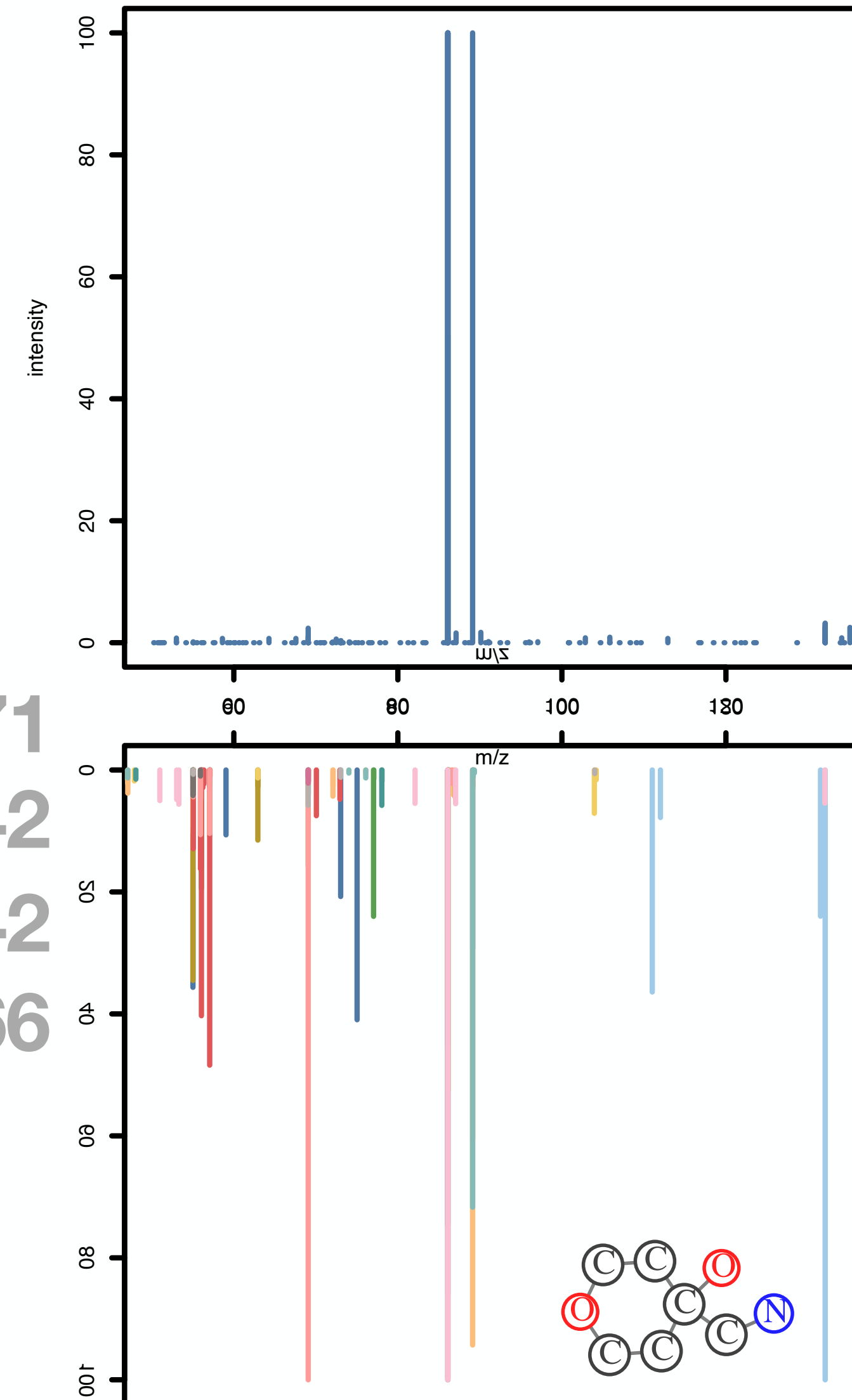


Situation actuelle



Nos méthodes
d'annotation ont évolué plus rapidement
que nos rapports

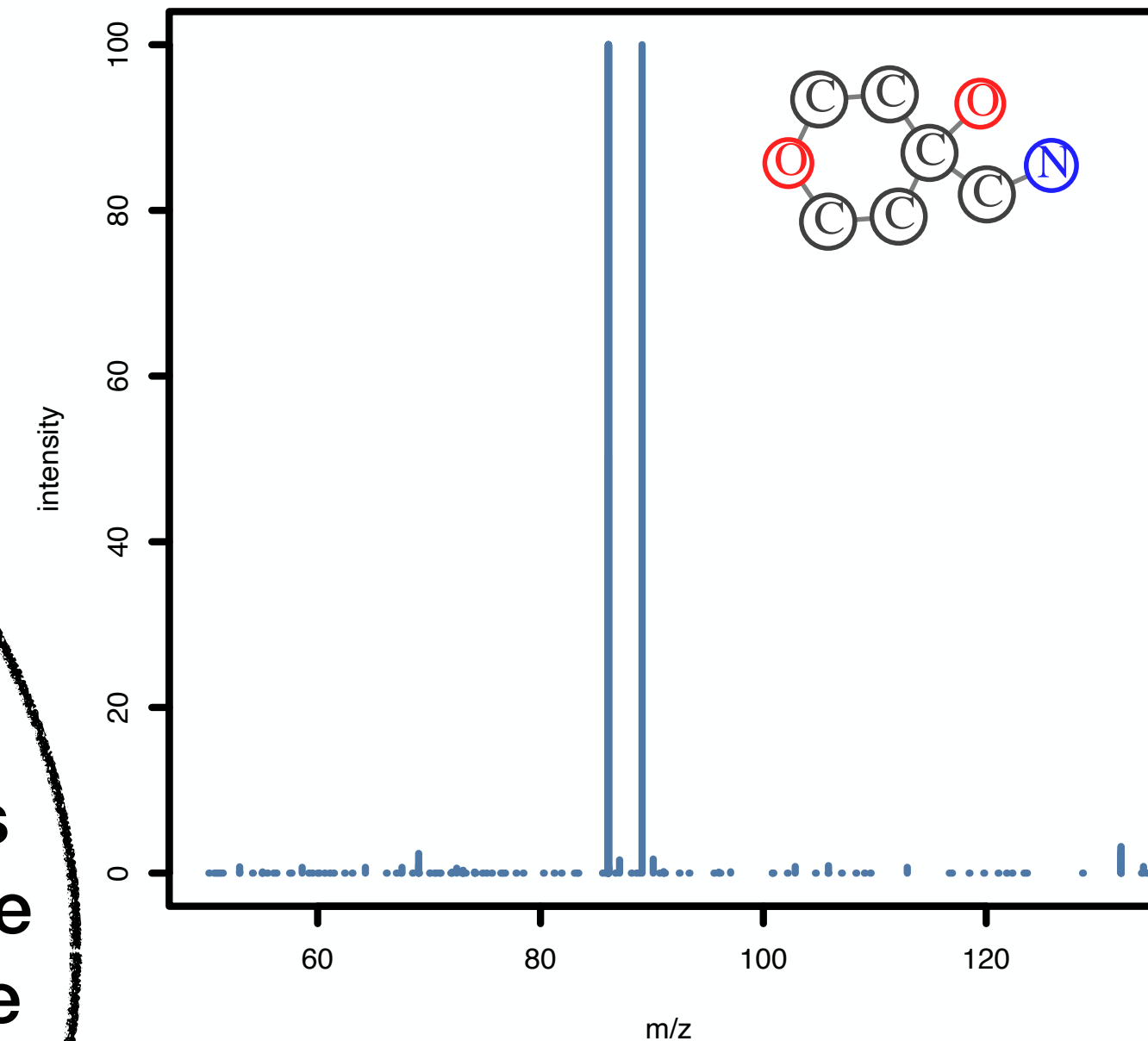
Cosine: 0.71
MS2DeepScore: 0.42
CSI Finger ID Score : -110.342
TIMA Score : 0.66



Situation idéale?



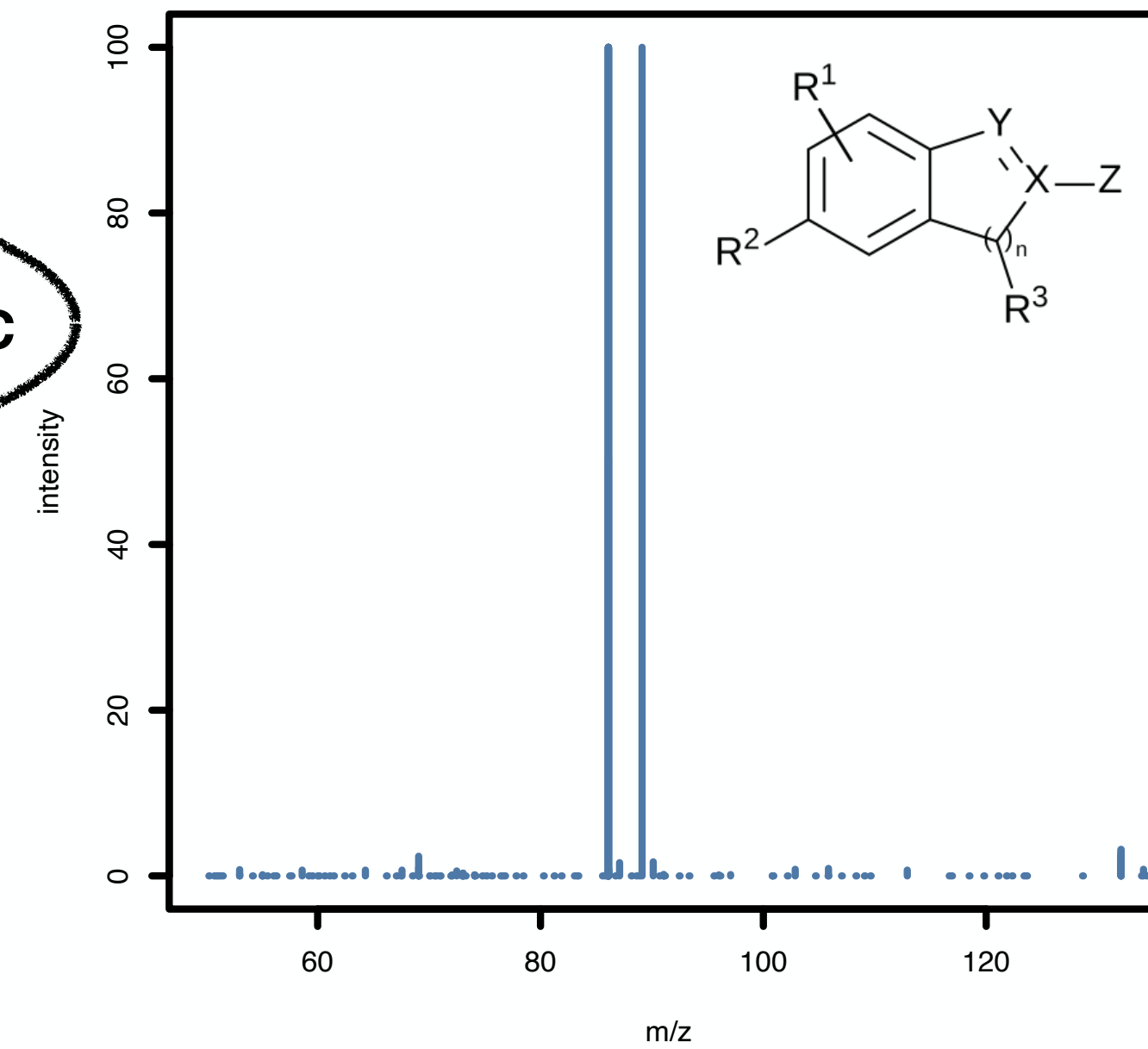
J'ai en effet annoté cette molécule parmi 4,312 autres, après avoir effectué un pré-traitement des spectres à l'aide de la méthode implémentée dans le logiciel X version 1.17, avec les paramètres cachés dans mon dossier. Cela a donné un score initial de 0,71, soit 0,01 au-dessus de mon seuil, qui a ensuite été complété par d'autres scores dont je ne me souviens plus, ni des bibliothèques utilisées. Je l'ai également comparée à une bibliothèque interne que je ne peux pas partager ; vous devrez donc me croire sur parole. Si vous êtes un informaticien ou un scientifique cherchant à tirer parti d'études antérieures, bonne chance pour comprendre ce que j'ai fait. Ah, et attendez... J'ai adapté les paramètres des autres en fonction de mon intuition (guidée mon microbiome).



Vraie situation idéale?



D'après les données du spectre, je peux vous dire que la structure présente le squelette suivant, avec les substitutions suivantes :



Le changement de paradigme

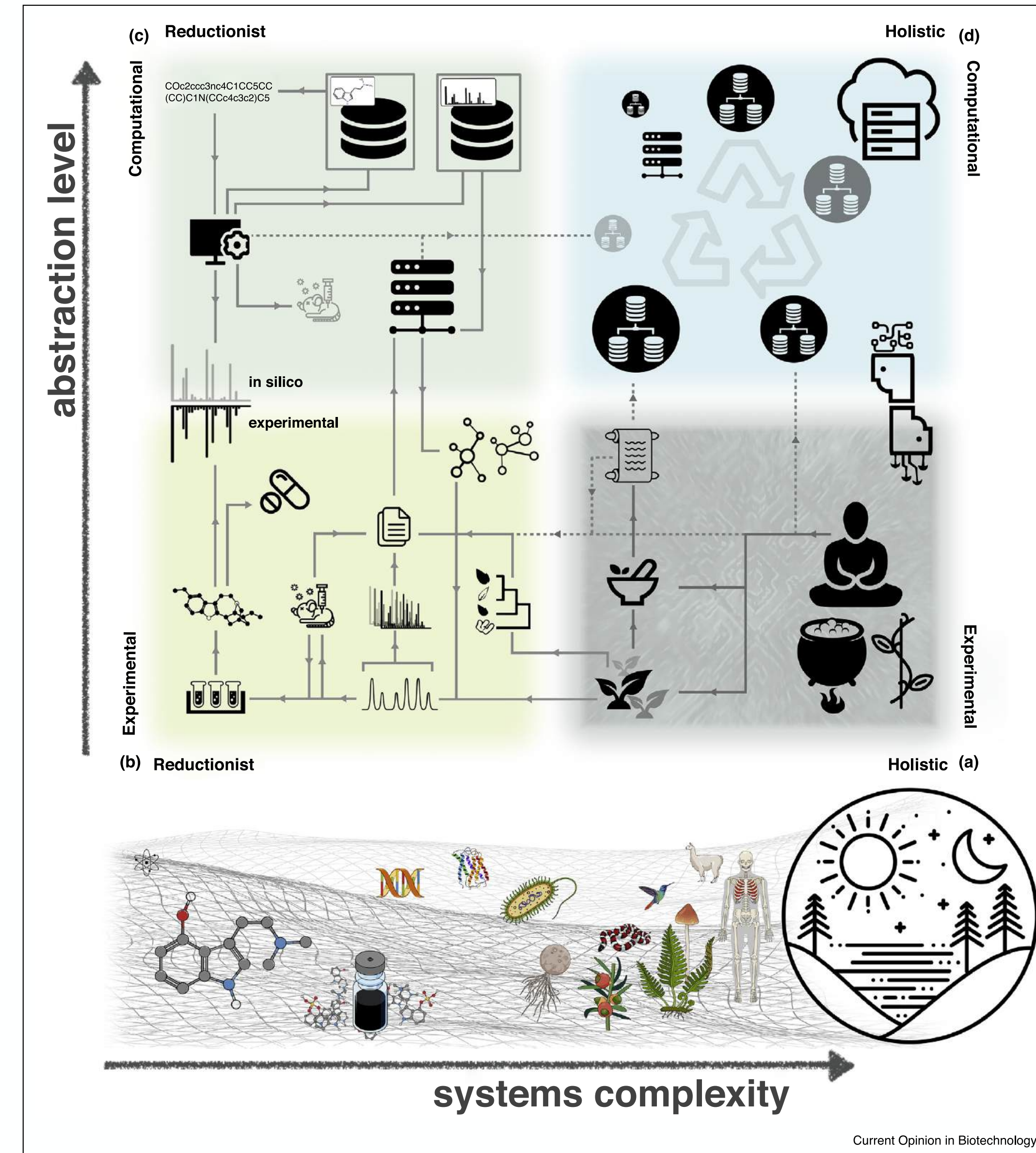
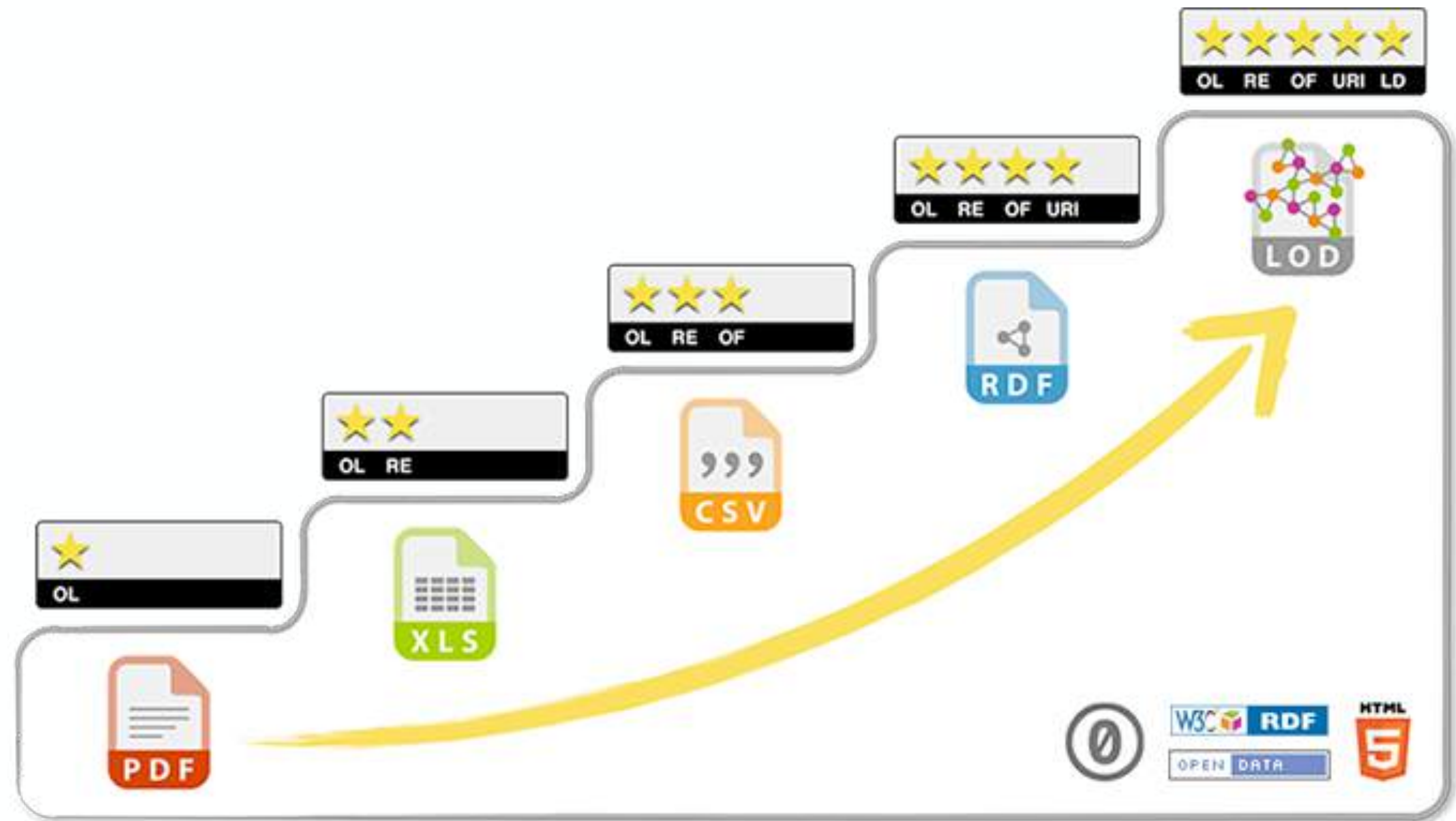


Figure from Allard et al. 2018. “Pharmacognosy in the Digital Era: Shifting to Contextualized Metabolomics.” Current Opinion in Biotechnology 54 (December): 57–64. <https://doi.org/10.1016/j.copbio.2018.02.010>

Figure from Allard et al. 2018. “*Pharmacognosy in the Digital Era: Shifting to Contextualized Metabolomics.*” Current Opinion in Biotechnology 54 (December): 57–64. <https://doi.org/10.1016/j.copbio.2018.02.010>

Données ouvertes liées

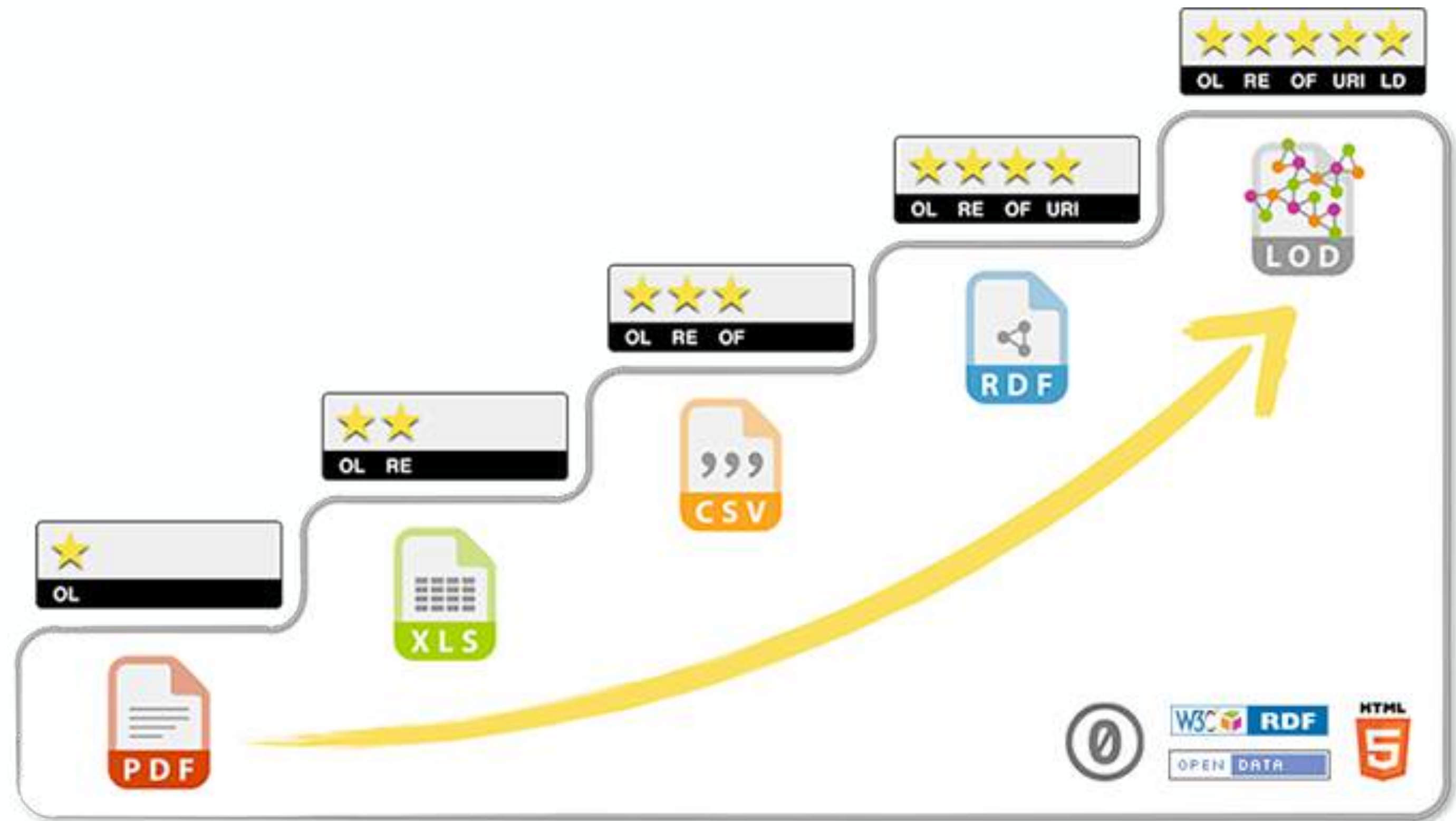


https://commons.wikimedia.org/wiki/File:5-star_deployment_scheme_for_Open_Data.png

<https://5stardata.info>

Données ouvertes liées

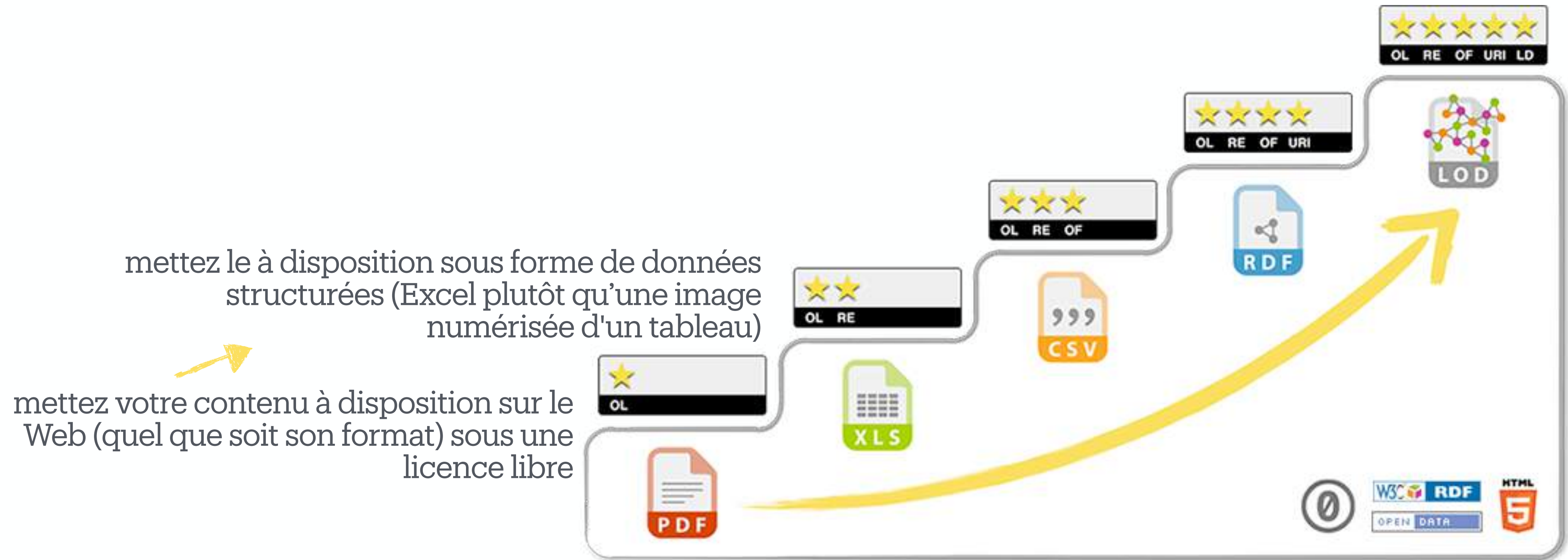
mettez votre contenu à disposition sur le Web (quel que soit son format) sous une licence libre



https://commons.wikimedia.org/wiki/File:5-star_deployment_scheme_for_Open_Data.png

<https://5stardata.info>

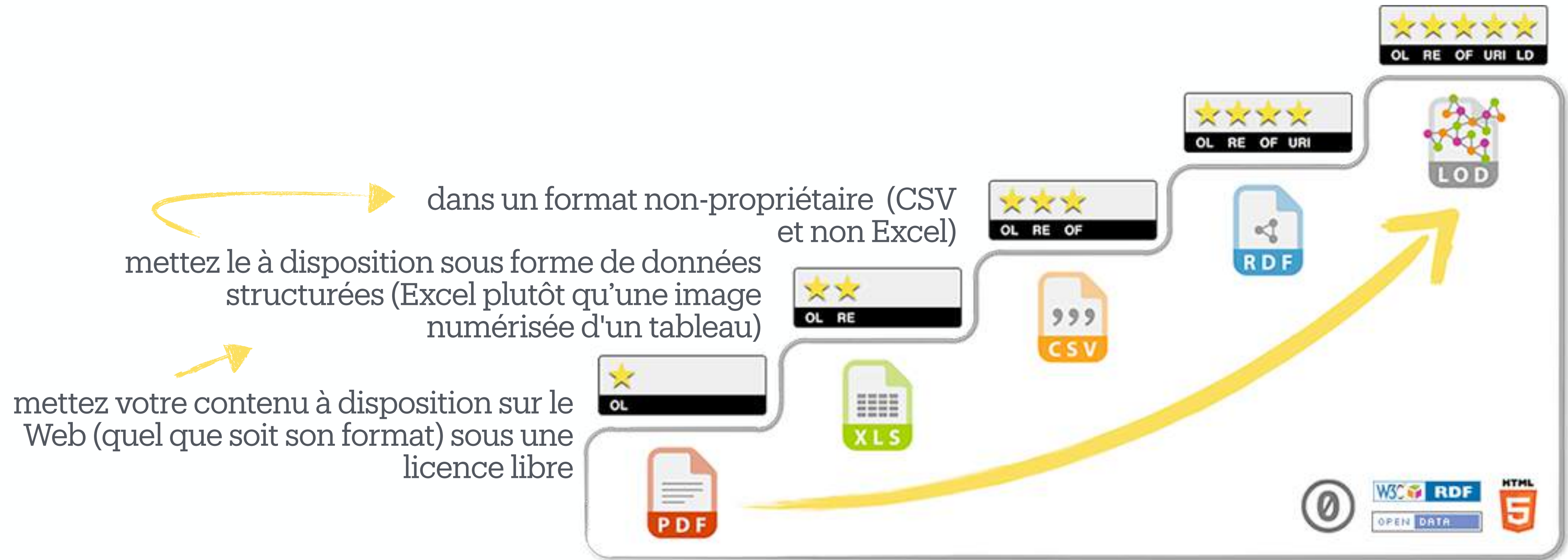
Données ouvertes liées



https://commons.wikimedia.org/wiki/File:5-star_deployment_scheme_for_Open_Data.png

<https://5stardata.info>

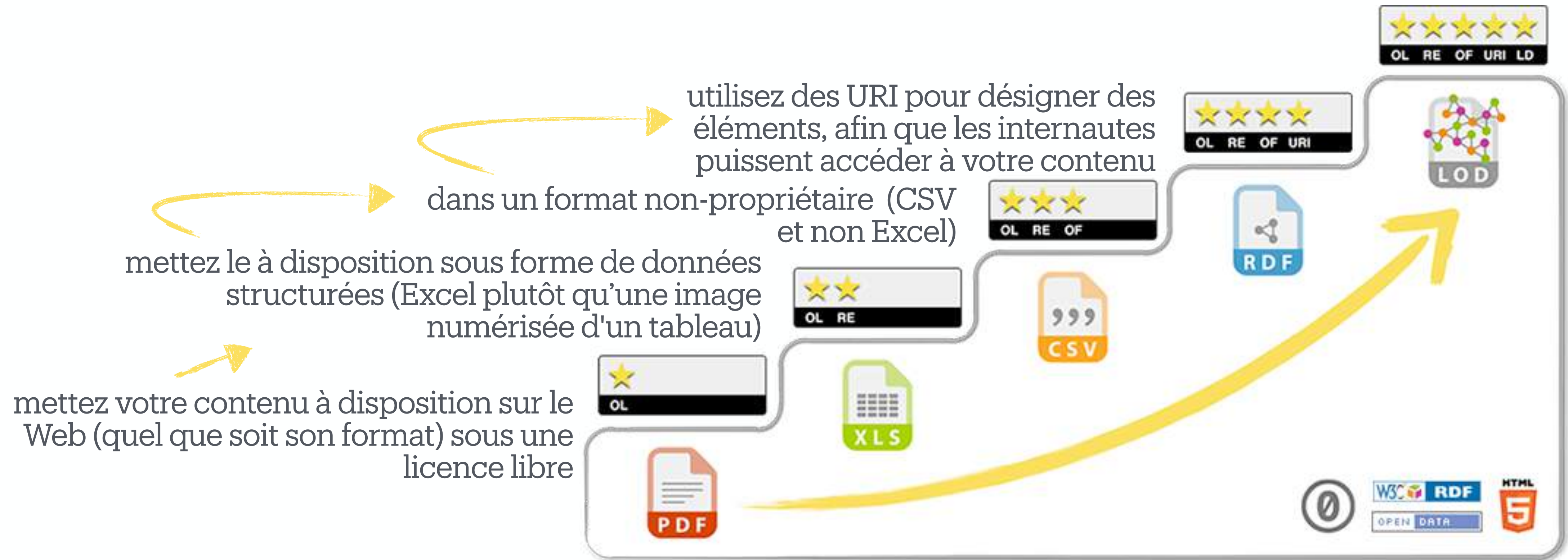
Données ouvertes liées



https://commons.wikimedia.org/wiki/File:5-star_deployment_scheme_for_Open_Data.png

<https://5stardata.info>

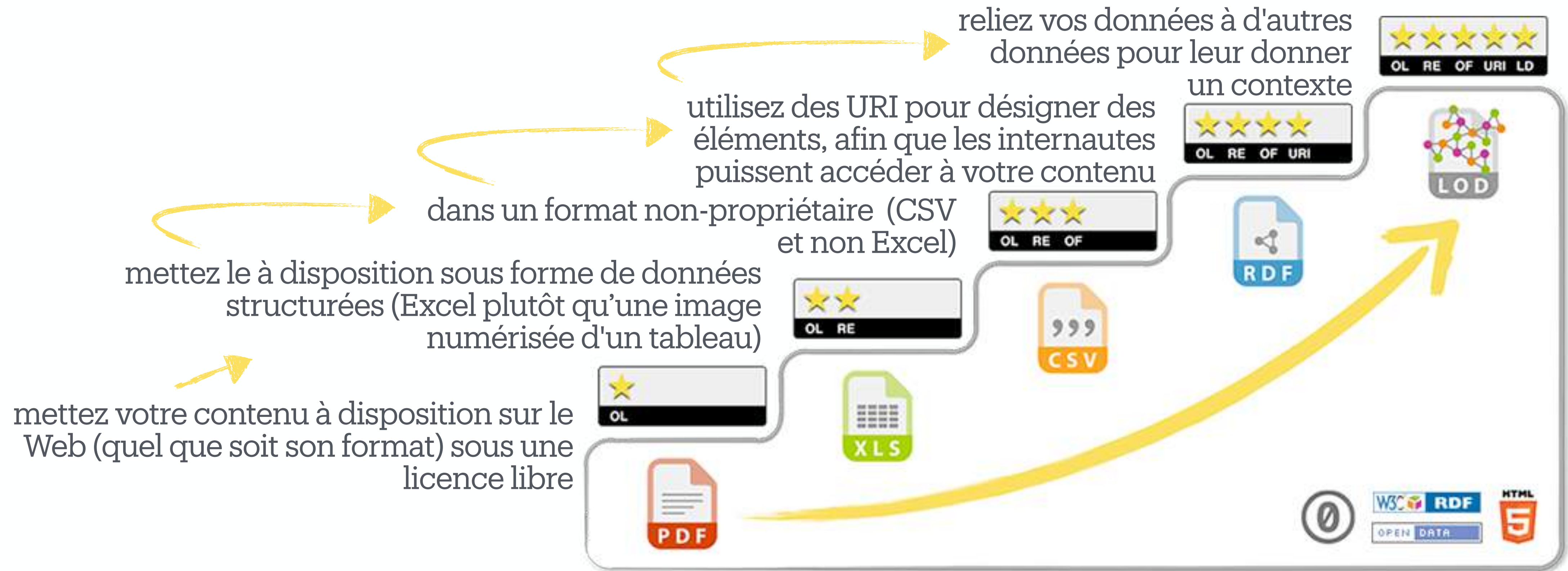
Données ouvertes liées



https://commons.wikimedia.org/wiki/File:5-star_deployment_scheme_for_Open_Data.png

<https://5stardata.info>

Données ouvertes liées



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<https://5stardata.info>

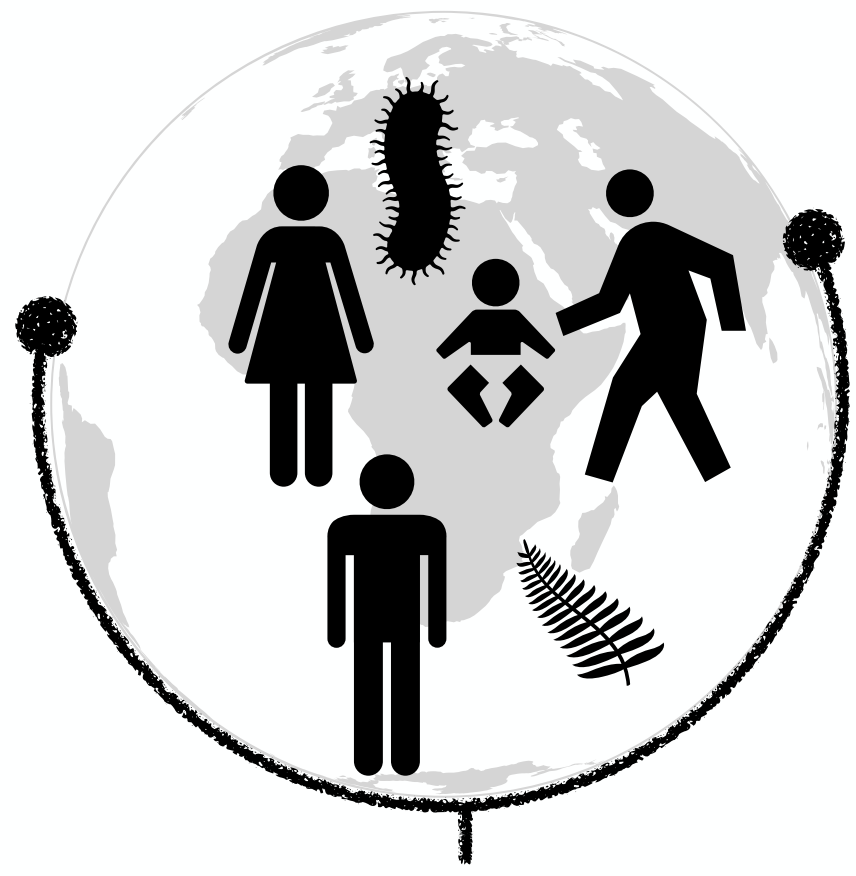
Données ouvertes liées



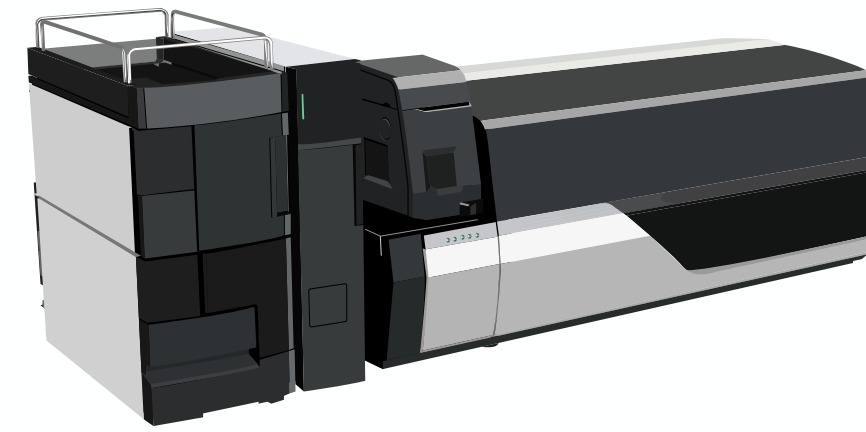
https://commons.wikimedia.org/wiki/File:5-star_deployment_scheme_for_Open_Data.png

<https://5stardata.info>

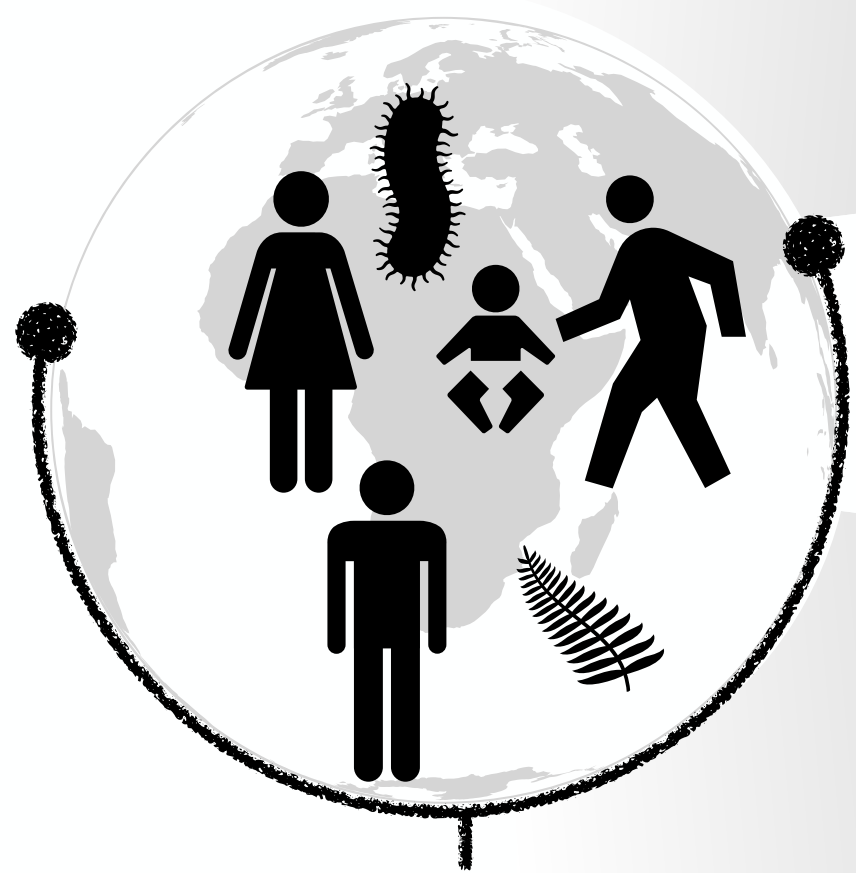
Les données ouvertes liées comme *macroscope* pour la métabolomique



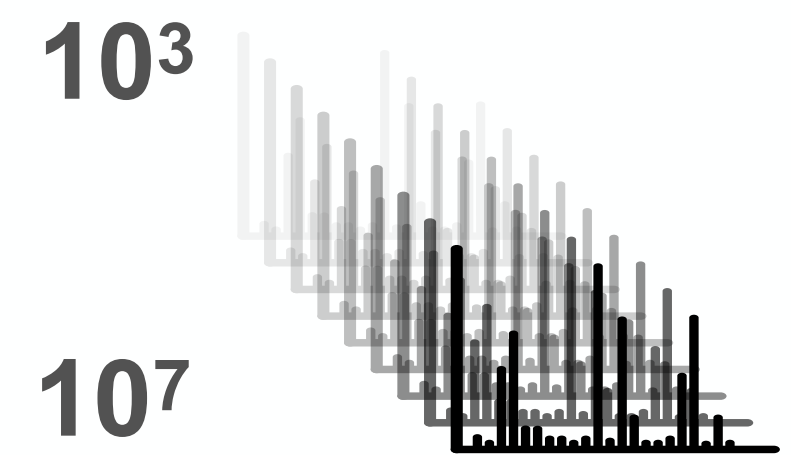
Adapté d'une version de Pierre-Marie Allard

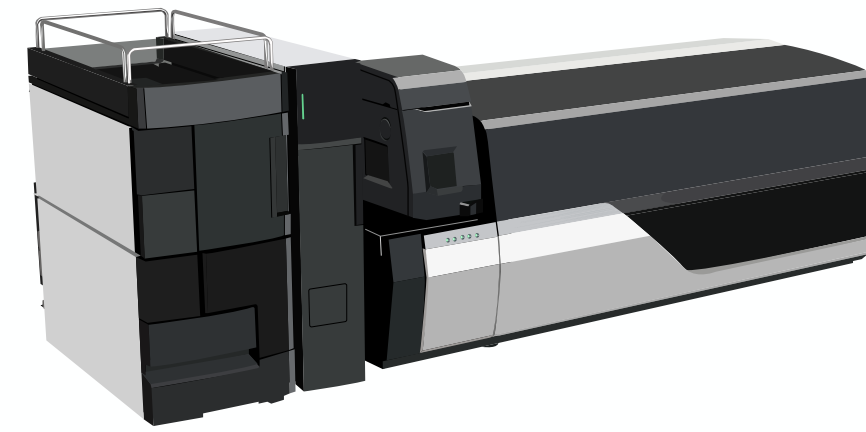


microscope



macroscope



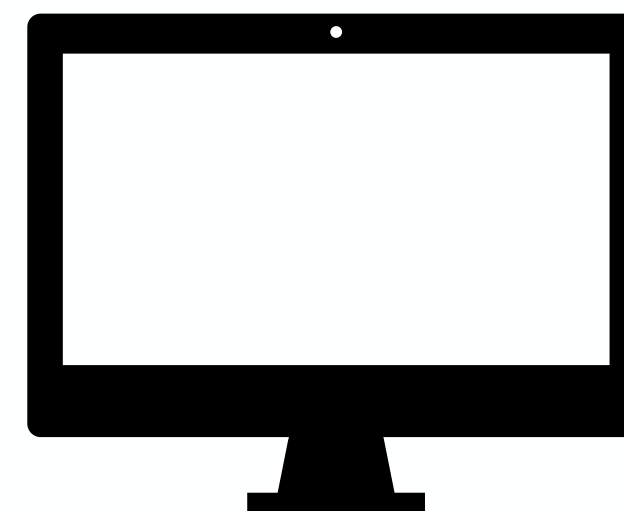
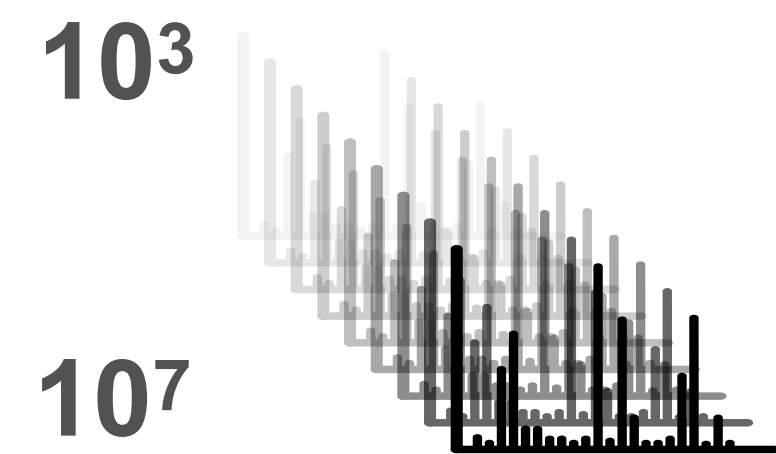
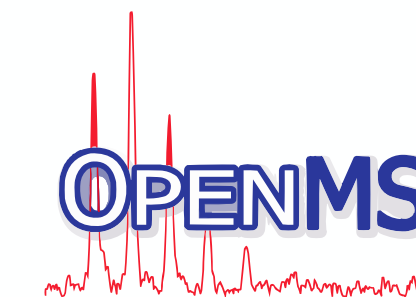


microscope

macroscope

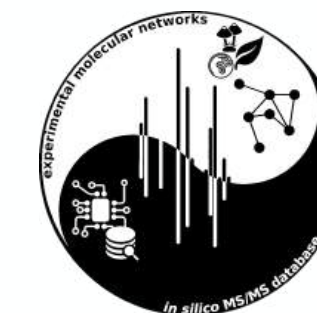


mzmine



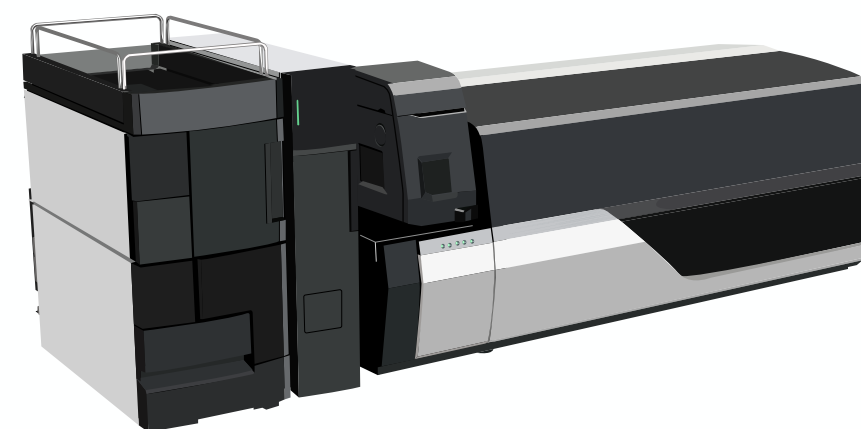
MassBank
High Quality Mass Spectral Database

MATCHMS



SIRIUS

MoNA - MassBank of North America



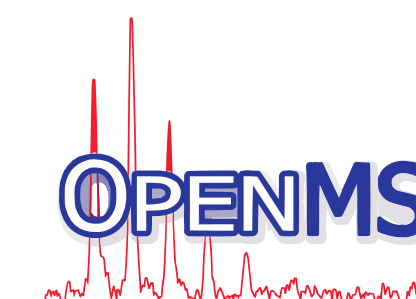
microscope



macroscope

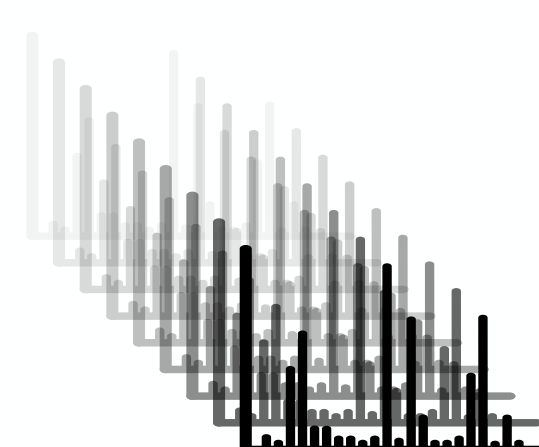


mzmine



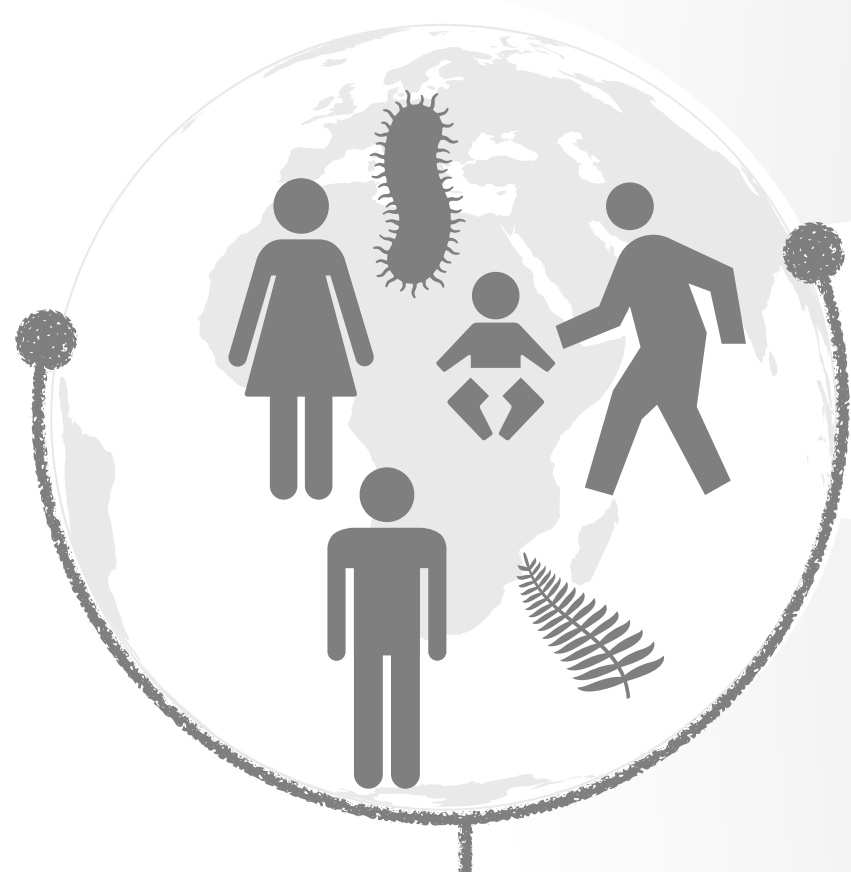
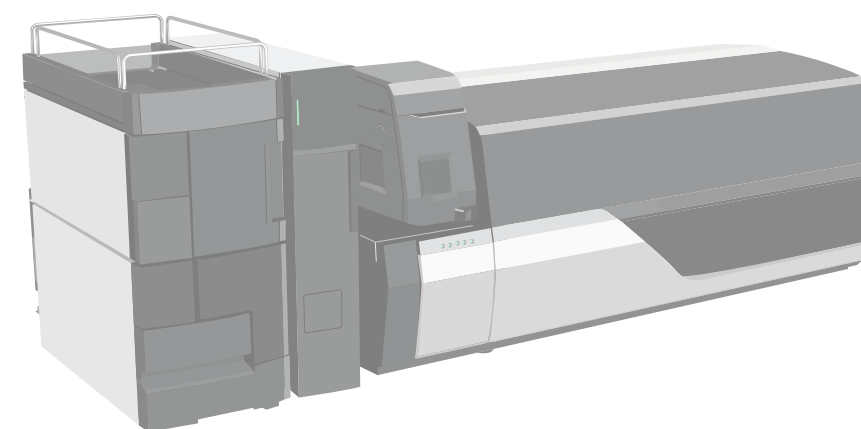
10^3

10^7



MoNA - MassBank of North America

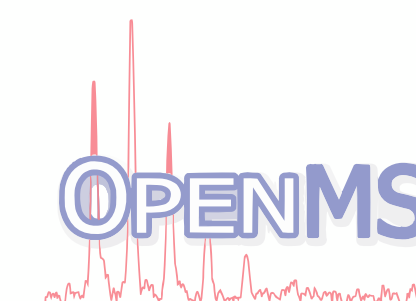
Adapté d'une version de Pierre-Marie Allard



microscope

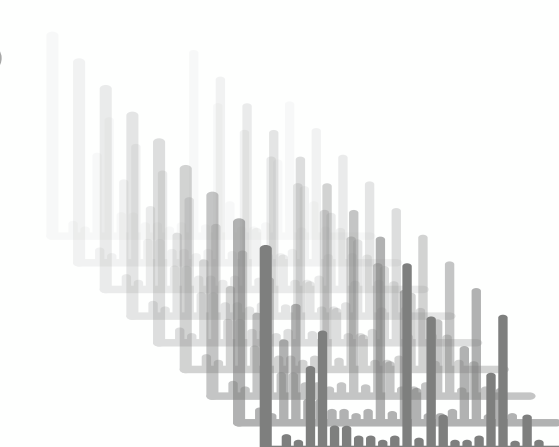


mzmine

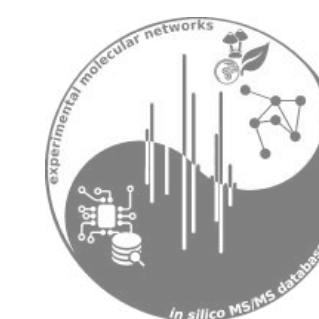


10^3

10^7



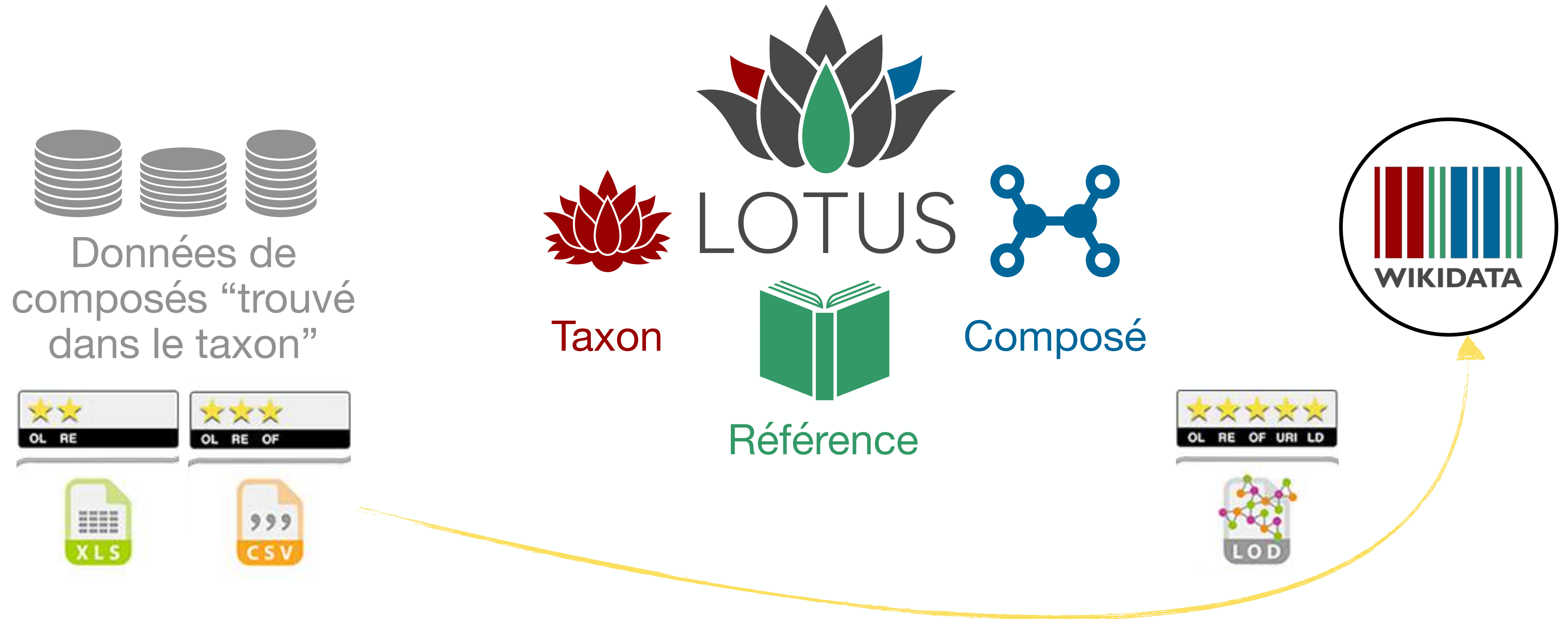
macroscope



MoNA - MassBank of North America

Adapté d'une version de Pierre-Marie Allard

The LOTUS Initiative



Rendre à César ce qui est à César...



FEATURE ARTICLE



SCIENCE FORUM

Wikidata as a knowledge graph for the life sciences

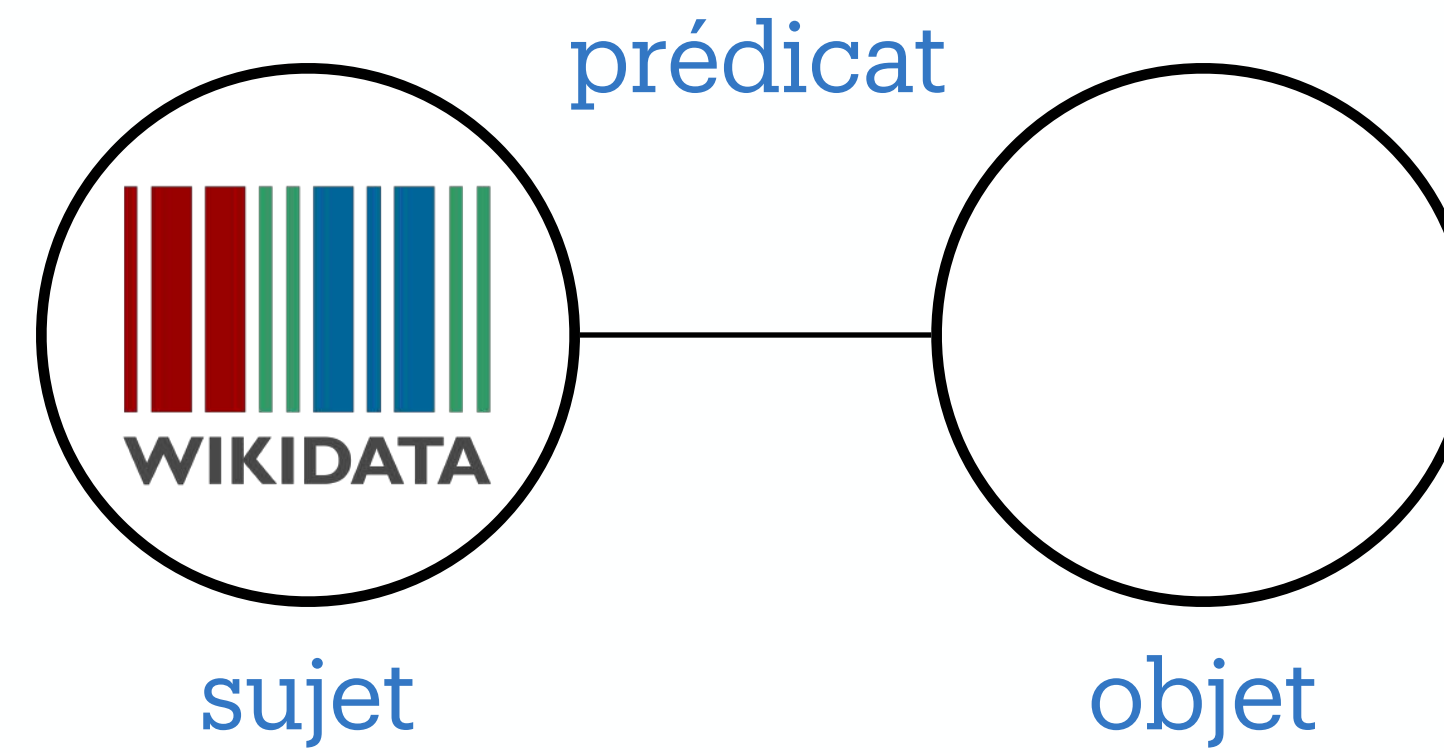
Abstract Wikidata is a community-maintained knowledge base that has been assembled from repositories in the fields of genomics, proteomics, genetic variants, pathways, chemical compounds, and diseases, and that adheres to the FAIR principles of findability, accessibility, interoperability and reusability. Here we describe the breadth and depth of the biomedical knowledge contained within Wikidata, and discuss the open-source tools we have built to add information to Wikidata and to synchronize it with source databases. We also demonstrate several use cases for Wikidata, including the crowdsourced curation of biomedical ontologies, phenotype-based diagnosis of disease, and drug repurposing.

ANDRA WAAGMEESTER[†], GREGORY STUPP[†], SEBASTIAN BURGSTALLER-MUEHLBACHER, BENJAMIN M GOOD, MALACHI GRIFFITH, OBI L GRIFFITH, KRISTINA HANSPERS, HENNING HERMJAKOB, TOBY S HUDSON, KEVIN HYBISKE, SARAH M KEATING, MAGNUS MANSKE, MICHAEL MAYERS, DANIEL MIETCHEN, ELVIRA MITRAKA, ALEXANDER R PICO, TIMOTHY PUTMAN, ANDERS RIUTTA, NURIA QUERALT-ROSINACH, LYNN M SCHRIML, THOMAS SHAFEE, DENISE SLENTER, RALF STEPHAN, KATHERINE THORNTON, GINGER TSUENG, ROGER TU, SABAH UL-HASAN, EGON WILLIGHAGEN, CHUNLEI WU AND ANDREW I SU*

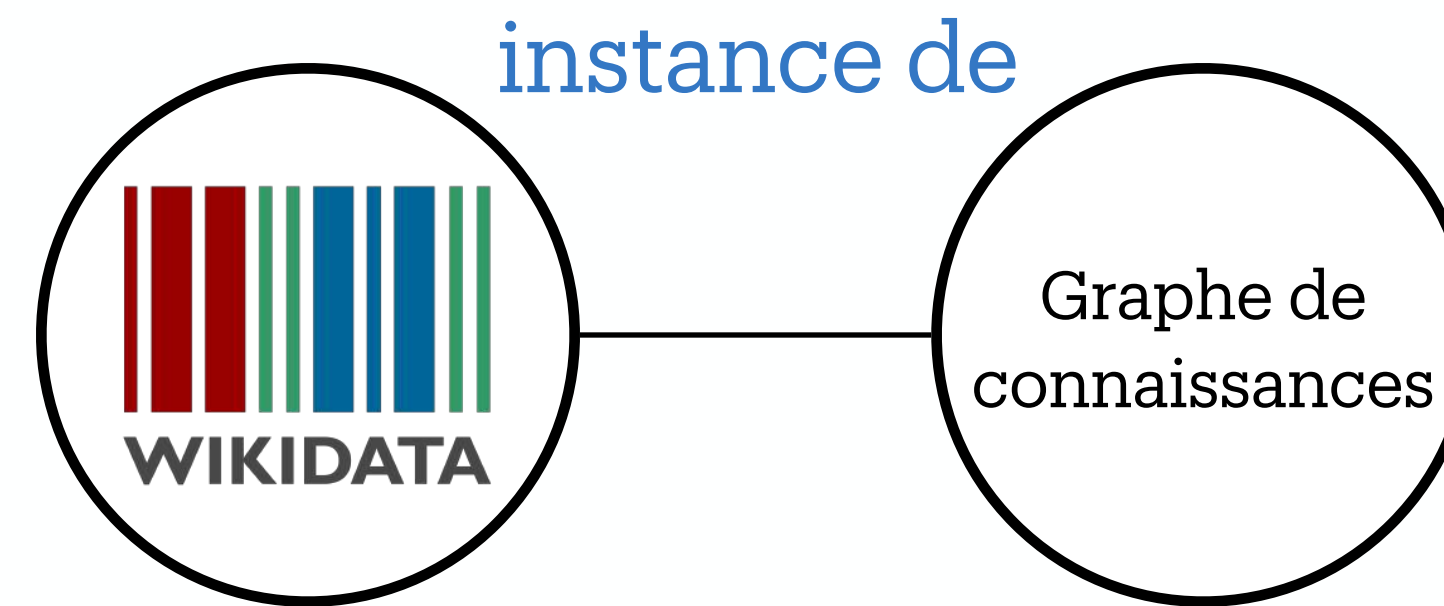
Wikidata



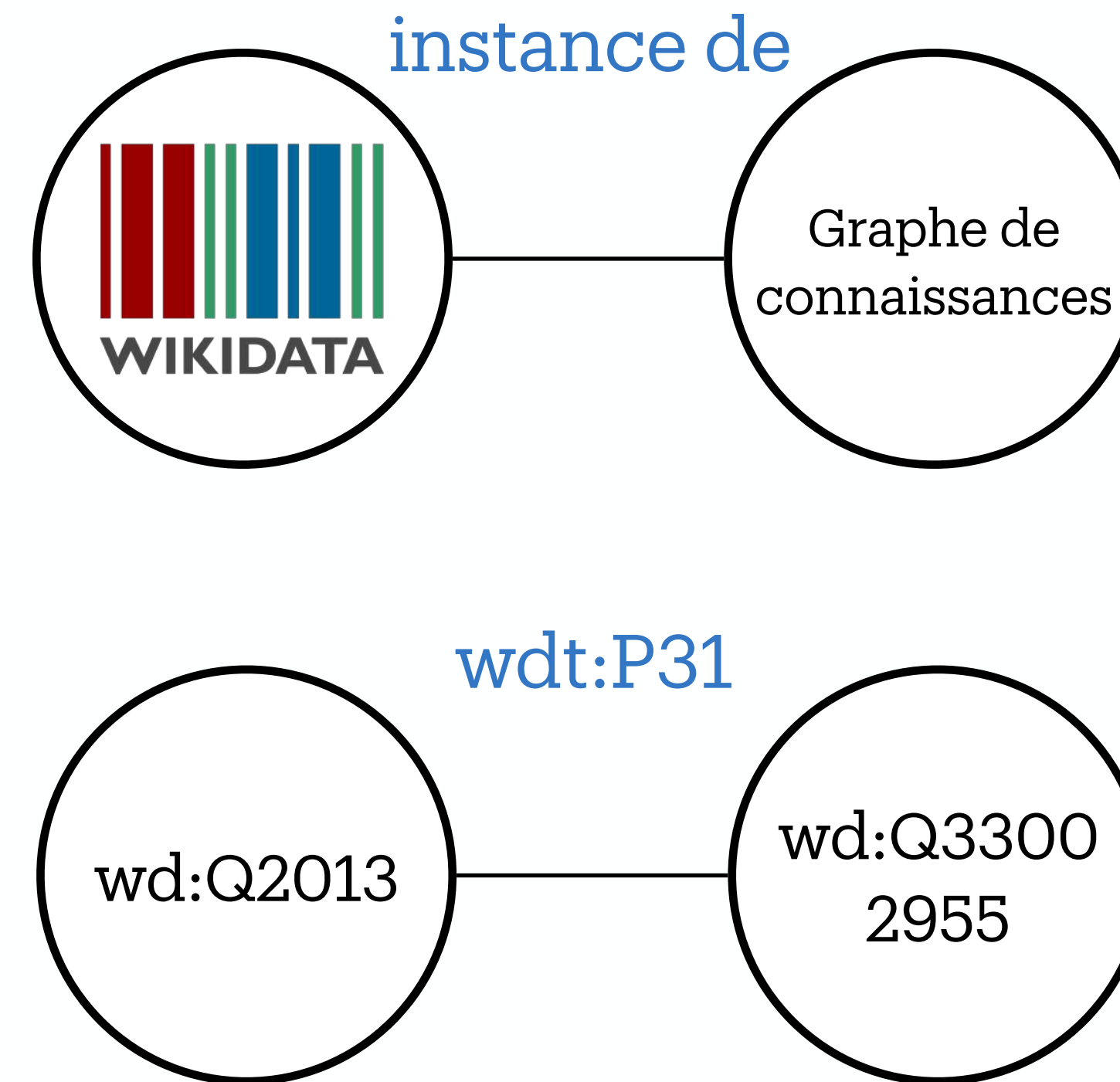
Wikidata



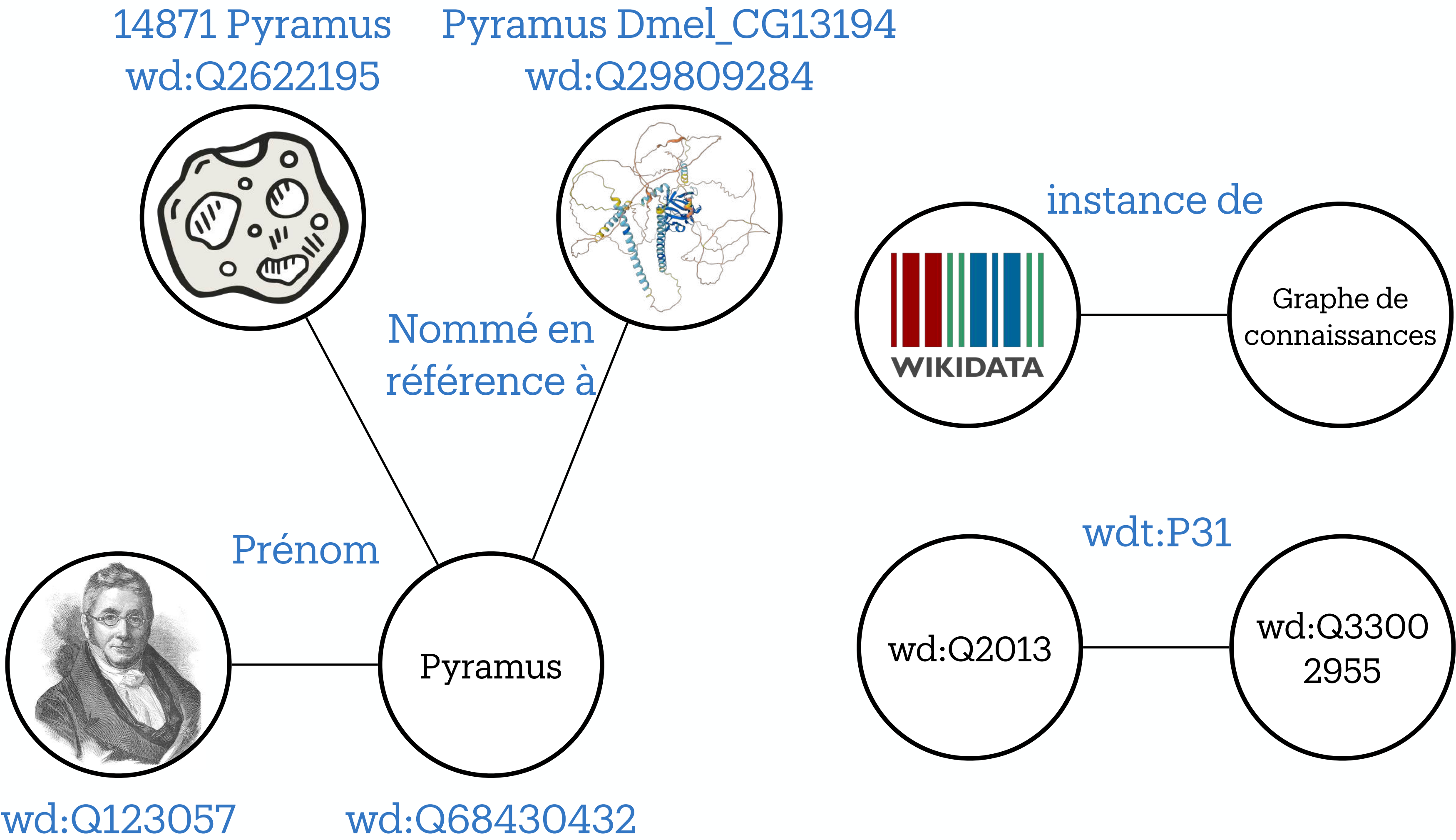
Wikidata



Wikidata

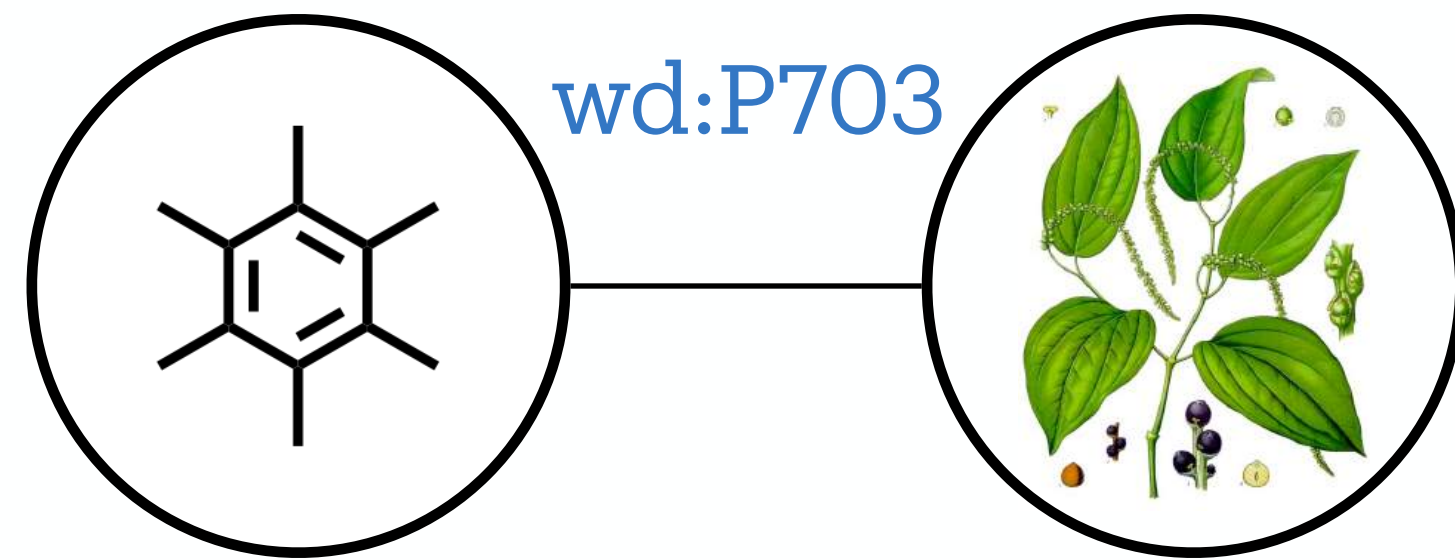


Wikidata



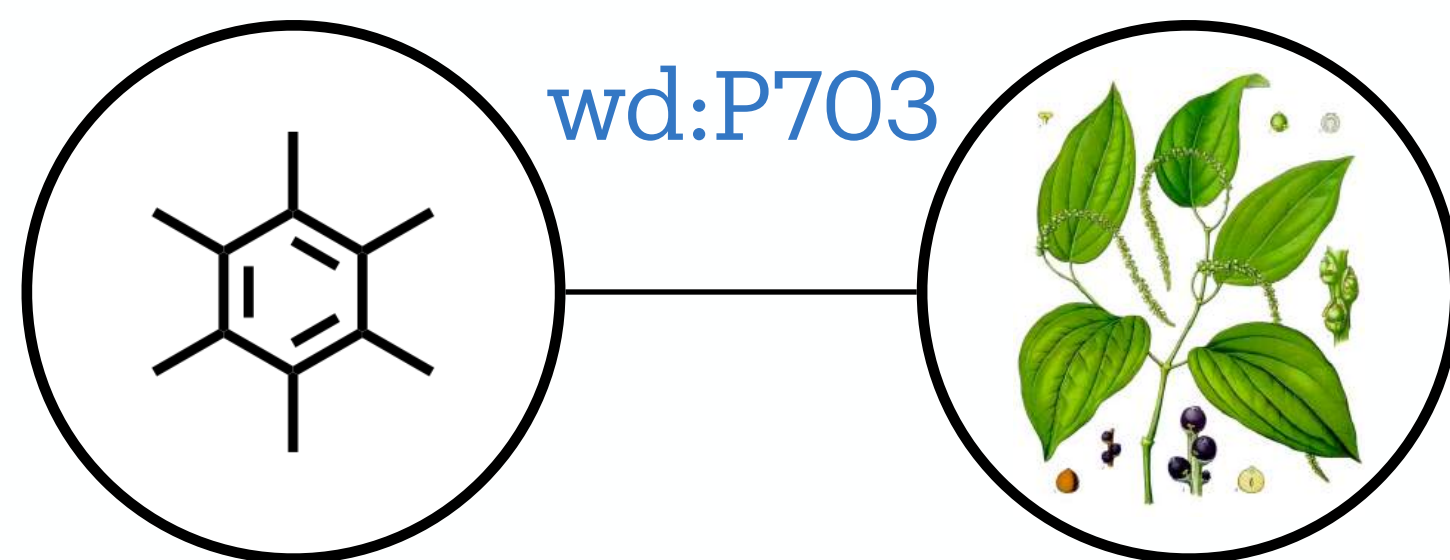
SPARQL

« Hey Wiki, quels composés ont été reportés dans *Piper nigrum* L.? »



SPARQL

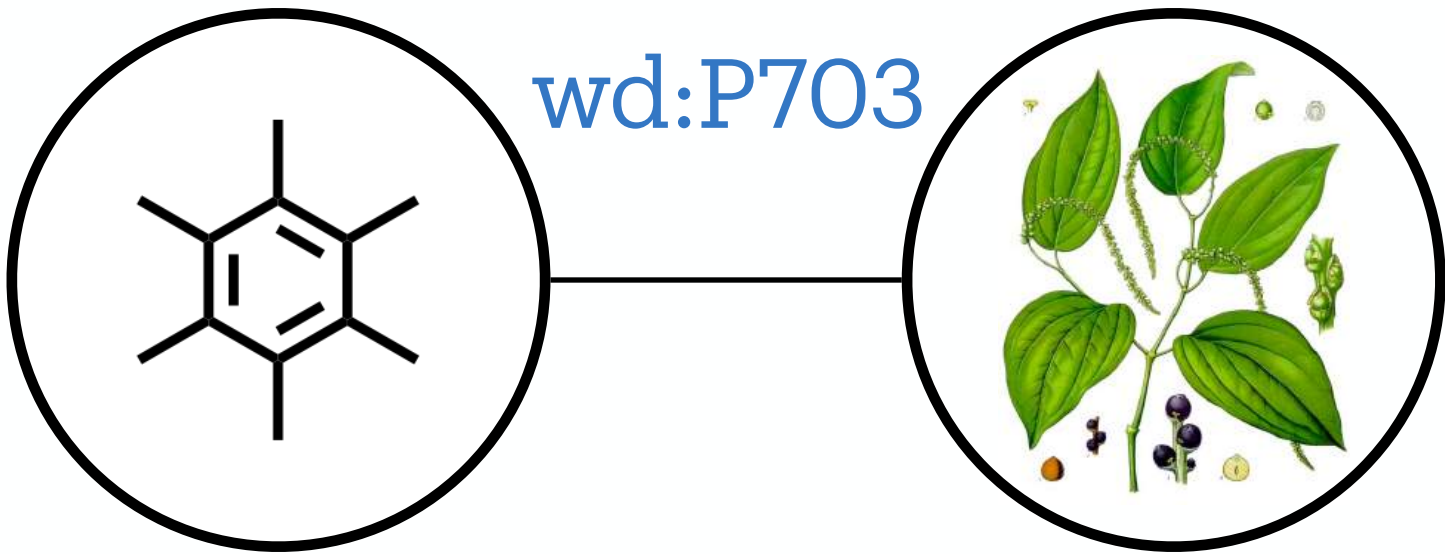
« Hey Wiki, quels composés ont été reportés dans *Piper nigrum* L.? »



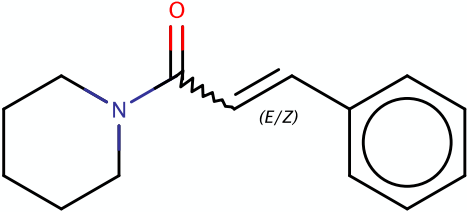
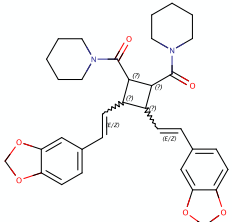
```
SELECT DISTINCT
?structure ?
structure_smiles ?
structure_inchikey WHERE
{
  VALUES ?organism {
    wd:Q43084 # Piper
nigrum
  }
  ?organism_child
(wdt:P171*) ?organism.
  ?structure wdt:P233 ?
structure_smiles;
  wdt:P235 ?
structure_inchikey;
  (p:P703/ps:P703) ?
organism_child.
}
```

SPARQL

« Hey Wiki, quels composés ont été reportés dans *Piper nigrum* L.? »



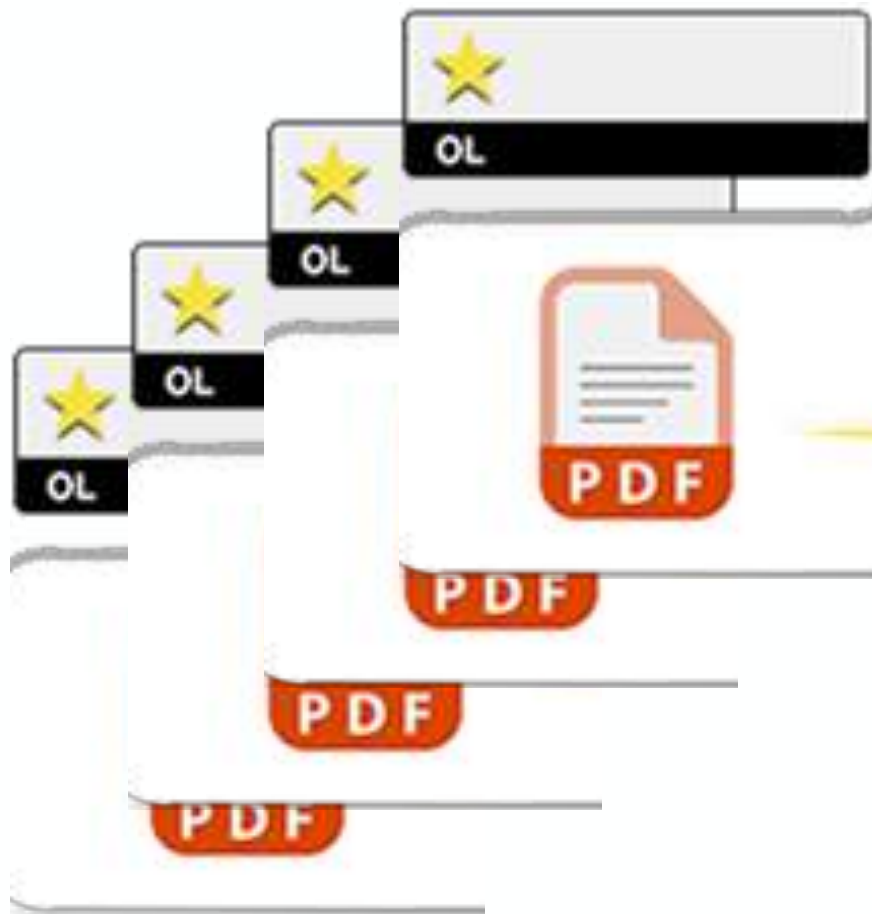
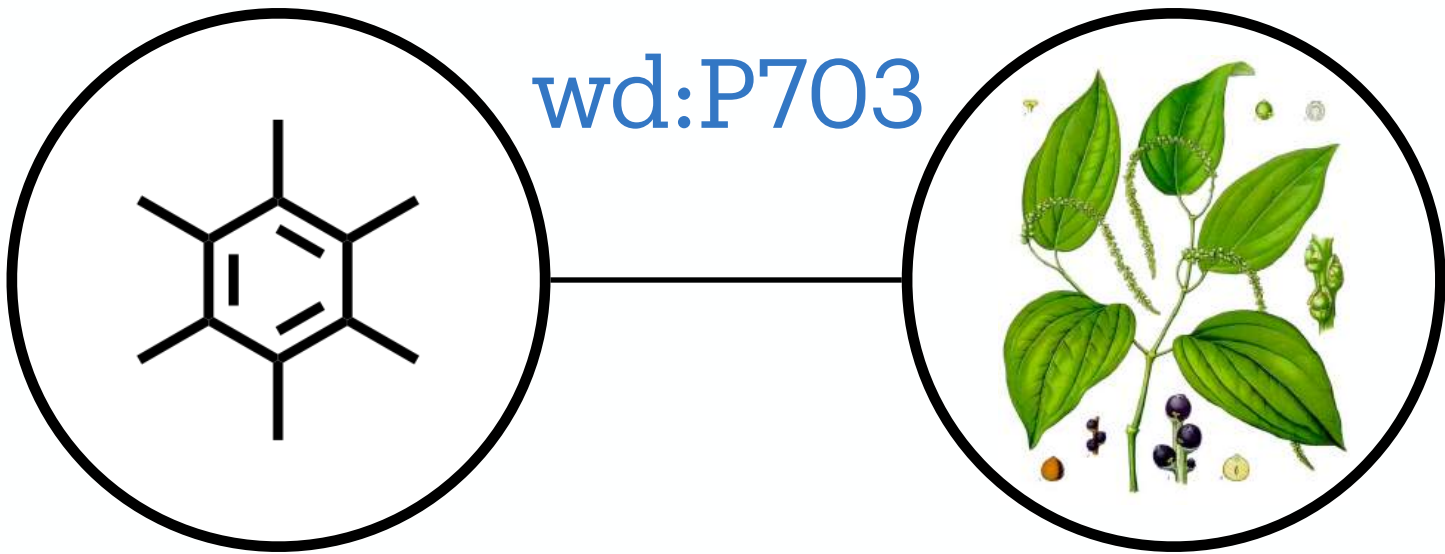
```
SELECT DISTINCT
?structure ?
structure_smiles ?
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{
  VALUES ?organism {
    wd:Q43084 # Piper
nigrum
  }
  ?organism_child
(wdt:P171*) ?organism.
  ?structure wdt:P233 ?
structure_smiles;
  wdt:P235 ?
structure_inchikey;
  (p:P703/ps:P703) ?
organism_child.
}
```

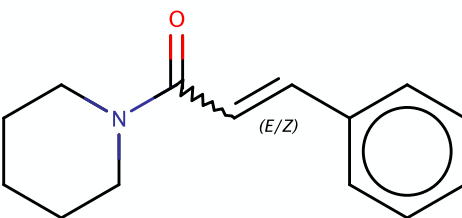
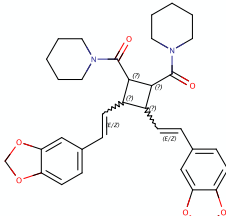
row	structure	structure_smiles	structure_inchikey
1	wd:Q105143500		KNOXUMZPTHELAO-UHFFFAOYSA-N
...	...	...	...
419	wd:Q105224845		QOESZPJYTVEXNN-UHFFFAOYSA-N

419 résultats en 0.5 seconde

SPARQL

« Hey Wiki, quels composés ont été reportés dans *Piper nigrum* L.? »



row	structure	structure_smiles	structure_inchikey
1	wd:Q105143500		KNOXUMZPTHELAO-UHFFFAOYSA-N
...	...	...	...
419	wd:Q105224845		QOESZPJYTVEXNN-UHFFFAOYSA-N

```
SELECT DISTINCT
?structure ?
structure_smiles ?
structure_inchikey WHERE
{
  VALUES ?organism {
    wd:Q43084 # Piper
nigrum
  }
  ?organism_child
(wdt:P171*) ?organism.
  ?structure wdt:P233 ?
structure_smiles;
  wdt:P235 ?
structure_inchikey;
  (p:P703/ps:P703) ?
organism_child.
}
```

419 résultats en 0.5 seconde

SPARQL

« Hey Wiki, quels composés ont été
reportés dans *H. sapiens*? »

SPARQL

« Hey Wiki, quels composés ont été reportés dans *H. sapiens*? »

1,886 résultats en 0.4 seconde

SPARQL

« Hey Wiki, quels composés ont été reportés dans *H. sapiens*? »

1,886 résultats en 0.4 seconde

« Hey Wiki, quels composés ont été reportés dans tous les primates? »

1,897 résultats en 0.2 seconde

SPARQL

« Hey Wiki, quels composés ont été reportés dans *H. sapiens*? »

1,886 résultats en 0.4 seconde

« Pardon Wiki, j'ai oublié combien y-a-t-il d'espèces de mammifères? »

19,379 résultats en 0.3 seconde

« Hey Wiki, quels composés ont été reportés dans tous les primates? »

1,897 résultats en 0.2 seconde

SPARQL

« Hey Wiki, quels composés ont été reportés dans *H. sapiens*? »

1,886 résultats en 0.4 seconde

« Pardon Wiki, j'ai oublié combien y-a-t-il d'espèces de mammifères? »

19,379 résultats en 0.3 seconde

« Hey Wiki, quels composés ont été reportés dans tous les primates? »

1,897 résultats en 0.2 seconde

« Ok Wiki, combien de composés amers ont été reportés dans ces espèces? »

59 results in 1.4 seconds

SPARQL

« Hey Wiki, quels composés ont été reportés dans *H. sapiens*? »

1,886 résultats en 0.4 seconde

« Pardon Wiki, j'ai oublié combien y-a-t-il d'espèces de mammifères? »

19,379 résultats en 0.3 seconde

« Hey Wiki, quels composés ont été reportés dans tous les primates? »

1,897 résultats en 0.2 seconde

« Ok Wiki, combien de composés amers ont été reportés dans ces espèces? »

56 résultats en 0.1 seconde



Quelques exemples additionnels

« Hey Wiki, j'ai vu cette revue l'autre jour, peux-tu faire la même? »



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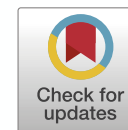


Actinobacteria in natural products research: Progress and prospects

Polpass Arul Jose ^{a,*}¹, Anjisha Maharshi ^a², Bhavanath Jha ^{a,b}^{**}

^a Marine Biotechnology and Ecology Division, CSIR- Central Salt and Marine Chemicals Research Institute, G. B. Marg, Bhavnagar, Gujarat, 364002, India

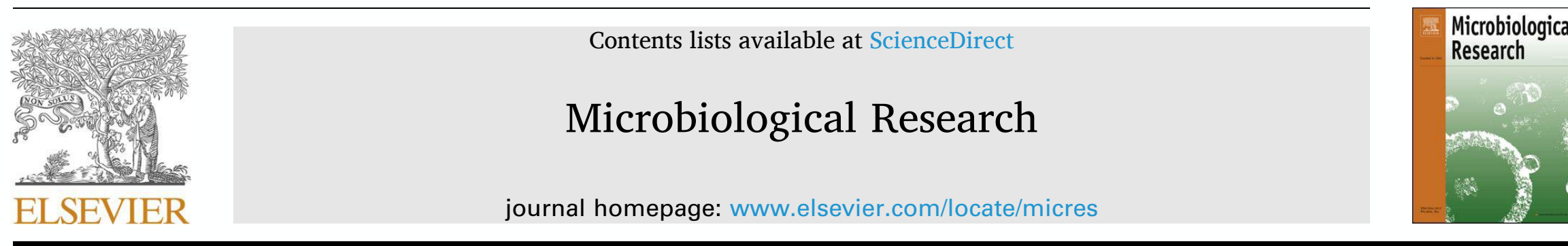
^b Academy of Scientific and Innovative Research (AcSIR), CSIR, India



<https://doi.org/10.1016/j.micres.2021.126708>

Quelques exemples additionnels

« Hey Wiki, j'ai vu cette revue l'autre jour, peux-tu faire la même? »



Quels composés avec une bioactivité ont été reportés dans une Actinobactérie (Q26262282), entre 2014 et 2019, avec les organismes et références reliés?

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Polpass Arul Jose^{a,*}, Anjisha Maharshi^{a,2}, Bhavanath Jha^{a,b,*}

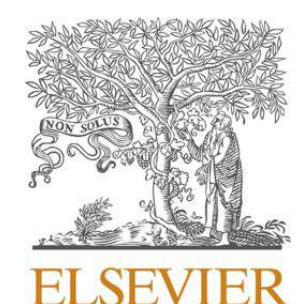
^a Marine Biotechnology and Ecology Division, CSIR- Central Salt and Marine Chemicals Research Institute, G. B. Marg, Bhavnagar, Gujarat, 364002, India

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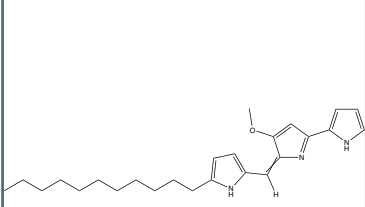
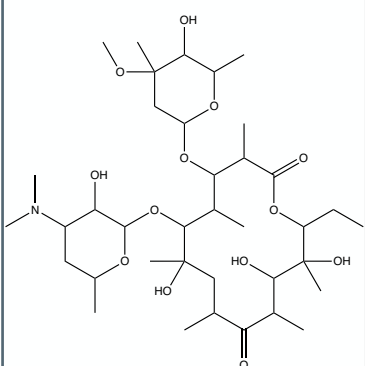
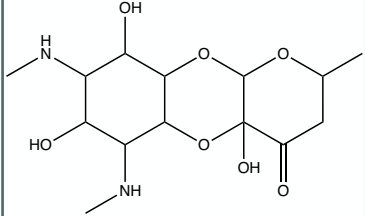
Table 1
Novel bioactive compounds isolated from Actinobacteria during last five years (2014-2019).

Organism	Compound	Bioactivity [#]	Reference
<i>Arthrobacter</i> sp. PGVB1	Arthroamide	f (anti-quorum sensing)	J Nat Prod. 2015;78:2827–2831
<i>Actinoallomurus</i> sp. ID145113 and 145,206	Paramagnetoquinones A–C (1–3)	(1/2) a; (3) a	J Nat Prod. 2017;80:819–827
<i>Actinoalloteichus hymeniacidonis</i> 179DD-027	Dokdolipid B	c	Mar Drugs 2019;17:237
<i>Actinomadura atramentaria</i> NBRC 14,695 ^{**}	Cinnamycin B	a	J Ind Microbiol Biotechnol. 2016;43:1159–65
<i>Actinomadura</i> sp. BCC 35,430 [*]	Actinomadurone	b	Tetrahedron Lett. 2017;58:3223–3225
<i>Actinomadura</i> sp. K13–0306	Sagamilactam	c, f (antiparasitic)	J Antibiot. 2016;69:818–824
<i>Actinomadura</i> sp. K4S16	Nonthmicin (1) & ecteinamycin (2)	(1,2) c	Org Lett. 2017;19:1406–1409
<i>Actinomadura</i> sp. KC191 [*]	Actinomadurol	a	J Nat Prod. 2016;79:1886–1890
<i>Actinosynnema pretiosum</i> HGF052::asm18 ^{**}	Actinosynneptide A (1) & B (2)	(1,2) c	Appl Microbiol Biotechnol. 2016;101:2273–2279
<i>Amycolatopsis</i> sp. IRD-009	Pradimicin-IRD	a, c	Natural Product Res. 2019;33:1713–1720

<https://doi.org/10.1016/j.micres.2021.126708>

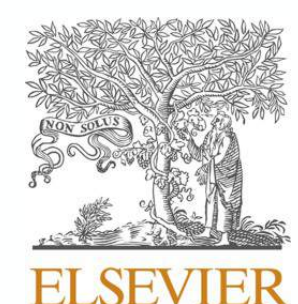
Quels composés avec une bioactivité ont été reportés dans une Actinobactérie (Q26262282), entre 2014 et 2019, avec les organismes et références reliés?

508 results in 4.8 seconds

Taxon name	Structure	Activity	Reference
<i>Streptomyces coelicolor</i>		Antibiotic immunosuppressive drug	10.1007/S11306-016-1025-6
<i>Saccharopolyspora erythraea</i>		Antibiotic protein synthesis inhibitors gastrointestinal agent	10.1021/ACSSYNBIO.8B00372
<i>Streptomyces venezuelae</i>		Antibiotic protein synthesis inhibitors	10.1016/J.JBIOTEC.2014.01.028

Quelques exemples additionnels

« Hey Wiki, j'ai vu cette revue l'autre jour, peux-tu faire la même? »



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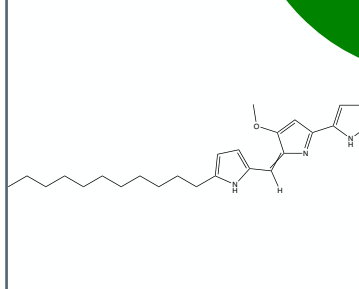
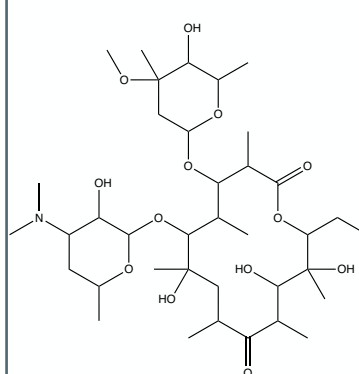
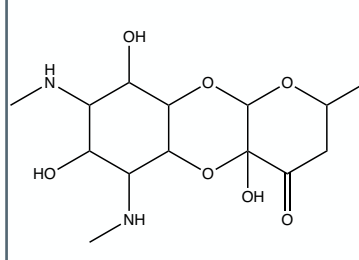
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<i>Arthrobacter</i> sp. PGVB1	Arthroamide	f (anti-quorum sensing)	J Nat Prod. 2015;78:2827–2831
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<i>Actinomadura</i> sp. K4S16	Nonthmicin (1) & ecteinamycin (2)	(1,2) c	Org Lett. 2017;19:1406–1409
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<https://doi.org/10.1016/j.micres.2021.126708>

Quels composés avec une bioactivité ont été reportés dans une Actinobactérie (Q26262282), entre 2014 et 2019, avec les organismes et références reliés?



08 results in 4.8 seconds

Taxon name	Structure	Activity	Reference
<i>Streptomyces coelicolor</i>		Antibiotic immunosuppressive drug	10.1007/S11306-016-1025-6
<i>Saccharopolyspora erythraea</i>		Antibiotic protein synthesis inhibitors gastrointestinal agent	10.1021/ACSSYNBIO.8B00372
<i>Streptomyces venezuelae</i>		Antibiotic protein synthesis inhibitors	10.1016/J.JBIOTEC.2014.01.028

Recherche multi-omique

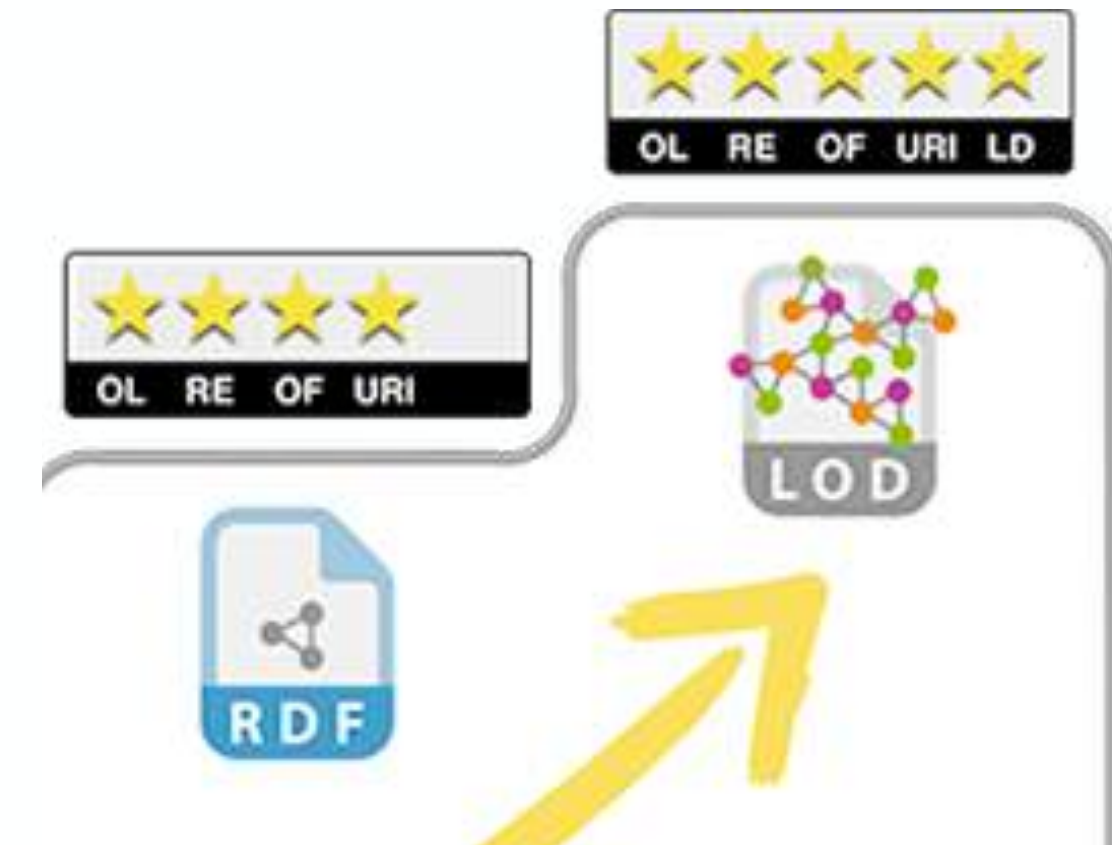
«Hey Wiki, pour combien d’organismes a-t-on au minimum 1 gène, 1 métabolite et 1 protéine disponibles?»

15 résultats en 4.7 secondes

	?taxon	?taxon_name	?genome	?metabolome	?proteome
1	Q10758650	Equus caballus	3	1	1
2	Q130888	Drosophila melanogaster	25,180	5	11,314
3	Q150162	Trypanosoma cruzi	21,465	1	22,939
4	Q158695	Arabidopsis thaliana	28	1,007	5
5	Q15978631	Homo sapiens	67,527	1,877	25,433
6	Q169444	Danio rerio	31,432	1	14,067
7	Q184224	Rattus norvegicus	52,931	20	15,805
8	Q184774	Gallus gallus	1	7	3
9	Q1950752	Leishmania donovani	1	2	3,963
10	Q2236682	Schizosaccharomyces pombe	2	8	4,061
11	Q25419	Escherichia coli	2	949	6
12	Q58697	Sus scrofa	1	4	2
13	Q719725	Saccharomyces cerevisiae	3	108	5
14	Q83310	Mus musculus	73,593	95	19,644
15	Q91703	Caenorhabditis elegans	47,274	765	17,673

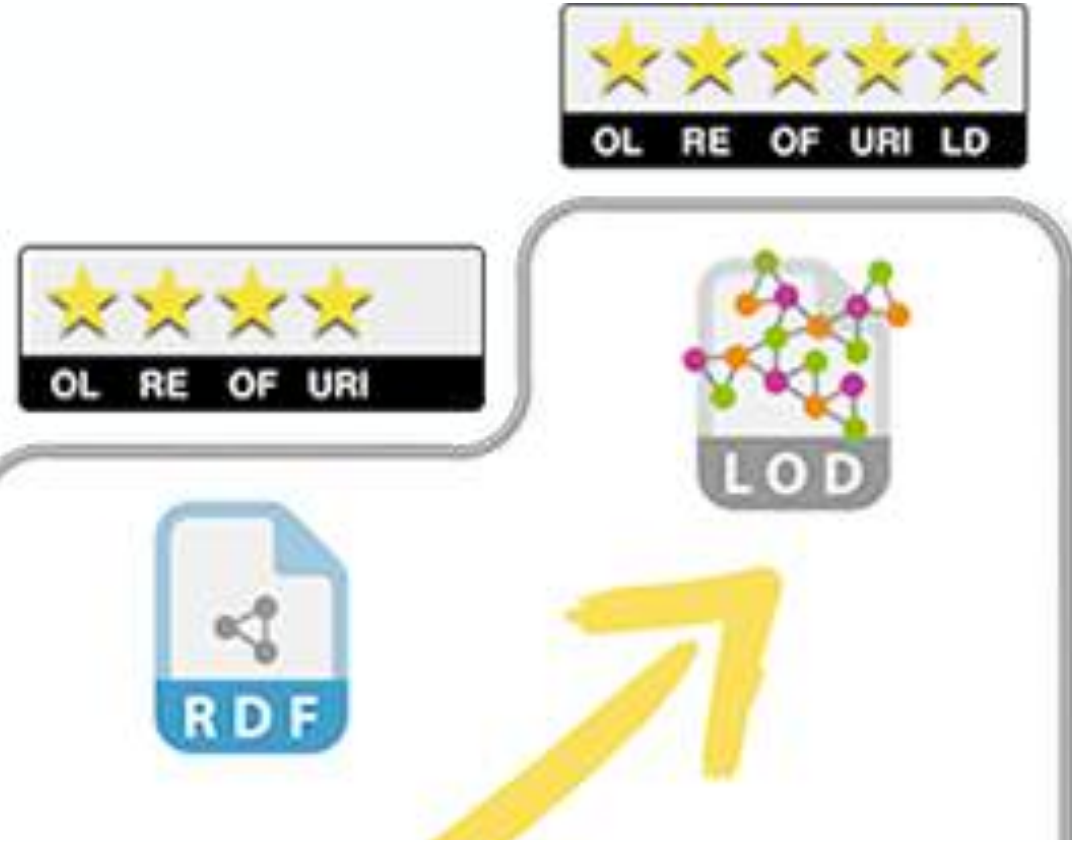
Fédération basée sur les données ouvertes liées

« Hey Wiki, dans quels genres ont été reportés des composés structurellement similaires à ceux responsable de la pseudo-chaleur du poivre noir? Donne moi les 5 premiers genres par nombre d'occurrences»



Fédération basée sur les données ouvertes liées

« Hey Wiki, dans quels genres ont été reportés des composés structurellement similaires à ceux responsable de la pseudo-chaaleur du poivre noir? Donne moi les 5 premiers genres par nombre d'occurrences»



5 résultats en 1.1 seconde

row	taxon	taxon_name	count
1	wd:Q159536	<i>Piper</i>	35
2	wd:Q163563	<i>Zingiber</i>	31
3	wd:Q133290	<i>Syzygium</i>	4
4	wd:Q2607361	<i>Aframomum</i>	4
5	wd:Q27583	<i>Bellis</i>	3

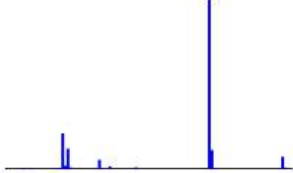
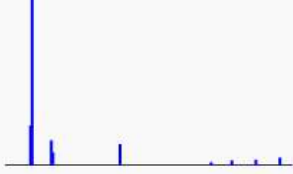
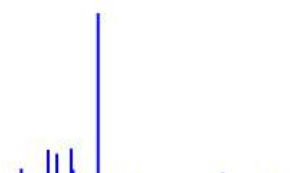


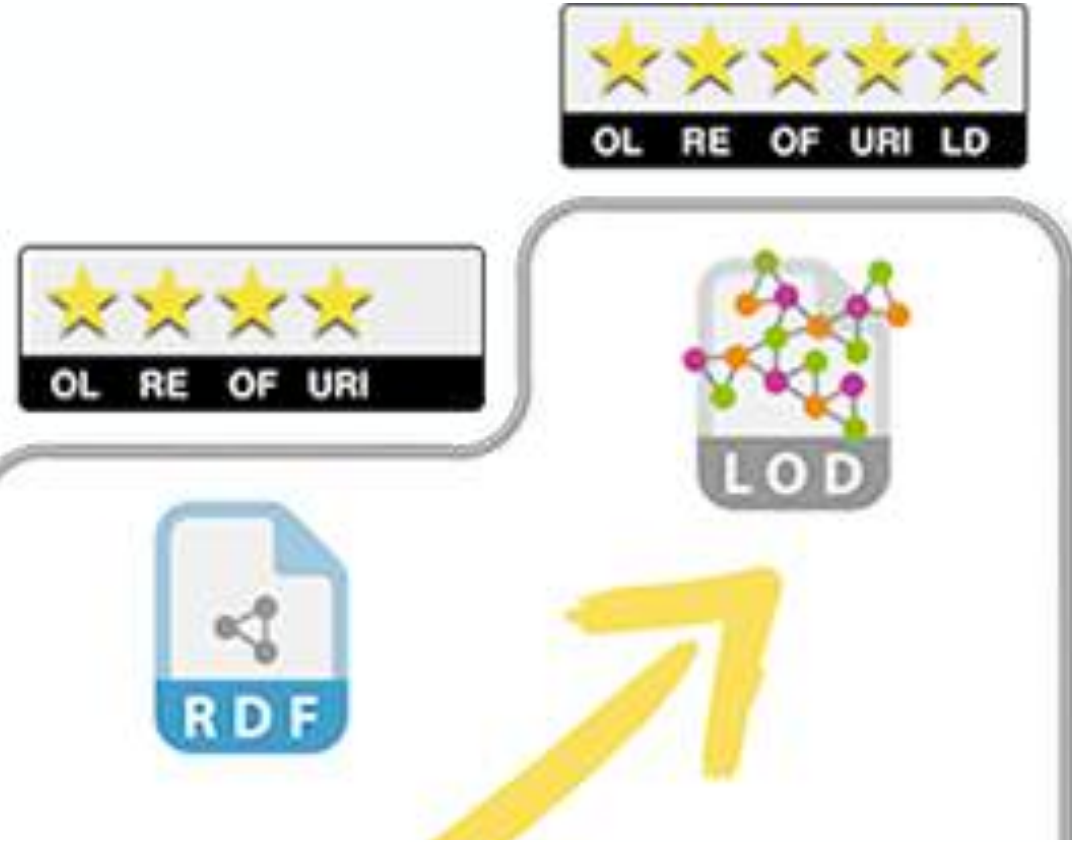
<https://doi.org/10.1186/s13321-018-0282-y>



Fédération basée sur les données ouvertes liées

« Hey Wiki, liste moi les spectres de masse de composés similaires à ceux ayant été reportés comme bio-actifs »

Table+ Response 1046 results in 173.318 seconds		Compact ☐	Filter query results	Page size:
	SPECTRUM	MSVALUE		
1	 mona:AU202803_MS	"188.1043:100"^^ ms:spectrum		
2	 mona:AU154503_MS	"188.1428:100 189.1459:8.509666 216.1378:8.315502 217.1415:1.020664 337.2283:14.375924 338.2312:2.634483"^^ ms:spectrum		
3	 mona:AU154504_MS	"117.0566:0.848243 117.0691:0.759028 120.0799:0.752058 130.0639:0.60778 132.08:20.82343 133.0831:2.088895 134.0956:11.968105 135.0988:1.113101 146.0954:5.500687 147.0992:0.595931 150.0903:1.632363 160.1112:1.145163 188.1428:100 189.1457:11.144257 190.1489:0.575021 216.1374:7.227144 217.1413:0.968823"^^ ms:spectrum		
4	 mona:JP005261_MS	"56:23.3 57:99.99 69:14.7 70:7.68 112:12.4 169:2 182:3.1 183:0.82 197:3.5 212:4.7"^^ ms:spectrum		
5	 mona:AU387304_MS	"117.0214:9.888005 119.0192:2.077807 129.0693:1.399941 141.0703:4.450339 145.0531:20.910698 147.0505:5.157678 152.0614:1.75361 153.0695:5.275567 154.0781:18.611849 155.0837:2.298851 167.0872:1.562039 169.1016:21.52962 170.1051:3.477748 171.0318:9.386973 172.0366:1.296788 173.03:1.886236 182.0716:1.370469 197.0964:100 198.0999:14.839375 200.0597:1.945181 233.0377:2.180961 251.0468:1.164162 327.0796:7.132331 328.0817:2.004126 384.1508:2.564103"^^ ms:spectrum		

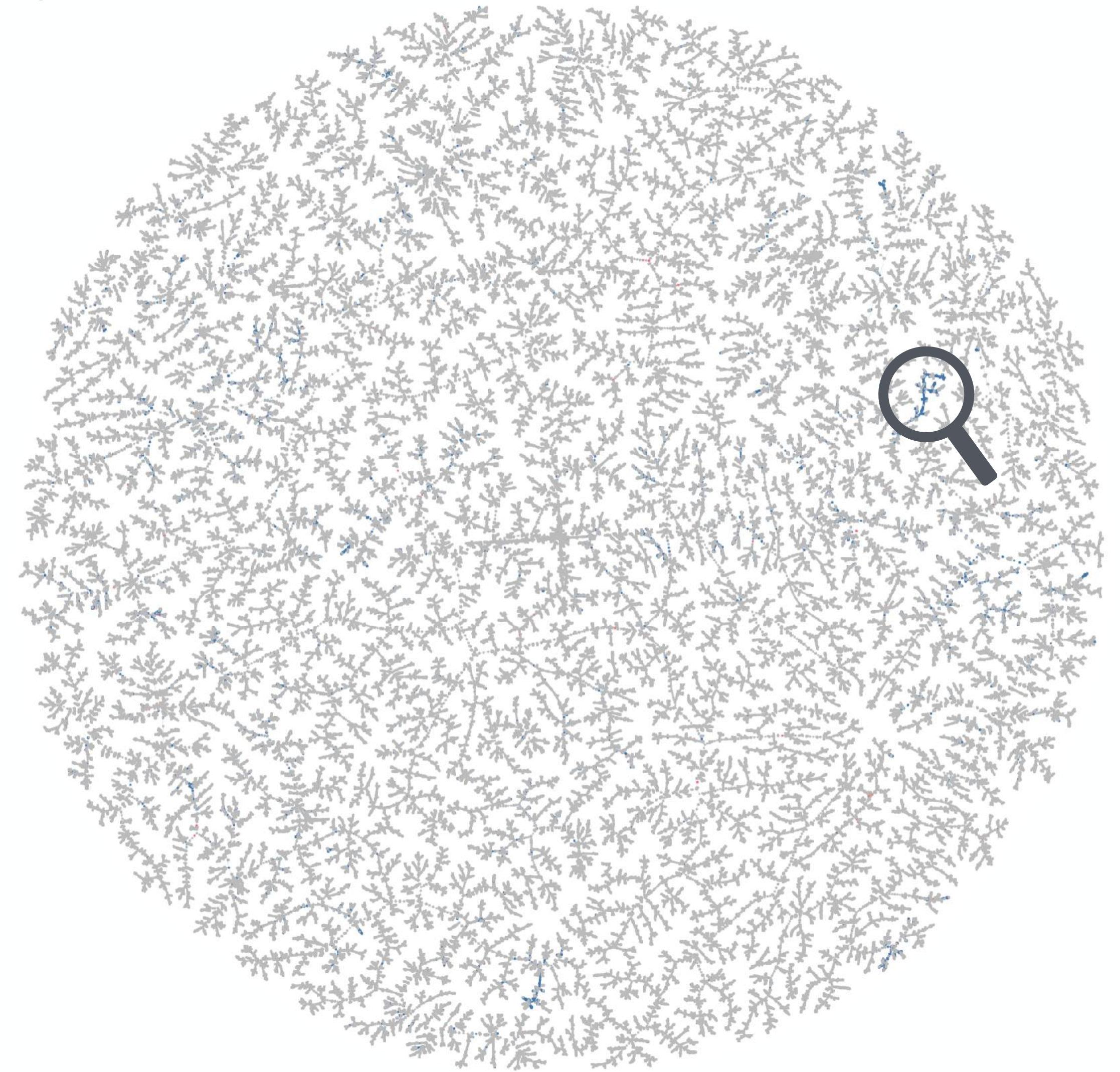
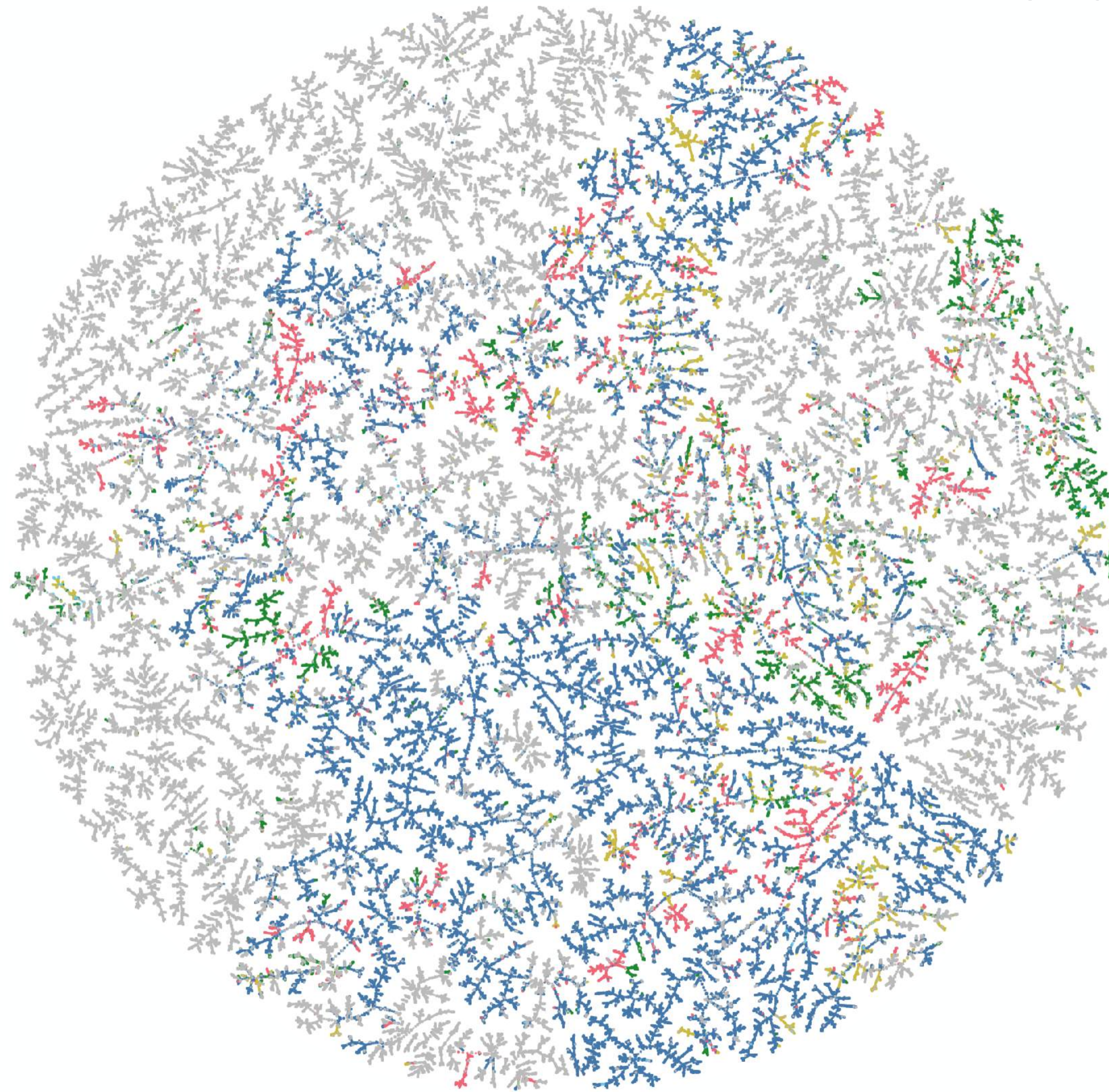


<https://doi.org/10.1186/s13321-018-0282-y>



Addendum

Contribuez au
monde ouvert!



● Plantae ● Fungi ● Bacteria ● Animalia ● Shared

● CID ● HCD ● EAD ● UVPD ● Shared

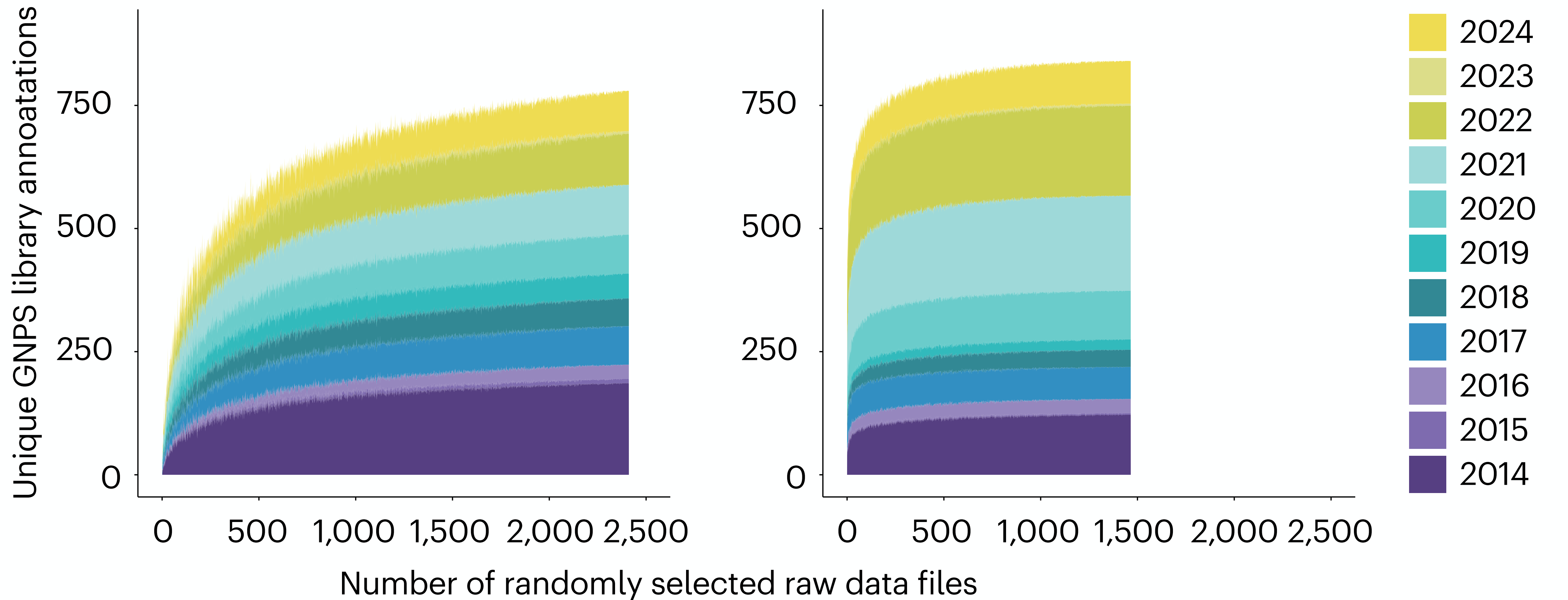
Addendum

Contribuez au
monde ouvert!



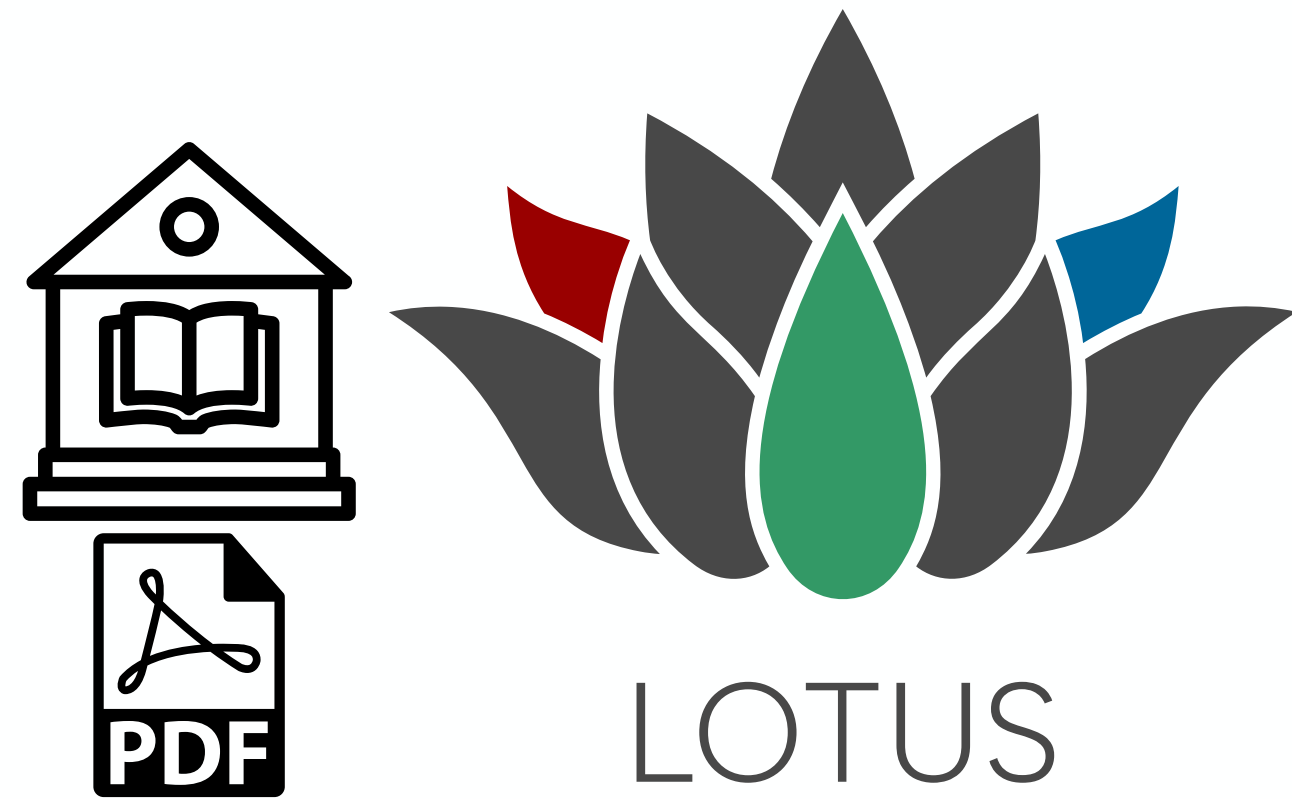
Fecal

Blood plasma



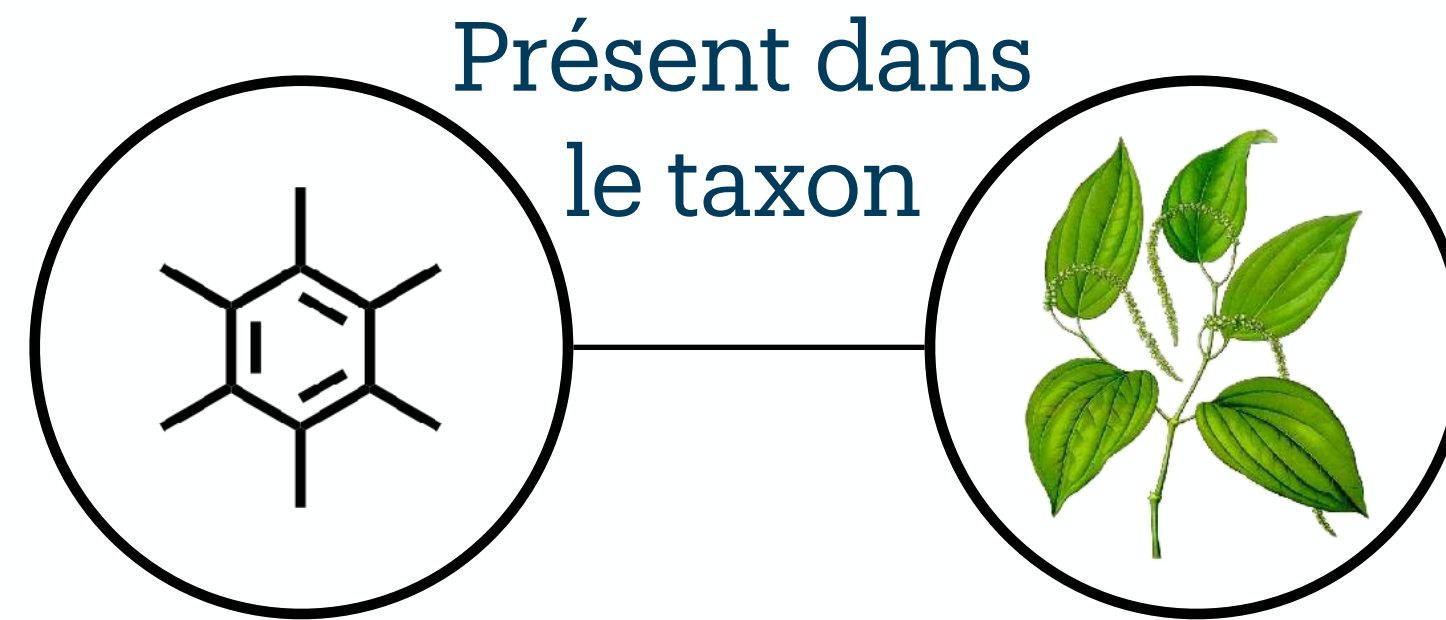
Addendum

Contribuez au
monde ouvert!



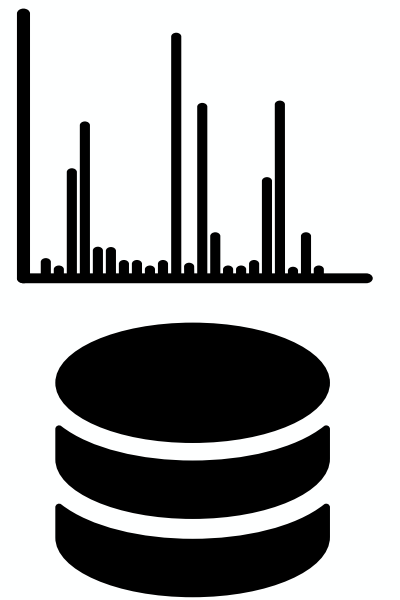
Connaissances préalables

<https://lotus.nprod.net/>



Connaissances futures

<https://www.earthmetabolome.org/>

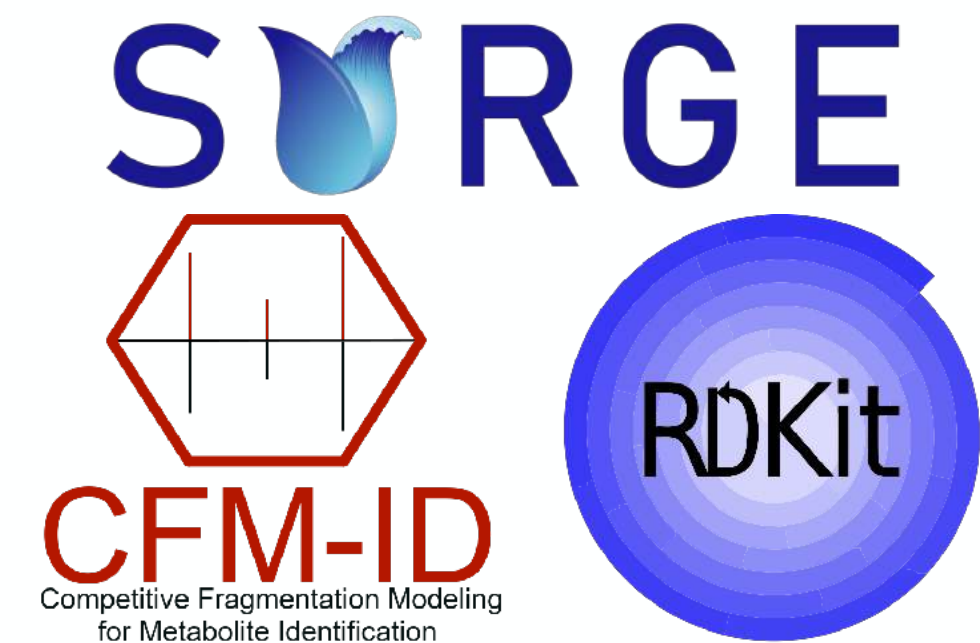


<https://www.metabolinkai.net/>

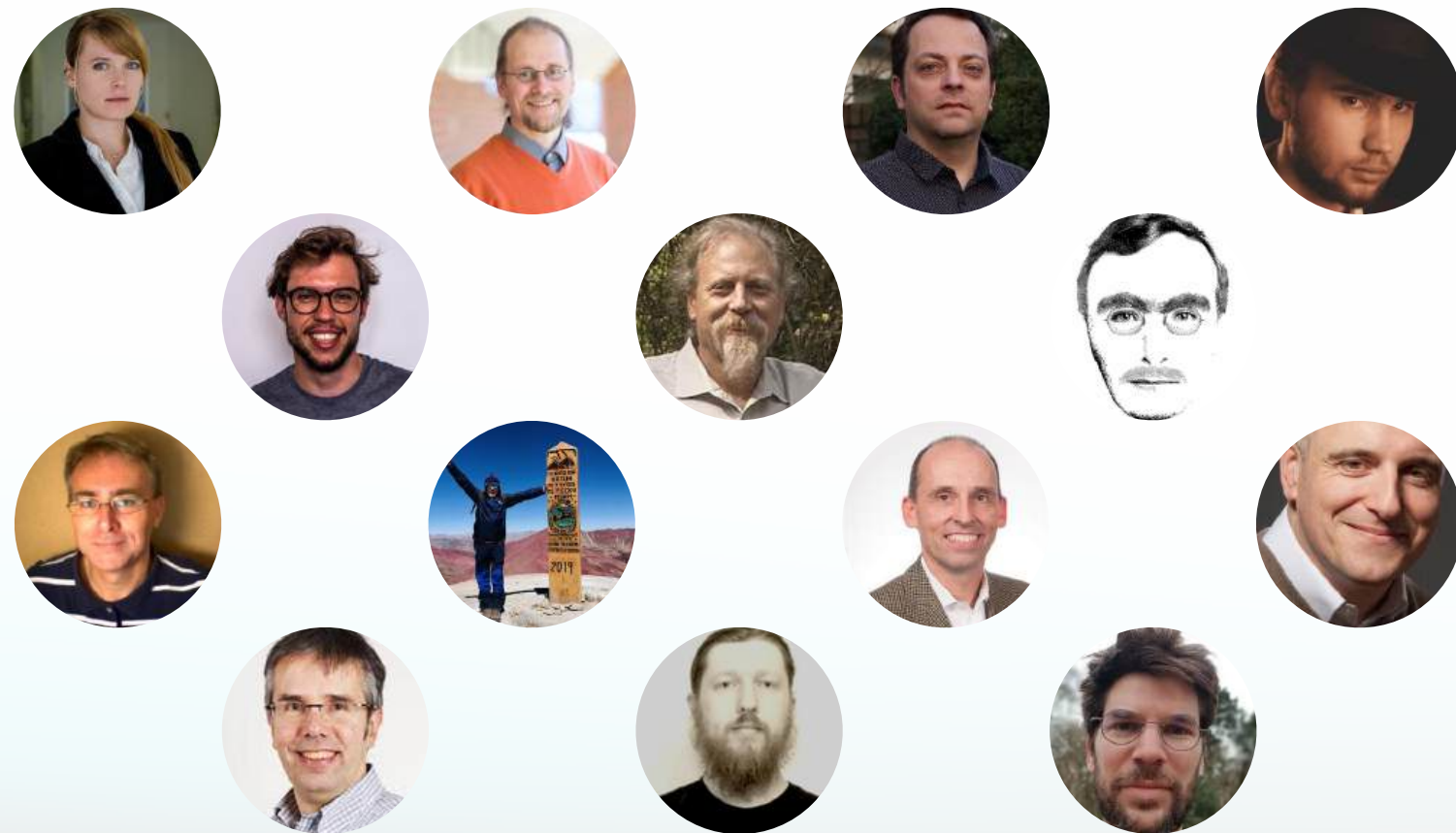
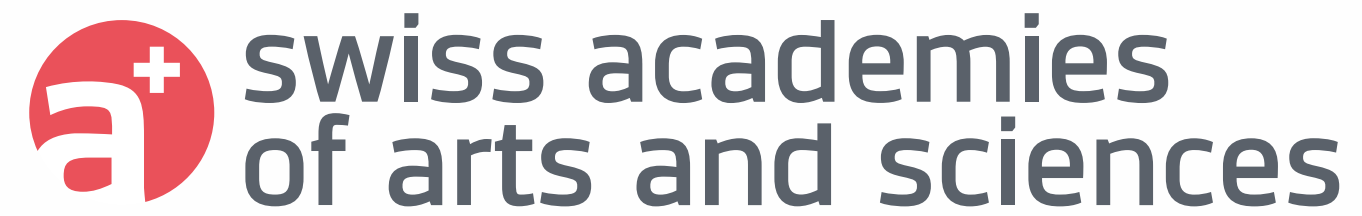
Utilisez des outils ouverts!



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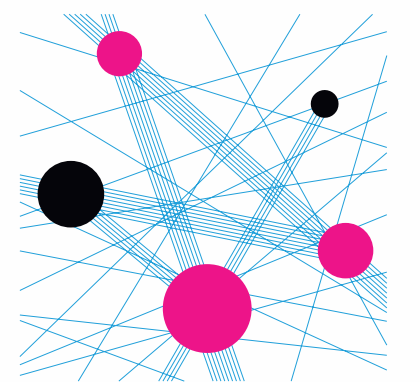
Remerciements



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